

SiGe-on-Insulator and Strained-Si-on-Insulator
for Strained-Si CMOS and Nanocrystalline-Ge Waveguides

by

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ABSTRACT

SiGe-on-insulator (SGOI) and strained-Si-on-insulator (SSOI) substrates combine both the benefits of a high-quality, monocrystalline SiGe or strained-Si surface layer with the advantages of an insulating substrate. In particular, many electronic and photonic devices are greatly enhanced by the use of such substrates. In this thesis, techniques were developed for the fabrication of SGOI and SSOI substrates for applications including strained-Si complementary metal-oxide-semiconductor (CMOS) field-effect transistors, and nanocrystalline-Ge waveguides and photonic devices.

A versatile fabrication technique was pioneered for the fabrication of low defect density SGOI and SSOI substrates, allowing for the creation of ultra-thin SGOI and SSOI, combining both the benefits of high-mobility strained-Si and SOI. The method pioneered employed wafer bonding of polished, relaxed SiGe virtual substrates to oxidized handle wafers. Layer transfer onto insulating handle wafers was accomplished using grind-etchback or delamination via implantation. Both methods were found to produce a rough transferred layer, but chemical mechanical polishing (CMP) was found to be unacceptable due to non-uniform material removal across the wafer and the lack of precise control over the final layer thickness.

To solve these problems, a strained-Si stop layer was incorporated into the bonding structure. After layer transfer, excess SiGe was removed using a selective etch process, stopping on the strained-Si. Within the context of ultra-thin SGOI and SSOI fabrication, this work describes recent improvements including metastable stop layers, low-temperature wafer bonding, and improved selective SiGe removal. Such ultra-thin SSOI substrates are ideal for fully-depleted CMOS, in comparison to some thicker SGOI substrates that were fabricated, which are required for partially-depleted CMOS.

SGOI substrates can also be used as a platform for novel photonic devices. In this work, one such example is presented, whereby nanocrystalline-Ge waveguides were fabricated using SGOI substrates. These waveguides contained Ge nanocrystals embedded in an oxide matrix and isolated from the underlying Si substrate by a buried silicon dioxide layer. Fabrication challenges revolved around the instability of SiGe-oxide, which was required as an initial material for the nanocrystalline-Ge fabrication

process. Nanocrystals were formed via the reduction of SiGe-oxide by annealing in a hydrogen ambient. Resulting nanocrystalline-Ge distributions were mapped as a function of hydrogen partial pressure, annealing temperature, and time. Based on these results, the process kinetics of the nanocrystal formation process was deduced, and a mathematical model was created based on the observations. Simulations generated predictions for nanocrystalline density as a function of temperature and hydrogen pressure which were consistent with experimental results.

Based on the demonstrations in this thesis, SGOI and SSOI were shown to be enhanced platforms for microelectronic and photonic devices. With the advent and development of the general wafer bonding method presented in this work, these substrates may serve as platforms for a multitude of future novel applications.

Thesis Supervisor: Eugene A. Fitzgerald

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Chapter 1 Introduction and Background

1.1 Motivation

Silicon-Germanium (SiGe) is on the verge of expanding its sphere of influence into the ubiquitous world of conventional Si microfabrication. The commercial interest in enhanced complementary metal-oxide-semiconductor devices (CMOS) fabricated with strained-Si on SiGe virtual substrates, dubbed ϵ CMOS, is a strong indicator of growth in the SiGe field.^{1,2} Commercial interest in SiGe for CMOS applications is due to the realization that aggressive transistor scaling, which has driven conventional bulk Si CMOS for the last four decades, may be reaching fundamental physical and economic barriers. In response, SiGe may soon be accepted into mainstream CMOS foundries, allowing for improved performance while minimizing additional capital investments. Once this occurs, a plethora of novel devices shall have the opportunity to enter mainstream fabrication environments.

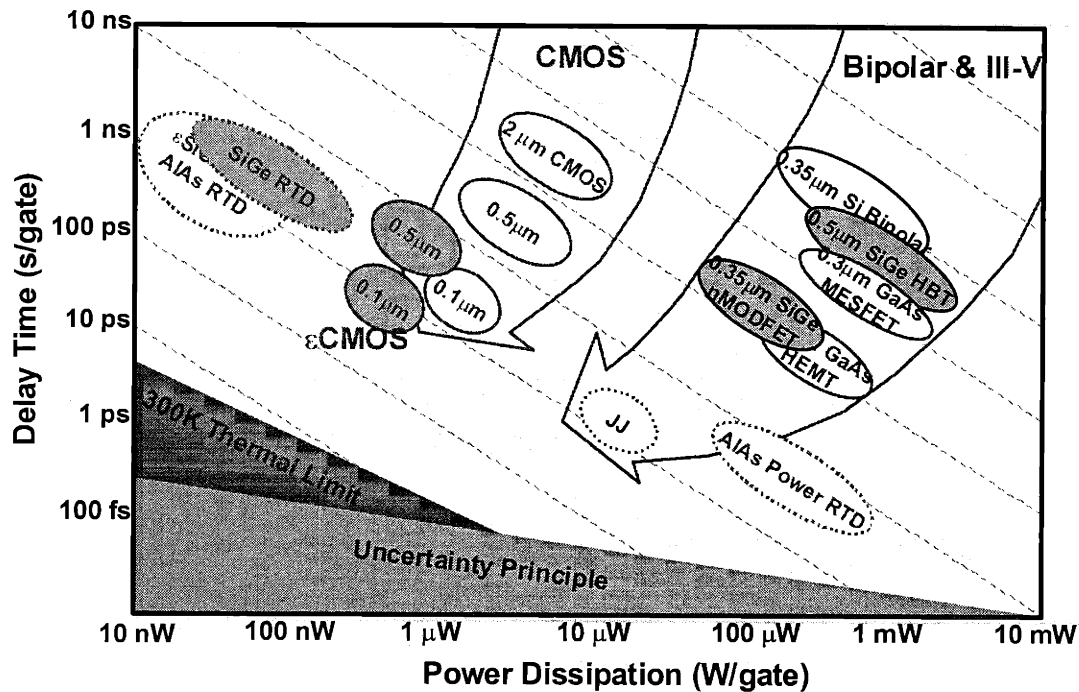


Figure 1.1 Power-delay map showing various SiGe-based electronic devices (in shaded bubbles), including SiGe HBTs, SiGe-based strained-Si CMOS (ϵ CMOS), and SiGe resonant tunneling diodes (RTDs).

SiGe has been an actively used in microelectronics research for two decades, with work ranging from research on SiGe heterojunction bipolar transistors (HBTs) to strained-Si CMOS (ϵ CMOS), as shown in Figure 1.1.³ SiGe-HBTs⁴⁻⁷ have long since been commercialized, and have been used in high-speed analog circuits,⁸⁻¹³ and mixed-signal applications employing monolithically integrated SiGe-HBTs and conventional Si CMOS (BiCMOS).¹⁴⁻¹⁷ Unfortunately, the integration of SiGe-HBT processing with CMOS is a complex task requiring twice the mask levels compared to standard CMOS, hence increasing costs significantly. In contrast, ϵ CMOS is a strong contender for the future evolution of conventional CMOS, because it allows for enhanced digital, analog and RF performance without any additional mask levels.¹⁸⁻²³

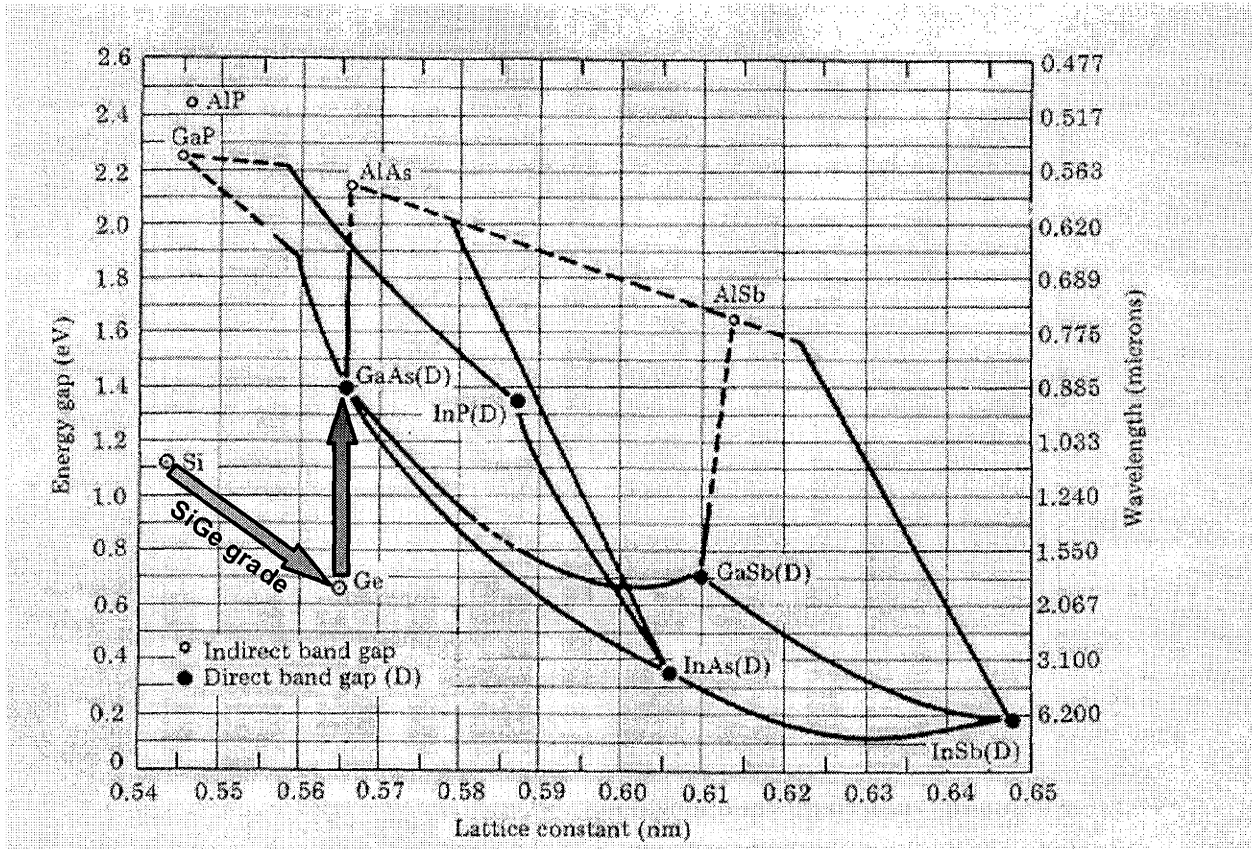


Figure 1.2 Bandgap versus lattice constant map for various semiconductors, showing how Ge on Si (formed using a SiGe graded buffer) matches the GaAs lattice constant.

Within the areas of optoelectronics and photonics, SiGe serves as an important material: (1) as an integration vehicle for direct bandgap III-Vs on indirect Si substrates,²⁴⁻²⁶ and (2) as an active material in integrated waveguide devices,^{27,28} and (3) in Ge-based photodectors.²⁹ SiGe graded buffers on Si substrates have been shown to be an effective method for the monolithic integration of low defect density Ge on Si.³⁰ Also, as shown schematically in Figure 1.2, Ge on Si can in turn be used as a substrate for the growth of lattice-matched, low defect density GaAs, allowing for the monolithic integration of a multitude of III-V opto-electronic devices on low cost Si substrates. GaAs based solar cells,²⁶ LEDs,²⁵ and lasers²⁴ on Si have already been demonstrated

using high-quality Ge on Si. SiGe has also been used to fabricate planar waveguide circuits and devices including wavelength filters,²⁸ electro-optical switches,³¹ directional couplers,³² and integrated waveguide-photodetectors.³³ SiGe-oxides serve as core materials for low-index and moderate-index contrast planar waveguide circuits possessing lower insertion and propagation loss compared to high-index counterparts.³⁴

In this work, novel SiGe-based structures and processes were studied, including (1) a study of the basic process of SiGe oxidation, (2) the stability of the mixed oxide, and (3) the synthesis of nanocrystalline Ge (ncGe) in oxide. In addition, novel structures were designed and fabricated, including (1) SiGe-on-insulator and strained-Si-on-insulator substrates intended for use in advanced ϵ CMOS, and (2) ncGe optical waveguides on Si substrates.

1.2 Overview of SiGe Fabrication Issues

Although similar to Si processing in many aspects, SiGe presents a new set of issues and restrictions. Both subtle and obvious differences make the blind application of Si microfabrication processes to SiGe structures somewhat problematic. Once a SiGe film is deposited onto a Si substrate, the remaining microfabrication steps can be performed with the standard set of Si fabrication tools, with slightly adapted recipes. Other than the additional cost of depositing SiGe on Si substrates, the only other concrete obstacle to SiGe adoption into mainstream Si foundries are SiGe fabrication issues stemming from SiGe material properties.

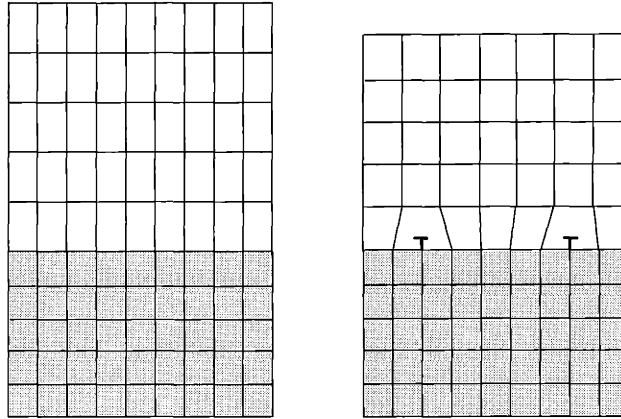


Figure 1.3 Lattice mismatched layers: (a) pseudomorphic, strained surface layer on relaxed substrate, (b) relaxed surface layer, showing misfit segments at the interface.

1.2.1 Lattice Mismatch and Dislocations

The central fabrication issue encountered when dealing with epitaxial deposition of SiGe alloys on Si substrates is lattice mismatch. Since Ge has a larger lattice constant than Si, a relaxed layer of $\text{Si}_{1-x}\text{Ge}_x$ also has a larger lattice constant, given by Vegard's law.³⁵ When a thin layer of SiGe is deposited on a Si substrate, the layer is pseudomorphic, implying that the in-plane lattice constants are equal, and hence the SiGe is fully strained, as shown in Figure 1.3(a).³⁶ If the SiGe layer is thick enough, strain in the SiGe layer can be relieved (or relaxed) via the generation of misfit dislocations running along the interface between the Si and the SiGe layer, as presented schematically in Figure 1.3(b).³⁷

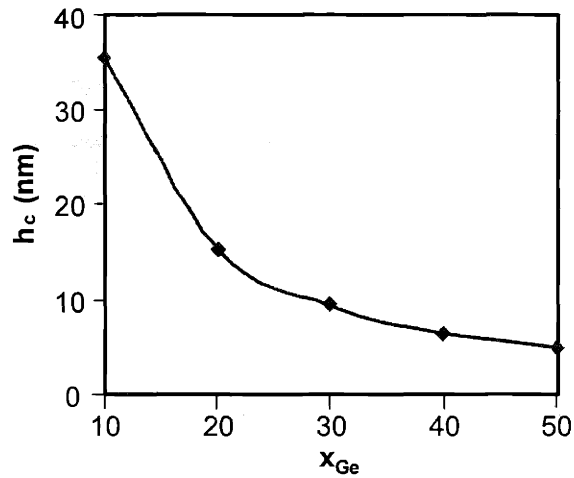


Figure 1.4 Critical thickness of $\text{Si}_{1-x}\text{Ge}_x$ deposited on Si substrate.

Being a distortion of perfect crystals, dislocations have an associated formation energy, and hence only form when the total energy of the system, comprising of strain energy and dislocation energy, is reduced by their introduction. From this thermodynamic standpoint, it can be shown mathematically that dislocation generation occurs when the SiGe layer is larger than an equilibrium critical thickness, h_c , plotted in Figure 1.4; beyond this thickness, strain is partially relieved via the formation of misfit dislocations.³⁸ In addition, the observed critical thickness at lower temperatures is significantly larger than the thermodynamic value due to kinetic barriers present in nucleating dislocations as shown in Figure 1.5.³⁹

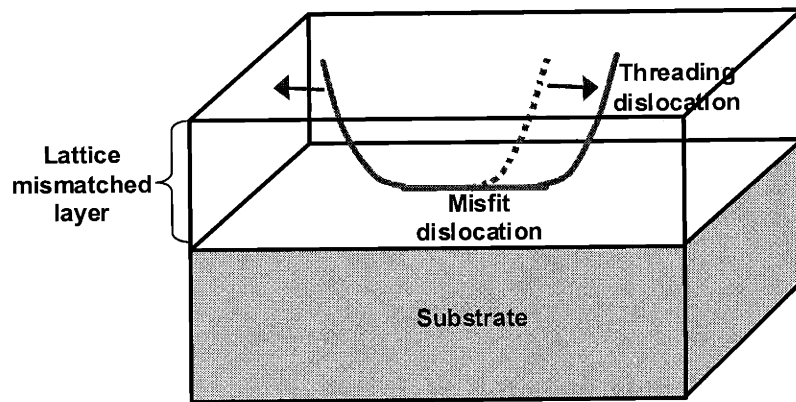


Figure 1.5 Diagram showing extension of a misfit segment via dislocation glide.

Since dislocations cannot terminate in the bulk of the crystal, misfit dislocation line segments, at the interface of the SiGe and the Si, must be accompanied by additional segments that terminate at the surface. Such segments are referred to as threading dislocations (shown in Figure 1.5) and can degrade device operation.⁴⁰ This is especially true for minority carrier electronic devices, where the threading dislocations can serve as electron-hole recombination sites, or in the case of lasers, as nucleation centers for dark-line defects.^{41,42} For majority carrier devices, dislocations act as scattering sites, and can lower electron and hole mobility if their density is high.⁴³ Finally, dislocations are also detrimental to device processing, since they can act as fast diffusion channels for dopants.

In the case of fully strained-SiGe layers on Si, namely layers where the SiGe thickness is less than the critical thickness, dislocations are not generated. In contrast, for microelectronic and opto-electronic devices requiring relaxed SiGe layers, threading dislocations must be aggressively minimized to achieve optimal device performance. The direct deposition of a highly lattice mismatched material on a substrate typically leads to high threading dislocation densities, which again is unacceptable for most electronic and opto-electronic applications.

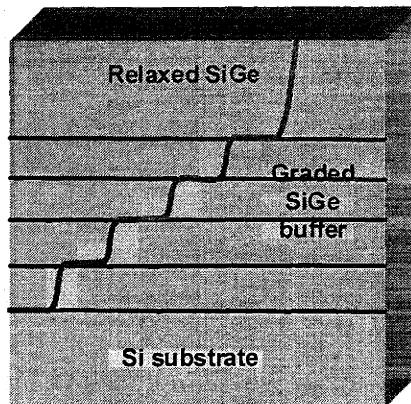


Figure 1.6 SiGe graded buffer showing the reuse of threading dislocation at each layer interface to relieve the strain of each newly deposited layer.

1.2.2 SiGe Graded Buffers

A relaxed layer of SiGe epitaxially deposited onto a Si substrate, generates a large number of threading dislocations, connected to short misfit segments at the growth interface. To decrease the threading dislocation density, compositionally graded buffer structures can be employed to deposit high-quality, relaxed SiGe layers on Si substrates.⁴⁴

As shown in Figure 1.6, the compositional grading approach consists of depositing SiGe layers of gradually increasing Ge content on a Si wafer. Typically, the grading rate is about 10% Ge per μm of material deposited, and the individual SiGe layers are around 200 nm. This graded structure continues until the desired Ge content is attained on the surface, and then a capping layer of relaxed SiGe with the final Ge fraction is deposited. The final SiGe cap is almost completely relaxed and has a minimal number of threading dislocations, typically low 10^5 cm^{-2} for 25% Ge, to low 10^6 cm^{-2} for pure Ge on Si. In contrast, for Ge deposited directly on Si, the threading dislocation density is around 10^9 cm^{-2} .

The principle way graded buffers reduce threading dislocation densities is by the re-use of existing threads to relieve strain due to newly deposited layers, as shown schematically in Figure 1.6. Since each layer is only slightly strained, few threads are generated to relax it, and via their re-use in subsequent layers, the total threading density is kept to a minimum. Since the efficient re-use of threads depends on their glide velocity (which increases exponentially with temperature), higher temperatures are usually preferred for graded buffer growths. Unfortunately, dislocation nucleation also increases with temperature and strain, however this effect can be kept at a minimum by using slow grading rates, which ensure the system only experiences low strain levels.⁴⁵

The final result of the graded buffer is the creation of a surface film of relaxed SiGe with a low threading dislocation density, typically on the order of 10^5 cm^{-2} for low Ge content films. Unfortunately, the surface is typically quite rough (with an rms of about 10 nm) due to the presence of strain-induced growth undulations due to underlying misfit dislocations. This surface roughness is termed cross-hatch roughness, and is typically removed by employing chemical mechanical polishing.²⁹ Irrespective of whether the surface is planarized, the final structure is many times referred to as a SiGe virtual substrate.

1.2.3 SiGe Processing Issues

Other than lattice mismatch, there are multitudes of other processing issues relevant to SiGe, including (1) wafer cleaning of the SiGe surface, (2) diffusion of SiGe into surrounding layers, and (3) oxidation of SiGe. Processing restrictions implied by the aforementioned issues are important examples of how SiGe and Si fabrication differ.

1.2.3.1 Cleaning

For Si microfabrication, Si wafers are typically cleaned with a standard RCA clean, consisting of two baths: Standard Clean-1 (SC1) which removes organic contaminants and consists of a 75-80°C solution of $\text{NH}_4\text{OH}:\text{H}_2\text{O}_2:\text{H}_2\text{O}$, and SC-2 which removes metallic contaminants and consists of a 75-80°C solution of $\text{HCl}:\text{H}_2\text{O}_2:\text{H}_2\text{O}$.⁴⁶ Although excellent for Si surfaces, using an RCA clean on a SiGe surface leads to disastrous results. SiGe is violently etched by SC-1 leaving behind a rough surface.⁴⁷ Etch rates for SiGe in SC-1 range from 3 nm/min for $\text{Si}_{0.6}\text{Ge}_{0.4}$ to a dramatic 4 $\mu\text{m}/\text{min}$ for pure Ge. Even bulk Si is etched in SC-1, at a rate of about 0.5 nm/min, but in this case, the uniform removal of surface material does not effect the final devices. In contrast, for SiGe structures, removal of significant amounts of surface material is unacceptable, because the first tens of nanometers of material may contain device channels. For example, for surface strained-Si CMOS, the strained-Si on the virtual SiGe substrate is typically only 10-30 nm thick.

Given these restrictions, it is crucial to find an acceptable replacement clean for the SC-1 bath that does not etch SiGe or Si significantly. The piranha clean meets this criteria. Consisting of $\text{H}_2\text{O}_2:\text{H}_2\text{SO}_4$ (1:3), the piranha clean removes organic surface contaminants, while at the same time only removing a small amount of SiGe.⁴⁸ For example, our estimates indicate that in the case of a relaxed $\text{Si}_{0.75}\text{Ge}_{0.25}$, a piranha clean oxidizes no more than 2 nm of the SiGe. The piranha clean can be employed as a substitute for the SC-1 bath in the RCA clean, yielding what is typically called a modified

RCA clean. The SC-2 bath is still usually employed since it removes any metallic contaminants, however its duration is often shortened.

1.2.3.2 Diffusion

Other than wafer cleaning, interdiffusion of SiGe layers imposes another important restriction on the thermal budget for the processing of SiGe structures. Unlike Si, which can withstand processing temperatures as high as 1100°C, SiGe layers interdiffuse quickly at this high temperature. Diffusivity increases rapidly with Ge content, making this a serious, though solvable problem, for high Ge content structures. In the case of a buried strained-Si layer surrounded by Si_{0.75}Ge_{0.25}, it is found that two hours at 800°C leads to only negligible (lattice) interdiffusion.⁴⁹

Interdiffusion along dislocation cores in SiGe may be significantly faster compared to lattice diffusion. This phenomenon is well-known in metals, but specific diffusion data for semiconductor dislocation core diffusion is sparse. Typically, in the case of metals, at $T = T_{\text{melt}}/2$, diffusion along a dislocation core (approximately 2 atoms in width) is about 5 orders larger than bulk diffusion.⁵⁰ Interdiffusion along dislocation cores could be a serious problem, especially in cases where the SiGe layers are only nanometers thick and the processing is extremely sensitive to higher Ge fraction that might be present in the dislocation core. Situations where this phenomena could create problems should be inspected closely to ensure that interdiffusion of Ge along dislocation cores is not taking place.

1.2.3.3 Oxidation

Silicon's dominance for microelectronic applications was made possible in part by one crucial material: silicon dioxide, SiO_2 . By oxidizing a Si surface in either a wet (H_2O) or dry (O_2) ambient, at elevated temperatures ranging from 700-1100°C, one creates a layer of thermal SiO_2 on the surface of the Si wafer. Silicon dioxide is both chemically stable, and possesses a very low interfacial state density, making it an ideal material for a gate dielectric. These properties have made SiO_2 the insulator of choice for gate dielectrics in metal-oxide-semiconductor (MOS) microelectronics, allowing for the creation of low power, digital, complementary MOS (CMOS), which has become the preferred technology for digital applications.

Unfortunately, SiGe oxides are not so well-behaved, and are much more complex. The blind application of Si oxidation recipes to SiGe leads to expulsion of Ge from the oxide.⁵¹ Ge expulsion is the first indication that SiGe-oxides are highly unstable, and are not favored thermodynamically. Only under low-temperature conditions is it possible to incorporate Ge into the oxide, however caution is still required when attempting to process the resulting oxide.⁵² Chemical instability is a serious issue, and the best applications of such oxides are those that employ this instability as an asset, rather than a liability.

1.3 Summary of Thesis Projects

Of crucial importance to the fabrication of many SiGe-based applications is SiGe oxidation, and the stability of such an oxide. Chapter 3 presents results on the oxidation of SiGe for various temperatures and Ge content. In particular, Ge incorporation into the oxide was investigated, in addition to oxidation rate enhancements compared to Si, and the potential for selective lateral oxidation based on this enhancement.

As presented in Chapter 4, the instability of GeO_2 can be used to precipitate Ge nanocrystals from mixed SiGe-oxide. Nanocrystalline Ge opens the door to fabrication of photonics devices employing the nonlinear optical properties of nanocrystals. Based on this concept, nanocrystalline Ge (ncGe) in oxide was formed by hydrogen reduction of SiGe-oxide under various annealing conditions, leading to an understanding of the process kinetics. In addition, the variation of bandgap versus nanocrystal size was used to explain the measured photoluminescence.

Chapter 5 presents results of ncGe planar waveguide design and fabrication. A major obstacle was the creation of an adequate silicon dioxide cladding between the ncGe core and the Si substrate. Three methods were explored: (1) oxidation of polycrystalline SiGe on thick SiO_2/Si , (2) high-temperature 1100°C oxidation of the underlying Si, and (3) oxidation of SiGe on thinned-SOI. Once a suitable fabrication method was developed, nonlinear directional couplers (NLDC) were designed, and a series of waveguides and NLDCs were fabricated. Based on the ncGe waveguide work, we realized that starting substrates of SiGe-on-insulator (SGOI) would have greatly simplified the waveguide fabrication process.

In addition, SiGe-on-insulator (SGOI), and strained-Si-on-insulator (SSOI) may soon be required as advanced CMOS substrates. To address this demand, we pioneered a novel fabrication process for SGOI and SSOI using wafer bonding of SiGe virtual substrates to oxidized Si substrates, as presented in Chapter 6. Two approaches for layer transfer were developed: (1) grind/etchback and (2) Smart-cut.TM In addition, a Si etch-stop layer was incorporated into the bonding structure allowing for the creation of smooth, uniform, thin SiGe or strained-Si layer on oxide. The proposed method was optimized using metastable strained Si stop layers and low-temperature wafer bonding. Finally, tests of various selective SiGe etches were conducted and optimized for SGOI and SSOI fabrication.

Chapter 2 SiGe Deposition and Characterization

2.1 Ultrahigh Vacuum Chemical Vapor Deposition

Chemical vapor deposition (CVD) is a widespread semiconductor deposition process, whereby source gases flow over a heated substrate at elevated temperatures, and deposit material via chemical reactions at the surface. For example, in the case of Si deposition, dichlorosilane (SiH_2Cl_2) or silane (SiH_4) are typical source gases. Originally, Si CVD was performed in atmospheric pressure reactors at very high temperatures. At atmospheric pressures, the presence of even low levels of oxygen or water vapor leads to the formation of native oxide, which impedes the growth of high-quality epitaxial semiconductor films. Fortunately, at extremely high temperatures this oxide is unstable and hence atmospheric pressure CVD (APCVD) is an acceptable deposition method. Typically, Si growth is performed using dichlorosilane diluted with hydrogen (H_2) at 1100°C .

The problem of oxygen and water vapor contamination reappears when attempting low-temperature epitaxy to avoid layer interdiffusion or dopant diffusion. In addition, in the case of SiGe layers, low temperatures are sometimes desirable if layers need to grown metastable, allowing for layer thicknesses greater than the equilibrium critical thickness. Such low-temperature growths are usually performed at temperatures of 650°C or lower. At such low temperatures, native oxide formed from oxygen and water contamination is stable, and hence is incorporated into the growing film. The need for high-quality epitaxial growth at low temperatures requires that the oxygen and water vapor content in

the reactor are suitably low. Ultrahigh vacuum chemical deposition (UHVCVD) accomplishes this by ensuring a base reactor pressure of 10^{-9} Torr, corresponding to oxygen and water vapor partial pressures of about an order of magnitude lower.⁵³ This ultra-pure background insures that there will be no significant oxygen incorporation into the deposited film.

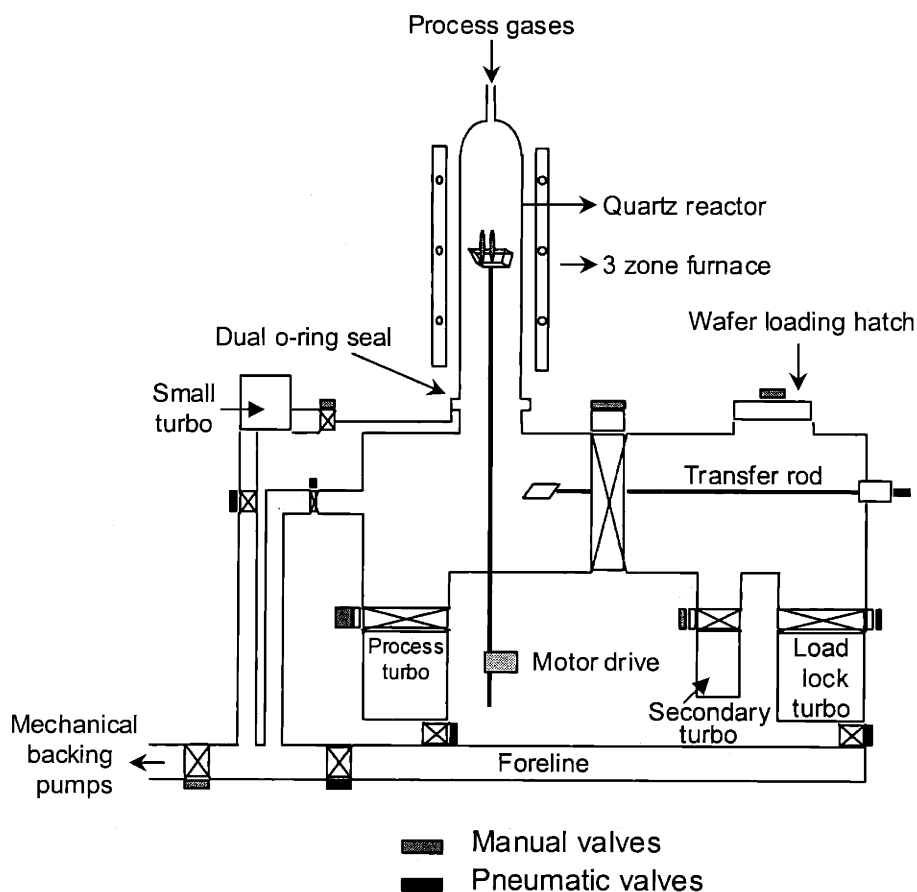


Figure 2.1 Schematic diagram of hot-walled UHVCVD reactor used for SiGe deposition.

2.1.1 UHVCVD Reactor

All the SiGe structures in this thesis were deposited using a research UHVCVD reactor, presented schematically in Figure 2.1. The reactor can accept wafers ranging in

size from 2" to 6", but all samples grown in this work were on 4" Si wafers with a vertical quartz boat containing up to 10 wafers. The UHVCVD reactor is composed of a load lock chamber, for wafer loading, and a growth chamber having a hot-walled quartz tube, heated by a three-zone furnace. Both the load lock and growth chamber are evacuated by independent turbo pumps backed by both mechanical and roots pumps, allowing the background pressure in the reactor to attain 10^{-9} Torr at 900°C. A dual o-ring seal between the quartz tube and the rest of the growth chamber allows for a UHV seal due to the evacuation of the inner o-ring by a small turbo pump.

Source gases for SiGe deposition consist of SiH_4 and GeH_4 , whereas B_2H_6 and PH_3 are employed for p-type and n-type doping. Source gases flow into the top of the vertical quartz tube and past the wafers situated in the mid-region of the tube. A gate valve above the growth chamber turbo allows the pumping rate during growth to be varied, providing a growth pressure ranging from 3 to over 25 mT. This flexibility in growth pressure, in conjunction with the ability to vary growth temperature from 400 to 900°C, allows the reactor to deposit both thick graded layers and thin sub-nanometer layers. Low temperatures are ideal for reducing interdiffusion of thin layers, and for avoiding GeH_4 gas phase nucleation present during the growth of high Ge content layers. On the other hand, high growth temperatures allow for fast growth rates, and hence reduce growth times for micron-thick graded buffers.

2.1.2 SiGe Deposition Procedure

Before epitaxial growth, wafers must be cleaned to remove any organic contaminants, or native oxide present on the surface. A 7-10 min piranha clean

consisting of a 3:1 solution of H_2SO_4 and H_2O_2 was employed to oxidize any organic contaminants present on the wafer surface. Cleaning times were scaled according to surface Ge content of the wafers; a 10 min clean was used for virgin Si wafers, whereas a shortened 7 min clean was employed for $\text{Si}_{0.75}\text{Ge}_{0.25}$ virtual substrates. The shortened time accounts for the higher reactivity possessed by Ge rich layers, hence possibly limiting the amount of surface roughening induced by the piranha clean on the SiGe surface. After the piranha clean, wafers were rinsed, and then dipped into a dilute HF solution for 30 s to 1 min to strip the native oxide from the surface, leaving behind a hydrogen-terminated hydrophobic surface.

After cleaning, the wafers are loaded into the reactor load lock and pumped down to UHV pressures, requiring a minimum of about two hours. Following pump-down, the wafers are transferred from the load lock to the growth chamber, where they attain an initial temperature of about 200°C . Wafers are left at this temperature for about 30 min, which desorbs any water vapor and organics. Typically, wafers are then raised to a temperature of 850°C and left at that temperature for 5-10 min to desorb any native oxide that may still be present. Next, the temperature is lowered to the desired growth set-point, and the growth pressure is calibrated by flowing Ar and adjusting the manual gate valve above the turbo. Once the temperature stabilizes, growth is initiated by flowing the required amounts of source gases to deposit the desired SiGe layers. If required by the structure, a multi-step growth can be performed, whereby multiple growth temperatures and pressures are used for the various layers. For example, a graded buffer can be grown

at high temperature and pressures, followed by a growth at lower temperatures and pressure to deposit thin Si and SiGe device layers.

2.2 Characterization Methods

A number of material characterization methods are required to analyze the quality of deposited SiGe and resulting structures formed after processing. Layer thickness, uniformity, surface roughness, defects, optical response, Ge content and strain are crucial properties that such methods must provide.

2.2.1 Transmission Electron Microscopy

Of utmost importance in any semiconductor thin-film research, especially in the case of epitaxy, is the ability to visually determine layer thickness and material quality. Transmission electron microscopy (TEM) fulfills that need, and in essence is to semiconductor epitaxy what optical microscopy is to medical science. No other technique has the needed high resolution, flexibility and maturity required to accurately image thin epitaxial layers.

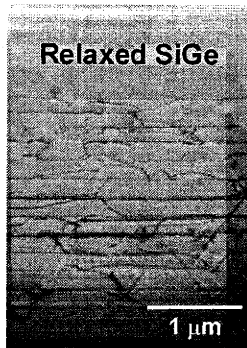


Figure 2.2 XTEM micrograph of relaxed SiGe graded buffer, showing threading and misfit dislocations.

Samples can either be viewed in cross-section TEM (XTEM) or plane-view TEM (PVTEM). XTEM allows for accurate determination of layer thickness, and nanoscale structures near the surface of the samples; an example of a XTEM micrograph of a SiGe graded buffer is shown in Figure 2.2. On the other hand, PVTEM is more aptly suited for examining defects present near the surface, in particular threading and misfit dislocations. Although ideal for viewing nanoscale features, TEM is far from perfect for the determination of large-area features. A classic example of such a case is inaccurate threading dislocations measured via PVTEM. Due to the small area sampled in TEM, the method can only be used to accurately measure threading dislocations if densities are on the order of 10^7 cm^{-2} or larger. Lower dislocation densities must be measured via techniques that reveal large surface areas.

The limitations of the TEM technique are not to be overlooked. Although TEM is unparalleled in resolution and for the information it provides, sample preparation is time consuming and very delicate. Sub-micron thick samples are required to achieve electron transparency, hence samples must be thinned mechanically and then milled (with an ion mill) for multiple hours. In addition, samples can sometimes be easily damaged during

this preparation process. Barring sample preparation time, TEM is the ultimate characterization technique for imaging thin films.

2.2.2 Scanning Electron Microscopy

Scanning electron microscopy (SEM) allows for the quick imaging of thin-film structures with excellent resolution for thickness of about 100 nm or greater. SEM is an ideal characterization method in cases when thickness and structure determination are of primary interest, and features sizes are greater than 100 nm. Sample preparation merely consists of cleaving a small piece of the wafer, and inserting it into the SEM, for either cross-section or plane-view imaging.

In this thesis, SEM was often employed to quickly determine deposition thickness after a growth, and hence allow for immediate planning of the following growth run. In addition, SEM was ideal for viewing large-area structures, especially after dry etching. The major limitation of the technique is the inability to ascertain epitaxial material quality, or to view nanoscale structures. For this reason, TEM was always employed as a final characterization method after an interesting sample was fabricated.

2.2.3 Etch-Pit Density

Large-area sampling of dislocations is crucial for the determination of epitaxial film quality. As mentioned, TEM is an acceptable method only in cases when the threading dislocation density is greater than about 10^7 cm^{-2} . As is the case for most high-quality SiGe films, the threading density is much lower, and etch-pit density (EPD) serves as an ideal method for dislocation counting. EPD employs the enhanced etching

properties of dislocations, which is typically due to either strain-related effects or chemical bonding effects due to dislocation core etching.

Irrespective of source, dislocations exposed to specific chemical etchants etch at a different rate compared to the bulk regions. This in turn implies that after etching, the sample surface has pits corresponding to dislocation threads, as shown in the example in Figure 2.3(a), and line segment trenches corresponding to misfit segments, shown in the example of Figure 2.3(b). These surface depressions can easily be viewed under differential interference (Nomarski) optical microscopy, and hence large areas of dislocations can be easily imaged and counted. EPD was often employed in this work to determine threading and misfit dislocation densities.

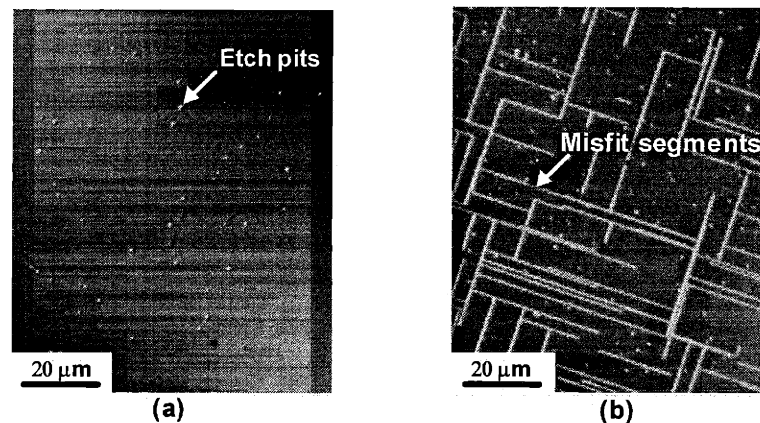


Figure 2.3 Nomarski optical images of SiGe surfaces, showing (a) etch pits due to threading dislocations terminating at surface, and (b) line segments due to misfit dislocations.

Major caveats concerning EPD are the inability of the chosen etchant to selectively etch dislocations, or the inaccuracy associated with measuring samples with very high dislocation densities. When threading densities are too high, etch pits can form from groups of multiple dislocations close to each other that can sometimes resemble one etch pit, hence

leading to an under-estimate of the total thread density. For this reason, EPD measurements should only be trusted if the threading density is on the order of 10^6 cm^{-2} or lower, where TEM measurements are less reliable.

Dilute Schimmel selective etches were used most frequently in this thesis to analyze dislocation densities for low Ge content SiGe layers. The etch consisted of a 5:4 solution of 0.3 molar chromic acid (aqueous CrO_3) and hydrofluoric acid (49% HF).⁵⁴ The etch rate of this solution for undoped, relaxed $\text{Si}_{0.75}\text{Ge}_{0.25}$ is about 18 nm/s, and decreases with increasing Ge content. If Ge-rich layers were being studied, a substitute for the Schimmel etch is needed due to the low Schimmel etch rate. An ideal substitute is the iodine etch, consisting of a 5:10:11 solution of HF acid, nitric acid (HNO_3), and acetic acid (CH_3COOH), mixed with 30 mg of iodine powder (I_2) in 260 mL of the solution.⁵⁵ Since this solution has a Ge etch rate of about 200 nm/s, an etch time of a few seconds is ideal for revealing etch pits in pure Ge.

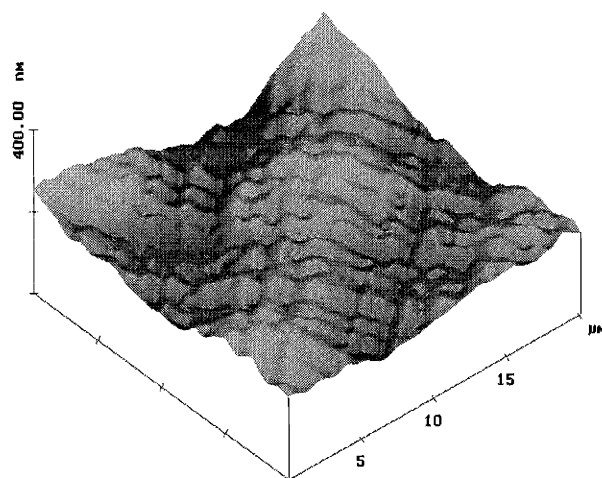


Figure 2.4 AFM surface scan showing crosshatch roughness of a SiGe graded buffer with a thick SiGe cap layer.

2.2.4 Atomic-Force Microscopy

In cases when surface roughness is of importance, atomic-force microscopy (AFM) is the optimal characterization technique. A quick surface scan with a stylus allows for accurate imaging of surface roughness, as shown in the example in Figure 2.4. Built-in analysis tools are employed to determine the root-mean-square roughness (rms), and the maximum peak-to-valley height. The main caveat associated with this technique is that it samples a very small area of the surface, typically a square area ranging from about $1 \times 1 \mu\text{m}^2$ to $20 \times 20 \mu\text{m}^2$. When comparing roughness from different samples, it is advisable to use rms roughness values determined using similar scan area sizes.

In addition, small scan areas are not necessarily representative of the entire surface of the sample, and hence results need to be interpreted with caution, especially when employing AFM to measure large-area properties. In particular, dislocation pits can be viewed using AFM, but the small scan area requires that to achieve accurate statistics, a multitude of scans be taken in many different areas on the sample.

2.2.5 Raman Spectroscopy

Peaks of Raman spectra correlate to bond vibrational states. From peaks locations, it is possible to extract information about both SiGe alloy composition and strain in thin films.⁵⁶ Raman spectroscopy has the ability to characterize films with film thicknesses as low as 10 nm, with a lateral resolution equivalent to the laser spot size. In contrast, X-ray diffraction typically requires thin films be at least 100 nm thick, and hence is unacceptable for the direct analysis of ultra-thin 10-nm thick strained-Si on SiGe. It has been found that when using high-resolution Raman diffraction gratings, the accuracy of

Raman spectroscopy for the determination of alloy composition and strain is equivalent to X-ray diffraction in the case of thick films, and superior for ultra-thin films.⁵⁶

In this work, Raman spectroscopy has been employed in both the characterization of Ge nanocrystals and for SiGe-on-insulator and strained Si-on-insulator strain and alloy determination. For the case of Ge nanocrystals in oxide, the presence of a strong Ge-Ge vibrational peak indicates the existence of Ge nanocrystals in the sample.⁵⁷ This is an ideal example where this conclusion would have been very difficult to ascertain quickly via any other characterization method. In all samples analyzed with Raman spectroscopy, no sample preparation was necessary, allowing for a quick survey of numerous samples. Samples with promising Raman spectra were then carefully imaged with TEM.

2.2.6 Photoluminescence

When optical properties are of interest, as is the case for opto-electronics or photonics applications, a characterization method that maps optical response is required. Photoluminescence (PL) is the method of choice for determining all possible emission wavelengths from a material. Laser light of a low wavelength (hence high energy) excites carriers in the material, and as they decay to lower states, they emit light of various wavelengths, yielding an intensity versus emission wavelength spectrum. PL spectra are characteristic of the materials' electronic states, and directly correlate to the band structure. Strain and alloy composition have a direct influence on the electronic states and hence vary the location of the PL peaks. Since thermal effects can interfere with photon emission via non-radiative mechanisms, like phonon creation, PL is often

carried out at low temperatures, to quench out non-radiative pathways, leading to a clear picture of the radiative mechanisms. Valuable information about bandgaps can be gleaned from low-temperature PL spectra.

Chapter 3 SiGe Oxidation

3.1 Thermodynamics of Si-Ge-O

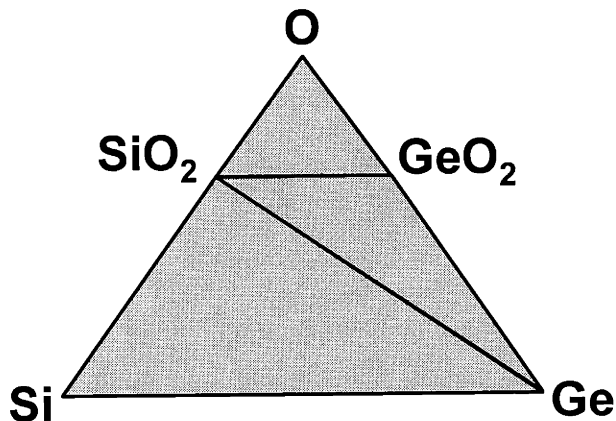


Figure 3.1 Ternary phase diagram for Si-Ge-O system, at 727°C and 1 bar. GeO_2 is not stable in the presence of Si, since a straight line connecting these two compounds is dissected by the SiO_2 -Ge tie-line.

Although pure germanium was one of the first materials used for transistor research, it did not achieve the wide-spread application enjoyed by Si. The main reason for this is due to the instability of GeO_2 ; for example, GeO_2 dissolves in room temperature water. GeO_2 is unstable in the presence of Si, as can be seen from the phase diagram in Figure 3.1.⁵⁸ This instability can be easily understood from the large enthalpy difference between GeO_2 and SiO_2 , namely 551 kJ/mol for GeO_2 and 910 kJ/mol for SiO_2 .⁵⁸ This implies that GeO_2 in contact with Si will reduce to form SiO_2 with expelled Ge.

In the case of dry oxidation of SiGe, the above thermodynamics are summarized by the two following reactions:





with an additional reduction reaction which is limited by the amount of Si that can be supplied to the interface through diffusion from the bulk:



For wet oxidation, similar equations apply, except that the O_2 is replaced by H_2O . In conclusion, for both wet and dry oxidation, this reduction reaction explains the instability of GeO_2 in contact with Si.⁵⁹

3.2 Oxidation Kinetics

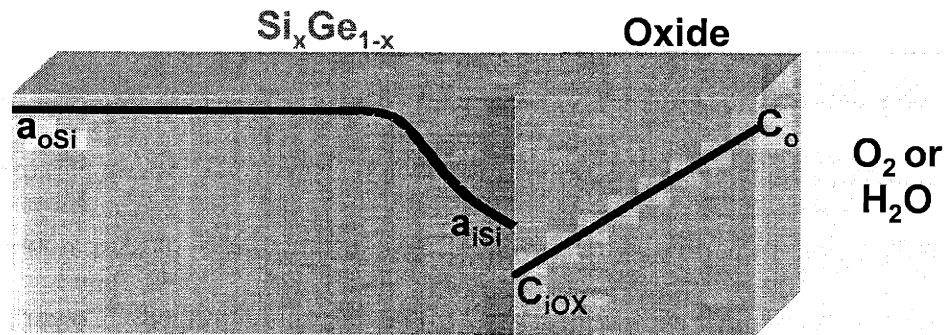


Figure 3.2 One-dimensional model for oxidation of SiGe for the case where Si can be supplied fast enough to the oxide interface so that GeO_2 reduction occurs.

A diagram of the SiGe oxidation process is shown schematically in Figure 3.2, where C_{ox} denotes the concentration of oxidant molecules, and a_{Si} denotes the activity of silicon. As was discussed earlier, as SiGe is oxidized, both GeO_2 and SiO_2 are created at the oxide interface, but since GeO_2 reduction consumes Si, one observes a drop in the Si concentration near the oxide interface, and a resulting buildup of expelled Ge.

This phenomenon is very sensitive to kinetics, and there are essentially two general temperature regimes.⁵² At sufficiently high temperatures (800°C or higher) the

diffusivity of Si in SiGe is large enough so that Si can be supplied at a fast enough rate for the GeO_2 to be completely reduced. This leads to the formation of pure SiO_2 with an expelled Ge layer ahead of the oxidation front.⁵¹ An alternative viewpoint is that at sufficiently high temperatures, the expelled Ge at the oxidation front diffuses away rapidly from the advancing interface, and hence is not reincorporated into the growing oxide.

On the other hand, at lower temperatures (700°C and lower), any expelled Ge diffuses too slowly away from the quickly advancing oxidation interface, and hence is re-oxidized and reincorporated into the growing oxide, thereby forming a mixed $\text{SiO}_2\text{-GeO}_2$ oxide.

3.2.1 Dry and Wet Oxidation

Depending on whether oxidation is wet (H_2O) or dry (O_2), kinetic parameters can be quite different. In particular, the reaction rate constants for Si oxidation (k_{Si}), and Ge oxidation (k_{Ge}) vary drastically with respect to the oxidant employed. For wet oxidation, it has been found that k_{Ge} is much larger than k_{Si} , which leads to a dramatic enhancement in the wet oxidation rates of SiGe in comparison to Si.⁶⁰ On the other hand, for dry oxidation, experiments indicate that the oxidation rates of SiGe are similar to those of Si, and hence k_{Ge} must be similar to k_{Si} .

It has been proposed that the cause of this difference is Ge-induced suppression of Si interstitial formation at the oxidation front.⁶¹ For rapid oxidation (*e.g.*: wet), the oxide front advances very quickly and Si interstitial diffusion away from the front is the rate-limiting step; Ge presence at the interface reduces the Si interstitial creation and hence

increase the reaction rate. On the other hand, for slow oxidation (*e.g.*:dry), the oxide front advances very slowly and Si interstitials have plenty of time to diffuse away from the front. In this case, the interstitial diffusion is not rate limiting, and therefore Ge suppression of their creation does not enhance the reaction rate.

3.2.2 Theory of SiGe Oxidation

In order to describe SiGe oxidation it is necessary to explore the various temperature extremes for the oxidation process; namely high and low temperature regimes (set by the Ge diffusivity in the SiGe alloy). At the low temperature end, Ge diffusion in the SiGe is very slow compared to the oxidation rate, whereas for higher temperatures, Ge diffusion in the alloy is fast enough so that expelled Ge is not overtaken by the moving oxidation front. Models for each regime are presented.

3.2.2.1 High-Temperature Model

To gain a better understanding of the physics for high-temperature SiGe oxidation, a one dimensional oxidation model is implemented numerically. The original model, developed by Hellberg,⁵⁹ is valid for temperatures and times when enough Si can be supplied to the oxide interface to reduce all the GeO₂ that has formed. This implies that the model is ideal for dealing with the initial phase of 800°C oxidation, during which pure SiO₂ forms due to Ge expulsion. In the current implementation, the model is extended to allow for the simulation of the subsequent mixed-oxide phase of the 800°C oxidation process, during which time all the Si at the interface has been depleted and Ge incorporation into the oxide begins. This is accomplished by simply setting the oxide-

front boundary condition such that the Ge fraction is unity, while allowing the Ge flux at the interface to vary. In contrast, during the initial oxidation phase, the Ge flux at the interface is set and the interface concentration is allowed to vary.

A diagram of the one-dimensional oxidation process is shown in Figure 3.2. From Fick's law, we can deduce that the flux of oxidant molecules reaching the front is given by

$$J_{OX} = D_{OX}(C_O - C_{iOX})/L_{OX} \quad \text{Eqn. 3.4}$$

where D_{OX} is the oxidant diffusivity, C_O is the oxidant solid solubility in the oxide, C_{iOX} is the oxidant concentration at the oxide front, and L_{OX} is the oxide thickness. The other fluxes required are those of Si and Ge driving the oxidation reactions



Corresponding Si and Ge oxidation fluxes at the front are given by

$$J_{Si}^{R1} = k_{Si} a_{iSi} C_{iOX} \quad \text{Eqn. 3.7}$$

$$J_{Ge}^{R2} = k_{Ge} (1 - a_{iSi}) C_{iOX} \quad \text{Eqn. 3.8}$$

where k_{Si} and k_{Ge} are the reaction rate constants for the oxidation reactions, and a_{iSi} is the Si activity at the oxide/SiGe interface. At steady state, the oxidant molecules must be consumed by the oxidation reactions implying that

$$J_{OX} = J_{Si}^{R1} + J_{Ge}^{R2} \quad \text{Eqn. 3.9}$$

Combining Eqn. 3.4 and Eqn. 3.9 gives

$$J_{\text{OX}} = \frac{[k_{\text{Ge}} + a_{\text{Si}}(k_{\text{Si}} - k_{\text{Ge}})]C_o}{1 + \frac{L_{\text{OX}}}{D_{\text{OX}}}[k_{\text{Ge}} + a_{\text{Si}}(k_{\text{Si}} - k_{\text{Ge}})]} \quad \text{Eqn. 3.10}$$

Since the above equations are for dry oxidation, to convert to the wet case, appropriate factors of 2 are required to account for the oxidant species having only one oxygen atom (H_2O for wet, instead of O_2).

To relate the oxidant flux to the growth rate of the oxide we use

$$\frac{dL_{\text{OX}}}{dt} = \frac{J_{\text{OX}}}{N_{\text{OX}}} \quad \text{Eqn. 3.11}$$

where N_{OX} is the number of oxidant molecules per unit volume in the oxide, and is different in the wet and dry case, namely $N_{\text{OX}}(\text{wet}) = 2 N_{\text{OX}}(\text{dry})$.

The final equation required is one that describes Si diffusion in the SiGe alloy.

Assuming ideal behavior, we approximate the Si activity with concentration, and arrive at the following diffusion equation

$$N \frac{\partial a_{\text{Si}}}{\partial t} = \frac{\partial \phi}{\partial z} = N \frac{\partial}{\partial z} \left[D(a_{\text{Si}}) \frac{\partial a_{\text{Si}}}{\partial z} \right] \quad \text{Eqn. 3.12}$$

where ϕ is the Si flux, $D(a_{\text{Si}})$ is the diffusivity (which varies with the local activity of Si), and N is the concentration of Si in the bulk.

The diffusion equation is solved numerically on a finite grid, using a zero-flux boundary condition on the bulk side, and $J_{\text{Si}} = J_{\text{Ge}}^{R2}$ (for dry) or $J_{\text{Si}} = J_{\text{Ge}}^{R2}/2$ (for wet) for the oxide front. This boundary condition assumes that all the GeO_2 is reduced by Si forming additional SiO_2 (in conjunction with Si that is directly oxidized), and hence is only valid during the first phase of SiGe oxidation when enough Si can be supplied to the

interface. During the second phase, when some Ge is incorporated into the oxide, the boundary condition used at the oxide front is $x_{\text{Ge}}=1$, and the flux is free to vary. The above model was implemented numerically, and simulations were performed to parameters.

3.2.2.2 Low-temperature Model

For the low-temperature regime, GeO_2 is reduced by Si adjacent to the oxidation interface, but since the oxide front advances faster than the diffusing expelled Ge, this Ge is re-oxidized and reincorporated into the mixed oxide. In this oxidation regime, the Ge diffusivity in the SiGe is irrelevant and one may simply describe the oxidation process using the analytic Deal-Grove⁶² model with an effective reaction rate constant

$$k_{\text{eff}} = x_{\text{Ge}}k_{\text{Ge}} + (1 - x_{\text{Ge}})k_{\text{Si}}, \quad \text{Eqn. 3.13}$$

where k_{Ge} and k_{Si} are, respectively, the rate constants for Ge and Si oxidation, and x_{Ge} is the Ge concentration in the original SiGe film. The model requires that in steady state, the oxidant flux, given by Eqn. 3.4, is equal to the combined Si and Ge flux

$$J = k_{\text{eff}}C_{\text{iOX}}, \quad \text{Eqn. 3.14}$$

which is easily derived from Eqn. 3.7 and Eqn. 3.8, with the approximation that Si and Ge activities are equal to their atomic fractions. Equating the oxidant flux to the above combined Si and Ge fluxes allows one to solve for C_{iOX} . Replacing this into Eqn. 3.14, and using the oxide thickness equation Eqn. 3.11, yields a simple first-order differential equation in the variable L_{OX} . Solving analytically gives

$$L_{\text{OX}}^2 + AL_{\text{OX}} = Bt \quad \text{Eqn. 3.15}$$

where $A=2D_{OX}/k_{eff}$ and $B=2D_{OX}C_o/N_{OX}$.

For early times, the oxidation is reaction rate limited and we have $L_{OX} \approx (B/A)t$, whereas for longer times, the process is diffusion limited and $L_{OX} \approx (Bt)^{1/2}$. Due to this limiting behavior, B/A is called the linear rate constant, and B is the parabolic rate constant.

From a mathematical viewpoint, low-temperature SiGe oxidation is exactly the same as Si oxidation. The only difference between the two systems is the values of the oxidation constants used in the Deal-Grove model. In particular, the largest differences arises from variations in k_{eff} , D_{ox} , and C_o due to the presence of Ge in both the oxide and in the SiGe at the oxide front.

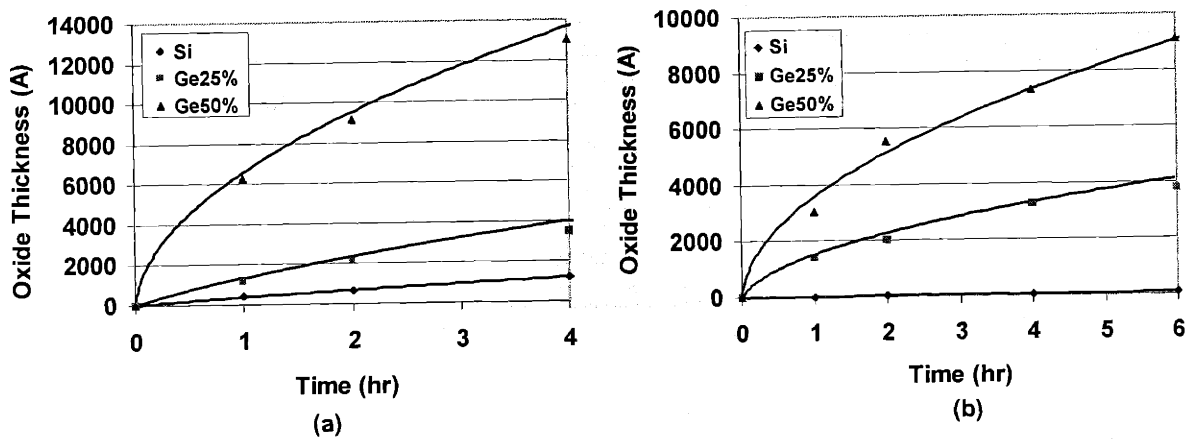


Figure 3.3 Experimental wet oxide thickness, for (a) 800°C, and (b) 700°C, for epitaxial Si(100), Si_{0.75}Ge_{0.25}, and Si_{0.5}Ge_{0.5} high-quality graded buffers. Curves are fits to Deal-Grove equation.

3.3 Wet Oxidation of High-quality Epitaxial SiGe

Experiments were carried out to determine wet oxidation rates for high-quality epitaxial SiGe alloys. To ensure high-quality low dislocation density films, SiGe graded

buffers were deposited on Si substrates using UHVCVD at 850°C, capped with 3 μm uniform layers of SiGe. In this experiment, both 25% and 50% Ge graded buffers were grown, to ascertain the variation of oxidation rates on Ge concentration. Once oxidized the samples were partially coated with wax, and etched in a concentrated HF solution, allowing oxide depth determination with profilometry. Samples were oxidized in a wet environment at both 700°C and 800°C, for various times.

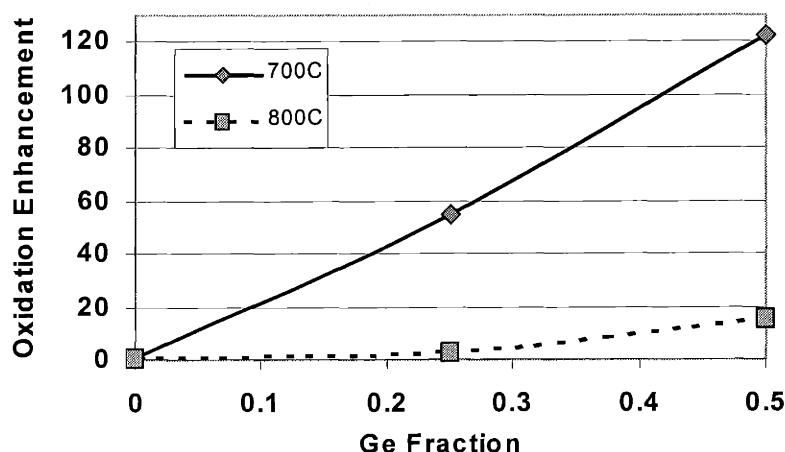


Figure 3.4 Oxidation enhancement of SiGe compared to Si for an oxidation time of 1 hr, where enhancement is defined as the ratio of the SiGe-oxide over the SiO_2 thickness.

Raw data from these experiments, presented in Figure 3.3, shows the general trend of increased oxidation thickness with increasing Ge fraction. Another important observation is that the oxidation enhancement of SiGe over Si is much larger at the low temperature of 700°C, than 800°C, as shown in Figure 3.4. Since the cause of enhancement depends on the oxide character, and since Ge incorporation into the oxide is dependant on temperature, the enhancement factors also vary greatly with oxidation

temperature. In addition, the nature of enhancement also depends on whether the oxidation is in its early (reaction limited) or late (oxidant diffusion limited) stage.

At early times, a common cause of enhanced oxidation, for both high and low temperatures, is a large Ge oxidation reaction rate constant k_{Ge} . In this regime, $L_{OX} \approx (B/A) t = (C_o k_{eff} / N_{OX}) t$, and hence the oxide thickness scales with k_{eff} , which in turn varies linearly with the Ge fraction at the oxide front. In addition, since the oxide formed at low temperatures is mixed (not pure SiO_2 , but rather $Si_xGe_{1-x}O_2$), one might expect that the parameters influencing oxidant behavior in the oxide might vary from those of SiO_2 . In particular, our data will be used to arrive at the conclusion that the oxidant solubility in the mixed SiGe-oxide must be significantly larger compared to pure SiO_2 . This, in addition to the large Ge reaction rate constant k_{Ge} , results in a further increase in the SiGe (low-temperature) oxidation enhancement when compared to Si.

On the other hand, for long times, the oxidation process is diffusion limited and is described by $L_{OX} \approx (Bt)^{1/2} = (2D_{OX}C_o/N_{OX})^{1/2} t^{1/2}$, for low temperatures like $700^\circ C$. In this limit, the oxidation is enhanced due to the increased diffusivity and solubility of oxidant in mixed SiGe-oxide. In contrast, for $800^\circ C$ the oxidation does not increase as dramatically in the long time regime, since the surface of the oxide is now almost pure SiO_2 and a large increase in oxidant solubility does not occur.

3.3.1 High-Temperature Simulation Fits

In the high-temperature limit, the data is analyzed using the extended Hellberg⁵⁹ theoretical framework presented in Section 3.2.2.1. First, both simulations and analytical

calculations (using the Deal-Grove model) were conducted to fit the Si(100) 800°C oxidation data, yielding the following results: $k_{Si}=1016$ nm/min, $D_{OX}=1.17 \times 10^6$ nm²/min, and $C_o=0.026$ nm⁻³. Both the solid solubility and the oxidant diffusivity agree very well with values from the literature,⁶³ although the Si reaction constant k_{Si} used to fit the data is slightly larger, possibly due to the actual oxidation temperature being slightly higher than the set value.

Once Si(100) oxidation values were determined, numerical simulations were performed and a value for the Ge reaction rate constant k_{Ge} was chosen so as to provide a best fit to the SiGe oxidation data. For our Si_{0.75}Ge_{0.25} data, a value of $k_{Ge}=4.3k_{Si}=4369$ nm/min, was found to yield oxidation thicknesses in agreement with the experimental data, as shown in Figure 3.5.

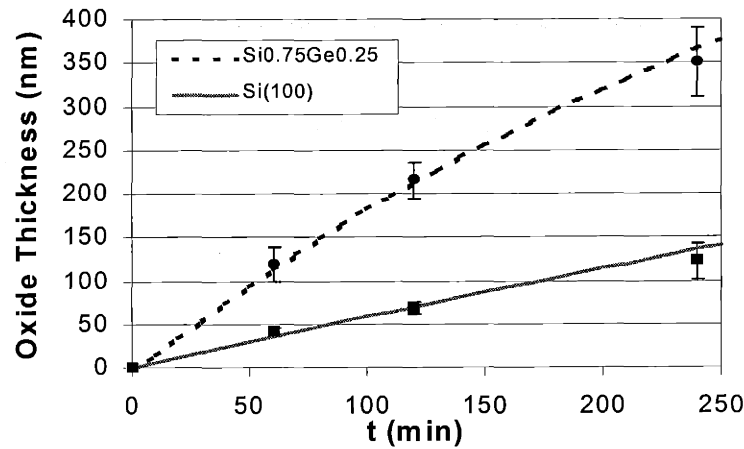


Figure 3.5 Simulations fits for Si(100) and Si_{0.75}Ge_{0.25} wet oxidation data at 800°C.

In addition to providing the oxide thickness versus time, the model can also be used to explore Ge pile-up ahead of the oxidation front. As explained in Section 3.2.2.1, this arises from GeO₂ reduction by Si leading to Ge expulsion from the oxide. Simulation

results showing this pile-up phenomena are presented in Figure 3.6(a), where the oxide front is located at $z=500$ nm for all times. Given the ideal approximation that the Si activity equals the Si concentration, then the Ge fraction is set by $1-a_{\text{Si}}$. Based on the simulation, this implies that as oxidation progresses, a Ge pile-up forms ahead of the oxidation front, until the Ge fraction at the interface reaches unity and sufficient Si cannot be supplied to the interface to reduce all the GeO_2 . In our system, simulations indicate that this occurs at an oxide thickness of approximately 100 nm, at which point, Ge is incorporated into the oxide (as GeO_2), as is shown in Figure 3.6(b).

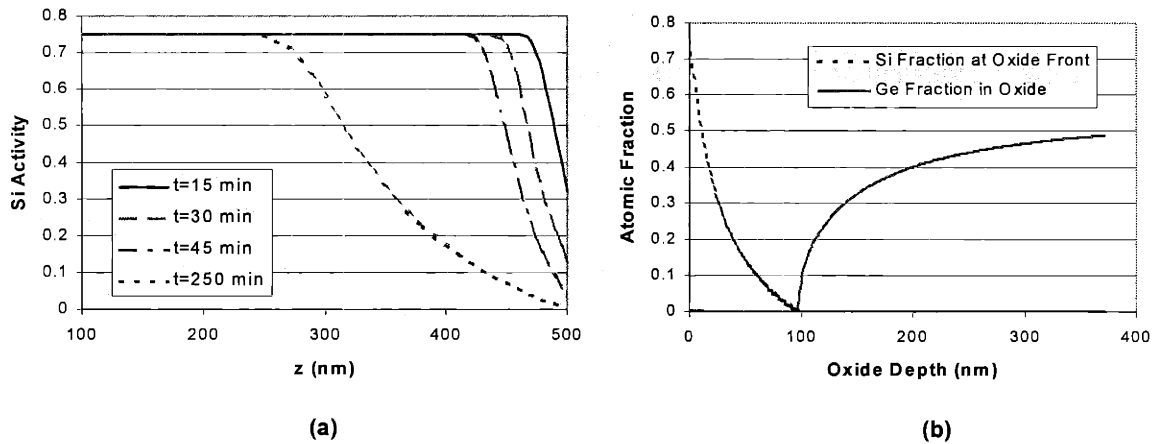


Figure 3.6 Simulation results for the $\text{Si}_{0.75}\text{Ge}_{0.25}$ oxidation: (a) Si activity profiles for various oxidation times at 800°C , where the oxidation front is at $z=500$ nm, (b) fraction of Si in the SiGe at the oxide front, and fraction of Ge incorporated in the oxide versus depth.

The onset of the Ge pile-up and incorporation into the oxide is very sensitive to the diffusivity function chosen for the simulation. A larger diffusivity, implies that the Ge piled-up at the interface can diffuse very quickly away from the interface, thereby delaying the onset of Ge incorporation. In our simulation, the diffusivity fit used by Hellberg⁵⁹ (and based on experimental data from McVay *et al.*⁶⁴) was employed.

Finally, note that just like Hellberg's results, our high-temperature oxidation simulation for high Ge concentrations (like $\text{Si}_{0.5}\text{Ge}_{0.5}$) does not match our experimental results. For high concentrations, the simulation predicts oxide thicknesses that are only slightly larger than the $\text{Si}_{0.75}\text{Ge}_{0.25}$ case, whereas our experimental results from Figure 3.4 indicate that the enhancement from 25% to 50% Ge should be approximately a factor of 5. This discrepancy may possibly be due to an earlier onset of Ge incorporation. If Ge incorporation commences when the oxide is very thin, then the oxide can no longer be approximated as pure SiO_2 , and as we shall show, this can dramatically increase the oxidation rates due to increased oxidant solubility and diffusivity.

3.3.2 Low-Temperature Model Fits

To analyze the 700°C wet oxidation data, Eqn. 3.15 was fit (using least-squares analysis) to the experimental oxidation thickness curves, generating both the A and B constants for each Ge concentration. These results, in conjunction with 700°C extrapolated data from Park⁶⁵, allow us to determine the parabolic rate constant dependence on Ge concentration, as is presented in Table 3.1.

x_{Ge}	0	0.2*	0.25	0.28*	0.47*	0.5
$B(\text{nm}^2/\text{min})$	200	390*	555	667*	2167*	2333

Table 3.1 Parabolic rate constant for various Ge concentrations in the SiGe films. Asterisks denote data from Park *et al.*⁶⁵

Using the relation for the parabolic rate constant (derived in Section 3.2.2.2), $B=2D_{\text{OX}}C_0/N_{\text{OX}}$, we deduce that the variation in B with x_{Ge} is due to C_0 and D_{OX} .

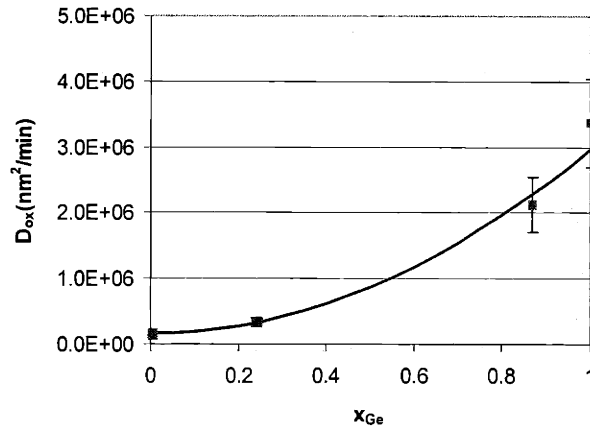


Figure 3.7 Water diffusivity at 700°C in SiGe mixed oxide as a function of the Ge fraction in the original SiGe layer.

To extract the variation of water solubility and diffusivity on x_{Ge} , we begin by using data from an extensive study of oxidant diffusion in SiGe-mixed oxides carried out by Schurman *et al.*⁶⁶ Their work showed that water diffusivity in mixed SiGe-oxides increases dramatically with GeO_2 content in the oxide. Using Schurman's diffusivity fits, we extracted water diffusivity in mixed SiGe-oxides at a temperature of 700°C. A plot of his data and our corresponding fit is presented in Figure 3.7. The fit was parabolic in x_{Ge} and given by

$$D_{OX}(x_{Ge}) = ax_{Ge}^2 + D_{OX}(x_{Ge} = 0) \quad \text{Eqn. 3.16}$$

where $D_{ox}(x_{Ge}=0)=D_{ox}(SiO_2)$ is the water diffusivity in pure SiO_2 at 700°C, given by $1.69 \times 10^5 \text{ nm}^2/\text{min}$, and x_{Ge} is the Ge fraction in the original SiGe film or equivalently, the fraction of GeO_2 in the oxide. Fitting the quadratic function to the data yields $a=2.8 \times 10^6 \text{ nm}^2/\text{min}$.

The oxidant diffusivity in SiO_2 was determined from the $x_{Ge}=0$ data in Table 3.1, in combination with $D_{OX}=N_{OX}B(x_{Ge}=0)/2C_o$, where the solid solubility in SiO_2 is set to

$C_o=0.026 \text{ nm}^{-3}$ and is approximately independent of temperature.⁶⁷ Setting $N_{OX}=44 \text{ nm}^{-3}$, a simple calculation yields $D_{OX}(\text{SiO}_2)=1.69 \times 10^5 \text{ nm}^2/\text{min}$, which is about an order of magnitude smaller than the value used to fit the 800°C data, as should be expected since the diffusivity varies exponentially with the inverse temperature, $D=D_o \exp(-E_a/kT)$.

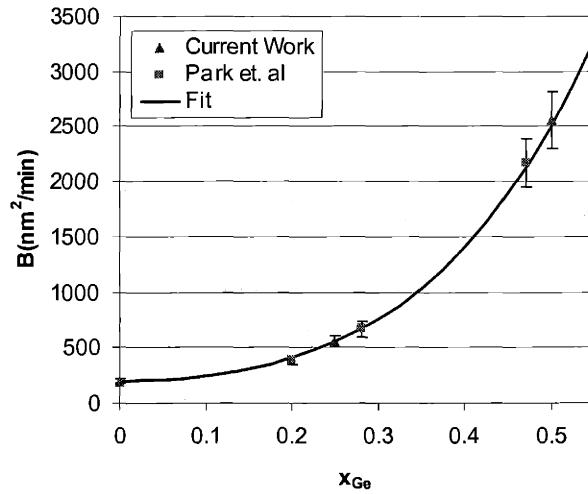


Figure 3.8 Parabolic rate constant at 700°C as a function of Ge fraction in the mixed SiGe-oxide.

By plotting the data in Table 3.1 as a function of the Ge content, it is possible to fit the $B(x_{\text{Ge}})$ data (as shown in Figure 3.8) by requiring that C_o varies quadratically with x_{Ge} . Extracting parameters from the fit yields

$$C_o(x_{\text{Ge}}) = ax_{\text{Ge}}^2 + C_o(x_{\text{Ge}}=0) \quad \text{Eqn. 3.17}$$

where $a=0.15 \text{ nm}^{-3}$, and $C_o(x_{\text{Ge}}=0)=C_o(\text{SiO}_2)=0.026 \text{ nm}^{-3}$. Figure 3.9 shows a plot of the water solubility versus the fraction of Ge in the mixed oxide composed of $(\text{GeO}_2)_x(\text{SiO}_2)_{1-x}$ where $x=x_{\text{Ge}}$.

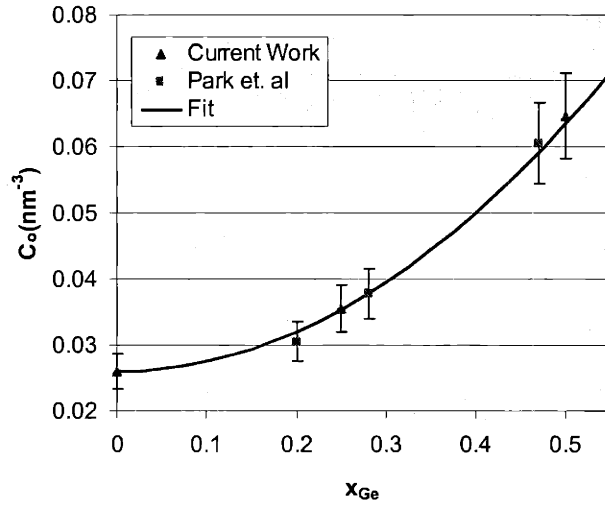


Figure 3.9 Water solubility in mixed SiGe-oxide.

An important final point is the origin of the oxidation enhancement seen for the samples consisting of 25% and 50% Ge, for the time scales used in our experiments (namely times greater than 1 h). For these times and Ge concentrations, the oxidation process is diffusion limited, and hence the parameter of importance is the parabolic rate constant $B=2D_{OX}C_o/N_{OX}$. This implies that both the oxidant solubility C_o and diffusivity D_{OX} influence the observed oxidation enhancement, and the reaction rate constant enhancement (k_{eff}), due to Ge at the oxidation front, does not have a significant effect on the oxidation rate.

On the other hand, for early times (much smaller than 1 hr), during which the oxidation thickness curve is approximately linear $L_{OX} \approx (B/A)t = (C_o k_{eff}/N_{OX})t$, the solid solubility and the reaction rate constant contribute to enhanced oxidation, whereas the oxidant diffusion plays a negligible role.

3.4 Two-Step Wet Oxidation

One serious drawback to the use of mixed SiGe-oxides for applications, is an order-of-magnitude increase in etch rates compared to SiO₂. In particular, for 30% Ge mixed oxide, the etch rate in 50:1 HF is increased by a factor of about 20 compared to SiO₂, and for 25% Ge mixed oxide, the etch rate increases by a factor of about 10. In addition, whereas pure SiO₂ is an excellent etch-stop for standard hot phosphoric, mixed Si_{0.7}Ge_{0.3}O₂, formed using 700°C wet oxidation, etches at a staggering rate of 26 nm/min. These large etch rate enhancements are to be expected given the highly unstable nature of pure GeO₂, which at elevated temperatures forms volatile GeO in the presence of pure Ge (via $\text{GeO}_2 + \text{Ge} \rightarrow 2\text{GeO(g)}$).⁶⁸ Another indication of the unstable nature of Si_{1-x}Ge_xO₂ is its dissolution at room temperature in water, for x=0.36 and higher (but not for x≤0.28).⁶⁹

Unfortunately, these etch enhancements make processing of exposed SiGe-oxide (with moderately large Ge fractions) highly impractical, and very sensitive to process variations like etch times or temperatures. If Si_{1-x}Ge_xO₂ with Ge fractions of 0.1 or more must be processed, then a scheme to protect the mixed oxide needs to be developed. An obvious approach would be to protect the SiGe oxide with a deposited SiO₂ layer, but such a method also requires a 900-1000°C anneal to densify the deposited oxide. This high-temperature anneal may be unacceptable from a thermal budget standpoint, and hence an alternative low-temperature process is desirable.

One such method involves the two-step oxidation of the SiGe. First, wet oxidation at 800°C is performed to oxidize the surface of the mixed oxide and create a protective

layer of pure SiO₂, with a high Ge region piled-up ahead of the oxidation front. This is then followed by a wet 700°C oxidation to create the mixed oxide. A warning when employing such a scheme is that a region of high Ge oxide is formed at the interface of the 800°C and 700°C oxides. The thickness of this layer depends on the amount of Ge pile-up and hence on the thickness of the 800°C pure SiO₂; a thinner pure SiO₂ surface layer implies less Ge expulsion.

When attempting to predict the oxidation rates for a two-step SiGe oxidation, it is important to account for differences in oxidant solid solubility and diffusivity of mixed SiGe-oxide versus pure silicon dioxide. In the previous section, it was deduced that the oxidant solubility for mixed SiGe-oxide was markedly larger than for pure SiO₂, and fitting the data to a quadratic polynomial gave $C_o(x_{Ge}) = 0.15 x_{Ge}^2 + 0.026 \text{ nm}^{-3}$. In addition, the diffusivity variation on Ge content was given by $D_{ox}(x_{Ge}) = 2.8 \times 10^6 x_{Ge}^2 + 1.69 \times 10^5 \text{ nm}^2/\text{min}$. This implies that SiGe capped with pure SiO₂ will oxidize at a much slower rate than equivalent SiGe capped with SiGeO₂. Such behavior is simple to predict based on the dependence of both the linear and parabolic rate constants with $C_o(x_{Ge})$ and $D_{ox}(x_{Ge})$. Estimates predict that the Si_{0.75}Ge_{0.25} mixed oxide thickness is thinner by about 200 nm due to the presence of a 150-nm SiO₂ surface capping layer, compared to the same system with a 150-nm Si_{0.75}Ge_{0.25}O₂ surface cap.

3.5 Wet Oxidation of Defective SiGe

Low-temperature oxidation (700°C or lower) of epitaxial SiGe alloys forms SiGe mixed oxide, since expelled Ge cannot diffuse away from the fast moving oxidation

front. In the case of these epitaxial films, only at higher temperatures does the oxidation form pure SiO_2 with a Ge pile-up ahead of the oxide front. In contrast, highly defective films, like polycrystalline SiGe, do not exhibit similar behavior, and in particular, complete Ge incorporation into the oxide is not observed at 700°C .

To explore the oxidation of defective SiGe films, two defective samples were grown: a uniform layer of $\text{Si}_{0.75}\text{Ge}_{0.25}$ on Si, and polycrystalline $\text{Si}_{0.75}\text{Ge}_{0.25}$ deposited on thermal SiO_2 . The uniform layer has a high threading dislocation density due to the large lattice mismatch between SiGe and Si. On the other hand, the polycrystalline sample consists of SiGe grains with associated grain boundaries.

Samples were oxidized in a wet ambient at 700°C for over 5 hours. As a control, a polished high-quality, epitaxial $\text{Si}_{0.75}\text{Ge}_{0.25}$ graded buffer was included, hence the samples in this experiment span the entire defect spectrum from polycrystalline, to high dislocation density, to very low dislocation density SiGe.

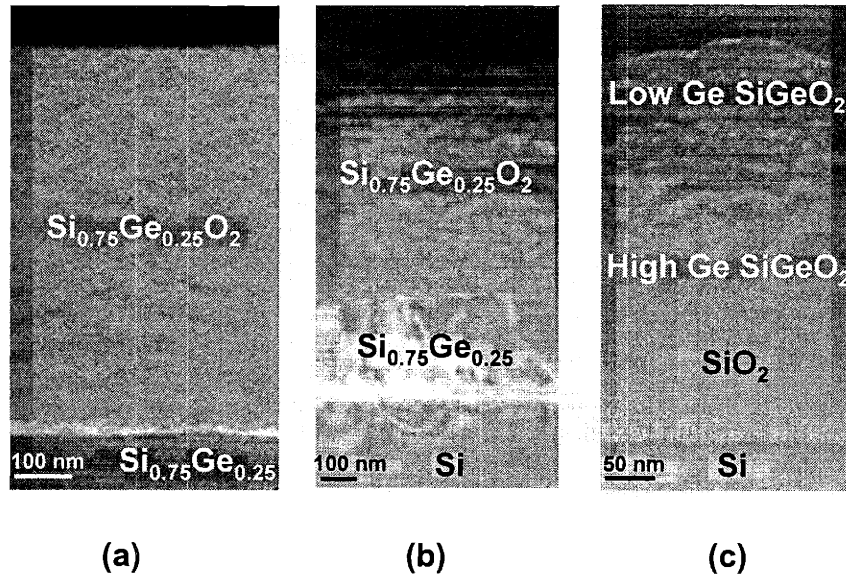


Figure 3.10 XTEM of 700°C wet oxidized samples of (a) high-quality, graded, low dislocation density Si_{0.75}Ge_{0.25}, (b) high dislocation density, uniform Si_{0.75}Ge_{0.25} on Si, and (c) polycrystalline Si_{0.75}Ge_{0.25} on SiO₂.

Cross-section TEM was employed to determine the structure of films after oxidation, and results are presented in Figure 3.10. All the samples have contrast striations present in the mixed oxide region, which correspond to regions of slightly higher GeO₂ concentration, as was verified via SIMS. The physical origin of these striations may be due to enhanced Ge lateral diffusion at the oxide/semiconductor interface, which due to the strain mismatch between pure Ge and SiGe causes Ge islanding, as is seen during epitaxial growth of Ge on Si. Such a mechanism is plausible, since diffusion at interfaces is usually many orders of magnitude larger than corresponding bulk diffusion. A system exhibiting similar behavior is the MOCVD growth of In_xGa_{1-x}P, which can lead to the formation of vertical contrast modulations

(VCM), due to enhanced surface diffusion during growth.⁷⁰ In both epitaxial samples (a) and (b), Ge is incorporated into the oxide forming GeO_2 , and the striations and hence the Ge are uniformly distributed throughout the depth of the oxide.

In contrast, oxide striations for polycrystalline SiGe sample are much more prominent at the bottom of the mixed oxide, indicating that lower regions consist of a high Ge content mixed oxide, whereas the upper portions of the mixed oxide have a much lower Ge content. This drastically different microstructure is due to enhanced diffusivity of Ge along grain boundaries.⁷¹ Although bulk lattice diffusion of Ge at 700°C is practically non-existent, grain-boundary diffusion at those temperatures can be orders of magnitude larger. For example, for a typical FCC metal at $T=T_{\text{melt}}/2$, diffusion along a dislocation core (approximately 2 atoms in width) is 5 orders larger than bulk diffusion, and grain-boundary diffusion is 7 orders larger.⁵⁰ To get a sense of these numbers, consider Ge diffusion in Si, for which the diffusivity at 700°C is approximately $1.1 \times 10^{-6} \text{ nm}^2/\text{min}$,⁴⁹ and hence the bulk diffusion length after 1 hour is $L=(Dt)^{1/2}=0.008 \text{ nm}$. On the other hand, assuming similar diffusivity enhancements as in a metal, then along a dislocation core the diffusion length is 2.5 nm, and along a grain boundary the diffusion length is 25 nm. This diffusion length is equivalent to bulk diffusion at about 1100°C . Given these dramatic enhancements, it is to be expected that Ge can diffuse quickly away from the oxide front, hence reducing the amount of incorporation into the oxide, as shown graphically in Figure 3.11.

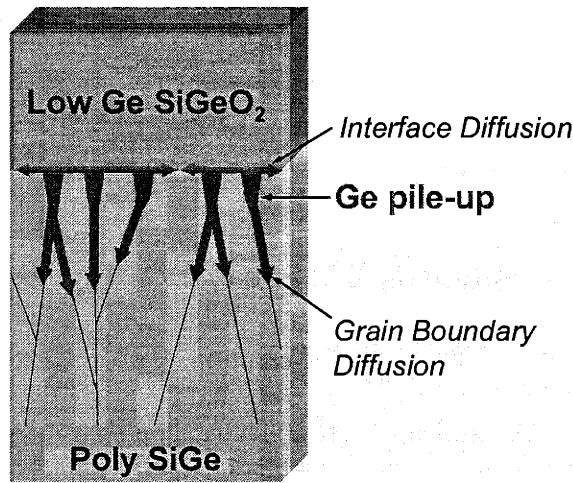


Figure 3.11 Fast diffusion of Ge away from the oxide front, along grain boundaries in polycrystalline SiGe.

In our polycrystalline sample, Ge is piled-up until it reaches the underlying SiO_2 , at which point, it is pinned, and the oxidation front overtakes the pile-up and oxidizes it forming the high GeO_2 region at the bottom of the mixed oxide.⁵² This experiment is a unique (albeit indirect) demonstration of enhanced Ge diffusion along grain boundaries.

3.6 SiGe Lateral Oxidation

A substantial amount of research has been carried out on the oxidation of SiGe wafers, and the most interesting result has been the enhanced oxidation of SiGe compared to Si.^{51,59,60} For example, at 700°C , we have found that the oxidation rate of $\text{Si}_{0.75}\text{Ge}_{0.25}$ after one hour is approximately 60 times larger than that of Si. Although this phenomenon has been extensively studied in the case of blank wafers, very little work has been pursued on the oxidation of more complex SiGe structures. In response to this lack of data, we conducted a series of oxidation experiments on different SiGe structures, with the goal of determining how preferential oxidation of SiGe over Si is altered in such

structures. Results of these experiments will be very useful from the point of view of device applications, because in many cases, structures being oxidized are much more complex than blank wafers.

3.6.1 Motivation: Alternate SOI Fabrication Technique

An interesting proposition is the creation of a buried oxide layer, with a monocrystalline semiconductor layer above it, created by employing the preferential oxidation of SiGe over Si at low temperatures. If a structure consisting of a buried SiGe layer surrounded by Si is etched such that the SiGe is exposed on the mesa sidewalls, then low-temperature oxidation might proceed laterally along the SiGe layer. Once complete, the process would lead to the formation of crystalline Si on a buried oxide layer. If successful, such a process might serve as an alternate technique for SOI fabrication. A simple experiment was performed to explore this possibility.

3.6.2 Lateral Oxidation of Buried Epitaxial SiGe

The lateral oxidation experiment consisted of the simple process shown in Figure 3.12. The first step was the UHVCVD deposition of a layer of strained-Si_{0.75}Ge_{0.25} on a Si wafer, capped with a thin layer of Si. The thickness of the Si_{0.75}Ge_{0.25} was chosen to be less than the critical thickness, and hence was pseudomorphic with the underlying Si, thereby not generating any dislocations.

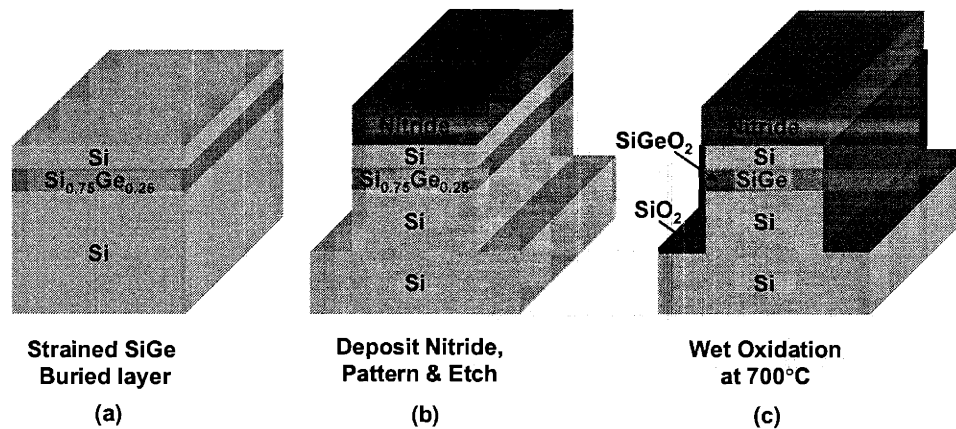


Figure 3.12 Process steps for lateral oxidation of SiGe buried layers, (a) UHVCVD deposition of strained-SiGe buried layer, (b) nitride mask deposition, lithographic patterning, and dry etching of trenches, (c) low-temperature wet oxidation.

A nitride mask was then deposited on the surface, which protects the surface of the Si from oxidizing. Next, lithographic patterning is employed to define trench structures with a width of a few microns, followed by dry etching. The final structure consists of strips with side-walls exposing the buried SiGe layer, as shown in Figure 3.12(b). Low-temperature wet oxidation at 700°C is used to laterally oxidize the buried SiGe layer, while hopefully causing only minimal oxidation of the exposed Si.

In the present experiment, the wet oxidation of a buried SiGe layer was conducted at 700°C for 5 h. The resulting oxidation front was imaged via cross-sectional TEM, as shown in Figure 3.13. As can be seen from the XTEM, the SiGe buried layer did not oxidize as much as would have been expected from our blank wafer data. Whereas we would have expected a blank 25% Ge wafer to have an oxide thickness of about 400 nm, our buried SiGe layer only shows a lateral oxidation of about 100 nm.

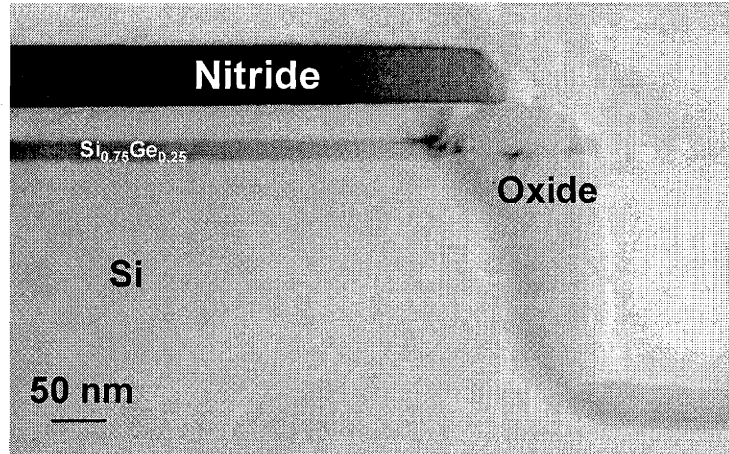


Figure 3.13 XTEM of buried SiGe trench structure, after wet oxidation at 700°C for 5 h.

This dramatic difference in oxidation rate is likely due to the enormous stress that forms in the buried oxide. SiGe-oxides exhibit a volume expansion of around 120%, but in our structure, this expansion cannot readily take place since the oxide is sandwiched between other material. This then leads to stress build-up at the oxide front, which limits the diffusion of oxidants to the front, and hence SiGe oxidation ceases to progress laterally.

It is interesting to note that such a lateral oxidation technique has also been used on AlAs layers sandwiched between GaAs, where only the AlAs oxidizes. In these experiments, the AlAs lateral oxidation was found to be self-stopping⁷² and the width depended on the thickness of GaAs above the buried layer. Also, below an AlAs thickness of 15 nm, oxidation did not proceed much laterally due to stress effects at the AlAs/GaAs boundary. Although one might be tempted to conclude that our SiGe buried layer would be similar, it is important to note that in the case of the AlAs the oxide contracts⁷³ by 20%, whereas in our scenario the oxide expands by 120%.

The conclusion of this experiment is that the lateral oxidation approach is not suitable for SOI fabrication using buried SiGe lateral oxidation, due to the inability of the oxidation front to completely proceed laterally due to stress effects. It is noteworthy to consider that the technique might be useful for applications where the selective lateral oxidation does not need to completely oxidize the entire buried SiGe layer. For example, our collaborative work with Lee *et al.* demonstrated a process by which selective oxidation of poly-Si_{0.5}Ge_{0.5}, deposited using our UHVCVD reactor, was used to create isolation layers for a vertical dual-gate transistor device.⁷⁴

3.7 Summary

Low- and high-temperature wet oxidation of SiGe showed an increase in SiGe oxidation rate compared to Si. In particular, the largest oxidation enhancement over Si was observed for SiGe wet oxidation at 700°C. At this temperature, the Ge is incorporated into the oxide forming a SiGeO₂ mixed oxide, whereas for 800°C, a Ge pile-up formed at the oxidation front. Oxidation models were developed and used to explain experimental results. Possible applications of enhanced oxidation of SiGe compared to Si were explored.

Chapter 4 Nanocrystalline Ge: Formation Kinetics and Photoluminescence

The past decade has seen a surge in research on group-IV, zero-dimensional nanocrystals,⁷⁵⁻⁷⁹ driven by a desire to create novel optical and opto-electronic devices, and to monolithically integrate these devices onto Si substrates.^{80,81} Apart from an understanding of the physics of nanocrystals, the main obstacle to commercial adoption of nanocrystals in devices is the compatibility and reliability of the fabrication process. To address these issues, a multitude of techniques have been explored to synthesize nanocrystals, including co-sputtering,⁸² colloidal suspension,⁸³ ion implantation into oxide,^{79,84,85} and reduction of metastable compounds.⁸⁶⁻⁸⁸

The last technique can be used to create nanocrystalline Ge (ncGe) in SiO₂ via H₂ reduction of metastable Si_{1-x}Ge_xO₂. An attractive feature of this approach is high wafer throughput, due to the ability to process a batch of wafers in parallel, compared to the serial processing required for many other techniques. In addition, in contrast to ion implantation, which creates Gaussian-shaped nanocrystal density depth profiles, H₂ reduction of metastable Si_{1-x}Ge_xO₂ creates sharp step profiles. Unfortunately, Ge nanocrystal formation via metastable Si_{1-x}Ge_xO₂ reduction is kinetically complex, which in turn complicates the prediction and control of nanocrystal size distributions. To address this issue, we have extensively explored the process kinetics of Ge nanocrystal formation via H₂ reduction of Si_{1-x}Ge_xO₂. The process was carried out for various annealing times, temperatures, and H₂ partial pressures, yielding a wide variety of nanocrystal size and spatial distributions, allowing for the extraction of information about the various kinetics processes involved in nanocrystal formation.

Other than structural information, optical data from either photoluminescence (PL) and/or absorption spectra is of particular importance, due to the desire to employ such nanocrystals in optical devices. PL data for Si nanocrystals is well understood, in contrast to the Ge system, which has seen a multitude of reports of blue-green PL by various researchers.⁸⁹⁻⁹¹ This visible PL peak has sometimes been attributed to the quantum confinement, but due to the invariance of this peak location on nanocrystal size, quantum-confined state transitions cannot be the source of this emission. Defects in the oxide matrix are a more plausible source of the blue-green PL peak.⁹²

This visible emission has been extensively studied, but the infrared spectrum has largely been overlooked. Given the multitude of potential telecommunication device applications operating in the infrared range of 1.3-1.6 μm , the infrared PL spectrum of nanocrystals is of particular interest. In this chapter, we also present measurements of the infrared PL spectra of nanocrystalline Ge in an oxide matrix, synthesized using the reduction of $\text{Si}_{0.75}\text{Ge}_{0.25}\text{O}_2$ under various annealing conditions. Samples chosen for PL measurements had numerous nanocrystals with small diameters (<15 nm), and were ideal for studying the effects of quantum confinement on the PL spectra.

4.1 Reduction of SiGe-Oxides

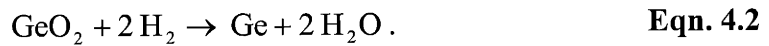
Given the thermodynamic instability of SiGe-oxides,⁶⁹ it is natural to expect that reduction will occur when a viable chemical pathway is present. In particular, SiGe-oxide reduction has been achieved with both Si and H_2 .^{69,87} From work on high-

temperature (800°C) oxidation of SiGe alloys,^{59,93} it is well known that when Si is in contact with GeO₂, the following reduction reaction occurs:



yielding a much lower free energy state. This reaction can occur if the SiGe-oxide layer is in contact with a Si or SiGe film. At the interface between the materials, Si from the substrate is available for the reduction reaction. Since the diffusivity of Si in oxide is quite small, Si can only be supplied to the oxide in close proximity (<10 nm) to the unoxidized film, and hence Ge precipitates only from at the interface and not throughout the entire mixed oxide. This has been verified experimentally with XTEM for vacuum annealing of SiGe-oxide on Si, at 900°C for 3 hours.⁶⁹

The other reduction pathway for SiGeO₂ is the reaction of the oxide with hydrogen at elevated temperatures, which according to the literature occurs via the following reaction^{87,88}



At first glance, it would seem to be a simple reduction mechanism, but the actual kinetics are somewhat more involved. Process kinetics of Ge nanocrystal formation via this reduction reaction depend strongly on annealing temperature and H₂ partial pressure, yielding a wide variety of nanocrystal size and spatial distributions.

4.2 Experiments

A hot-walled, ultra-high-vacuum chemical-vapor-deposition (UHVCVD) reactor was used to deposit a $\text{Si}_{0.75}\text{Ge}_{0.25}$ graded buffer on Si substrates, at 900°C , with a grading rate of 10% per μm . A 2 μm thick capping layer of $\text{Si}_{0.75}\text{Ge}_{0.25}$ was deposited, and then polished using chemical-mechanical-polishing (CMP). Polishing eliminates any crosshatch roughness arising from misfit dislocation strain fields. Samples were then oxidized hydrothermally in a conventional oxidation tube at 700°C for 5 h, producing 660 nm of oxide. The low oxidation temperature ensures that Ge in $\text{Si}_{0.75}\text{Ge}_{0.25}$ is incorporated into the oxide,⁹⁴ and does not form a pile-up at the oxidation front, as was verified by secondary ion mass spectroscopy (SIMS). In contrast, high-temperature (*e.g.* 800°C) oxidation has been shown to create SiO_2 with a Ge pile-up at the oxide front.⁵¹

After oxidation, the $\text{Si}_{0.75}\text{Ge}_{0.25}\text{O}_2$ on $\text{Si}_{0.75}\text{Ge}_{0.25}$ samples were annealed at atmospheric pressure in various ambients, including pure N_2 , dilute H_2 (6% H_2 : 94% N_2), and pure H_2 (100% H_2). The H_2 anneals were performed at 700°C , 800°C , and 875 - 900°C , for various times. For the highest temperature anneals in a pure H_2 ambient, samples were inadvertently exposed to a temperature range of 875 - 900°C , due to temperature fluctuations caused by the flow of pure H_2 . Since the samples consisted of a layer of mixed oxide on a $\text{Si}_{0.75}\text{Ge}_{0.25}$ film, any Ge liberated by the reduction of the mixed oxide was free to diffuse within the oxide until making contact with the $\text{Si}_{0.75}\text{Ge}_{0.25}$. Any such Ge that made contact with the underlying $\text{Si}_{0.75}\text{Ge}_{0.25}$ interface would bond and be immobilized, and therefore should be detectable via XTEM. This implies that the

underlying $\text{Si}_{0.75}\text{Ge}_{0.25}$ allows us to make important observations about the kinetics of precipitated Ge, which would otherwise be undetectable if the underlying film consisted of an oxide layer (like SiO_2).

Cross-sectional TEM, using a JEOL 200 TEM, was used to characterize many of the samples to determine the extent of nanocrystal formation. In conjunction with TEM, Raman spectroscopy, using an excitation wavelength of 785 nm, was employed to confirm that the nanocrystals observed with TEM were composed of Ge. In all samples with large nanocrystals, a strong 299 cm^{-1} Raman peak was observed, correlating to Ge-Ge bond vibrations in nanocrystals. In addition, a small frequency downshift, due to phonon confinement, was observed in samples with very small nanocrystals.⁵⁷ Dilute HF etch-rates for various samples, including a SiO_2 control, were determined by etching samples in a solution of 1:50 HF:H₂O. Also, samples that were determined to have small nanocrystals (<15 nm), via XTEM, were then analyzed using PL, at a temperature of 4K, with an Ar⁺ ion laser ($\lambda=488\text{ nm}$) at an output power of 260 mW.

4.3 Non-Reactive Ambient Annealing of SiGe-Oxide

In the past, there has been some debate as to whether a non-reactive N_2 annealing ambient could induce the formation of Ge nanocrystals throughout the entire oxide,⁹⁵ but the current consensus has refuted this claim.⁸⁶ To ensure that non-reactive atmospheres do not cause Ge crystallization in the bulk of the $\text{Si}_{0.75}\text{Ge}_{0.25}\text{O}_2$, a N_2 anneal was performed at 800°C for 1h. XTEM of as-oxidized and N_2 annealed samples reveals that no obvious Ge nanocrystals have formed after the N_2 anneal, as shown in Figure 4.1. The

XTEM of the as-oxidized sample, presented in Figure 4.1(a), reveals horizontal contrast striations in the oxidized region, corresponding to regions of slightly higher GeO_2 concentration.

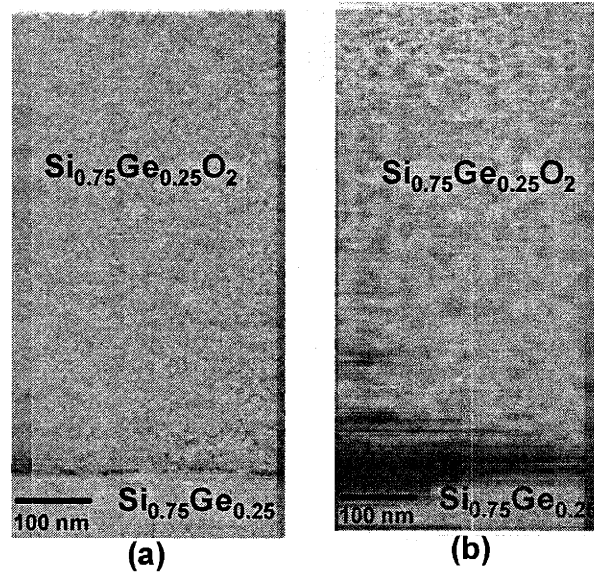


Figure 4.1 XTEM micrographs for SiGe-oxide: (a) as-oxidized, and (b) after 1 h N_2 anneal.

The origin of the striations may be due in part to enhanced Ge lateral diffusion at the oxide/semiconductor interface, usually many orders of magnitude faster than the corresponding bulk diffusion.⁵⁰ This effect implies that any Ge present at the oxidation front can quickly diffuse along the oxide/SiGe interface. In addition, due to the strain mismatch between $\text{Si}_{0.75}\text{Ge}_{0.25}$ and pure Ge, there exists a driving force for Ge islanding, as is seen during epitaxial growth of Ge on Si.⁹⁶ During the SiGe oxidation process, this driving force, combined with increased interfacial diffusion, may lead to the creation of stratified regions of slightly higher GeO_2 concentration. At extremely low oxidation temperatures, these non-uniformities would not be present, since the Ge diffusivity along the interface would be lower.

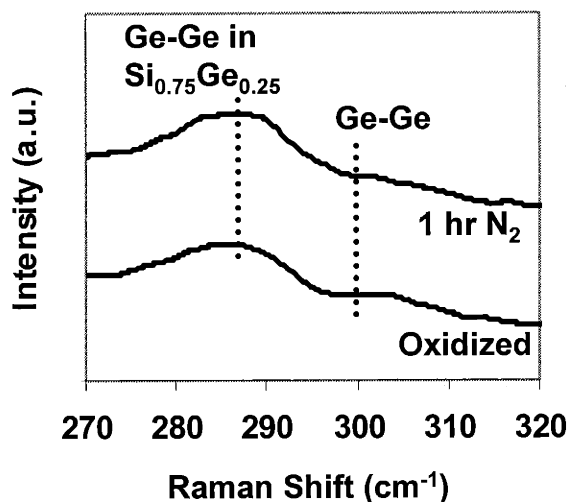


Figure 4.2 Raman spectra for the as-oxidized and N₂-annealed SiGe-oxide, showing the absence of a significant Ge-Ge peak.

After N₂ annealing of the Si_{0.75}Ge_{0.25}O₂, XTEM reveals a slight change in the stratified contrast, likely due to the local redistribution of GeO₂ via diffusion, as shown in Figure 4.1(b). Although this change in contrast might be mistakenly interpreted as the onset of Ge nanocrystal formation, Raman spectra of the as-oxidized and N₂-annealed samples are practically identical, which indicates that no Ge nanocrystals have formed. As shown in Figure 4.2, neither sample has a strong peak located at 299 cm⁻¹, which would correspond to Ge-Ge vibrational modes in Ge nanocrystals. Both samples have a peak at 286 cm⁻¹, due to Ge-Ge bonds in the underlying Si_{0.75}Ge_{0.25} film.⁵⁶ Our data supports the current consensus that non-reactive ambients, like N₂, do not reduce SiGe-oxide, and do not lead to the formation of Ge nanocrystals.

4.4 Nanocrystal Formation in Pure Hydrogen

Due to its large free energy, GeO_2 tends to decompose and form Ge in the presence of a reducing agent, during which process the O_2 is transferred to the agent. In the past, SiGe-oxide reduction has been achieved using Si or H_2 as reducing agents, forming crystalline Ge precipitates in a SiO_2 matrix.^{69,86-88} In this section, we present a detailed analysis of the Ge crystalline microstructure formed by annealing in a pure H_2 atmosphere at various temperatures, for a range of times.

The central process leading to Ge nanocrystal formation is the reduction reaction presented in Eqn. 4.2, and although simple in principle, the specifics of the process kinetics are quite involved. Nanocrystal formation can be separated into the following processes: (i) GeO_2 reduction leading to the creation of elemental Ge atoms; (ii) diffusion of liberated Ge in the oxide matrix; (iii) nucleation, due to Ge-Ge collisions causing the formation of immobile Ge clusters; (iv) growth, whereby diffusing Ge atoms bond to existing Ge clusters; and (v) coarsening (*i.e.* ripening) of nanocrystals, whereby small nanocrystals shrink, and large ones grow.⁹⁷

4.4.1 700°C Anneals

To illustrate these individual processes, XTEM results for the 700°C, pure H_2 anneals for various times are presented in Figure 4.3. The 1-min annealed sample in Figure 4.3(b) reveals the formation of Ge nanocrystals near the surface and a layer of tiny nanocrystals at the SiGe interface. The lack of nanocrystals in the mid-region of the oxide indicates that water out-diffusion is the rate-limiting step for the GeO_2 reduction

process. Due to the extremely fast diffusion of H_2 in oxide, the reduction reaction initially occurs equally at all depths in all regions, but as time progresses the water vapor byproduct accumulates and the reaction reaches equilibrium. Only after water diffuses out of a given region does further GeO_2 reduction occur, liberating more Ge. Since water near the surface out-diffuses quickly, the top region of the oxide has more liberated Ge and hence forms nanocrystals sooner than deeper regions.⁸⁸

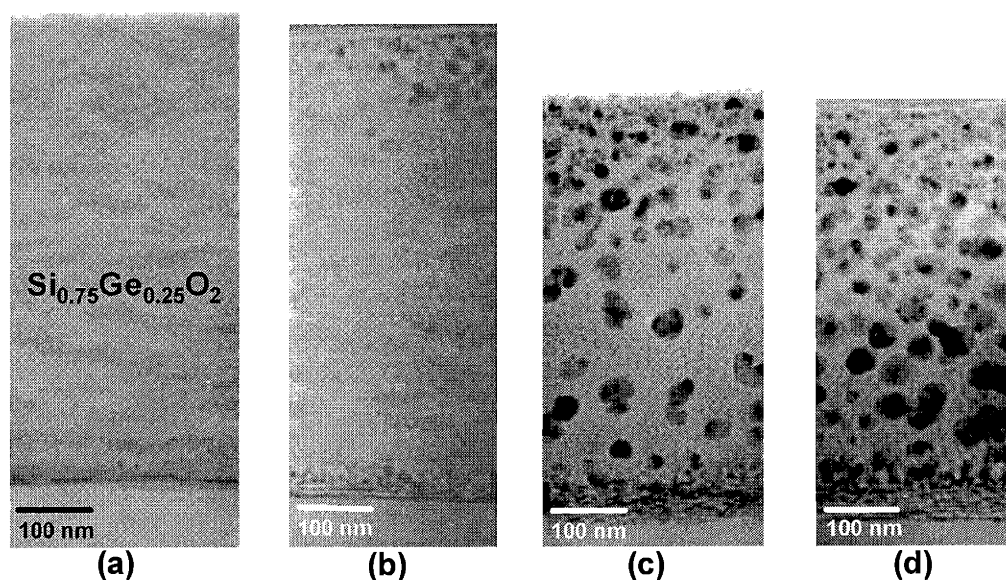


Figure 4.3 XTEM micrographs for anneals at 700°C in pure H_2 : (a) as-oxidized, (b) 1 min anneal, (c) 5 min anneal, and (d) 15 min anneal.

In addition, tiny nanocrystals at the oxide/ $Si_{0.75}Ge_{0.25}$ interface may initially form due to the reduction of GeO_2 (adjacent to the $Si_{0.75}Ge_{0.25}$) with Si (from the $Si_{0.75}Ge_{0.25}$),⁶⁹ or due to the bonding of liberated Ge, via the H_2 reduction of GeO_2 , onto the $Si_{0.75}Ge_{0.25}$ interface. Irrespective of their origin, these crystallites grow via the adsorption of additional liberated Ge onto their surface, which is the same growth mechanism as for nanocrystals in the bulk of the oxide.

The 5-min annealed sample, shown in Figure 4.3(c), contains nanocrystals throughout the entire oxide. The thickness of the oxide has decreased significantly, indicating the progression of the reduction reaction throughout the oxide. The XTEM micrograph also reveals larger nanocrystals present in deeper regions, as opposed to smaller, more numerous nanocrystals near the surface. This size distribution, thought to be caused by the presence of the water byproduct, is consistent with increased water concentration in deeper regions, as implied by the water out-diffusion profile. In the presence of increased water vapor, one would expect an increase in many relevant processes,⁸⁷ including both elemental Ge diffusion and nanocrystal coarsening. In regards to coarsening, water vapor may also lead to surface oxidation of Ge nanocrystals, forming a thin GeO_2 shell around a Ge core. These compounds could then react via $\text{GeO}_2 + \text{Ge} \Rightarrow 2\text{GeO}(\text{g})$,⁶⁸ where the GeO gas is extremely volatile and diffuses rapidly, possibly adhering to other nanocrystals, and allowing for a water-dependant coarsening mechanism.

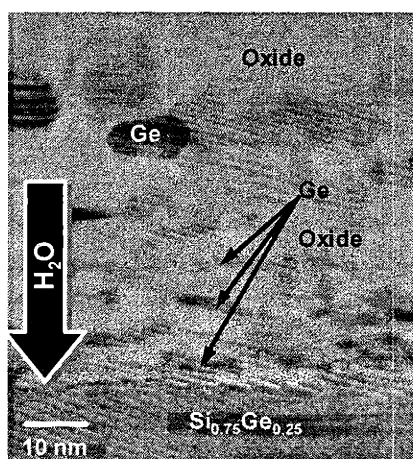


Figure 4.4 Lattice image of oxide/ $\text{Si}_{0.75}\text{Ge}_{0.25}$ interface after annealing at 700°C in pure H_2 for 5 min.

Also note that, although difficult to see in the current micrograph, high resolution XTEM reveals the existence of numerous small nanocrystals in the middle region of the oxide (with the majority of these having diameters in the 5-15 nm range). As annealing time increases, coarsening eliminates these small nanocrystals in favor of larger sized crystallites, as can be seen from the XTEM of the 15-min anneal in Figure 4.3(d). Furthermore, providing evidence of water production, a layer of new oxide can be seen to form at the $\text{Si}_{0.75}\text{Ge}_{0.25}$ interface. In Figure 4.4, lattice imaging of this layer reveals that it contains stratified layers of crystalline Ge expelled during the inadvertent wet oxidation of the underlying $\text{Si}_{0.75}\text{Ge}_{0.25}$ by the water byproduct.

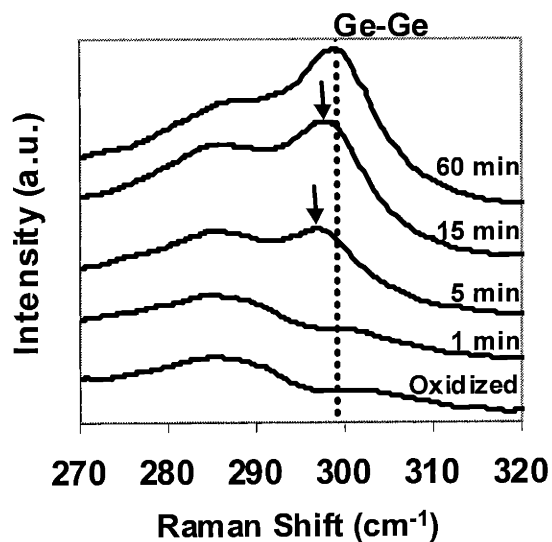


Figure 4.5 Raman spectra for SiGe-oxide annealed in pure H_2 at 700°C , for various times, showing the appearance and shift (due to nanocrystalline confinement) of a Ge-Ge peak.

Figure 4.5 shows the Raman spectra for the 700°C anneals. The first observation is the formation of the Ge-Ge peak at around 299 cm^{-1} , due to the creation of Ge nanocrystals. The location of the Ge-Ge peak in the 60-min sample is 299.1 cm^{-1} , but the

peak is downshifted to 297.9 cm^{-1} for the 15-min anneal, and to 297.0 cm^{-1} for the 5-min anneal. The downshift of the Raman peak is due to phonon confinement in Ge nanocrystals,⁵⁷ and hence is more predominant for smaller sized crystallites. This is evidenced by the 5-min annealed sample, which contains a large number of nanocrystals with diameters less than 15 nm.

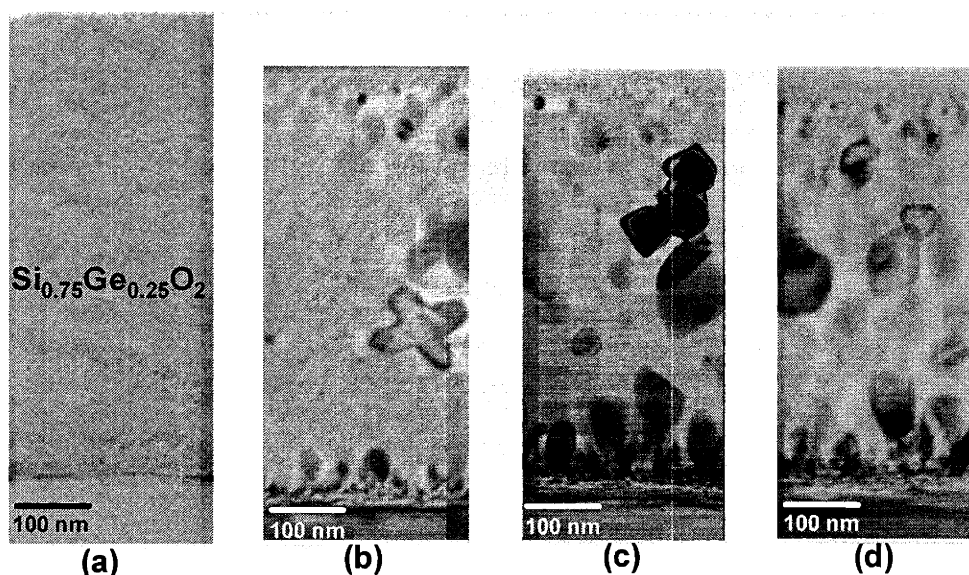


Figure 4.6 XTEM micrographs for anneals at 800°C in pure H_2 : (a) as-oxidized, (b) 1 min anneal, (c) 15 min anneal, and (d) 60 min anneal.

4.4.2 800°C Anneals

Figure 4.6 shows XTEM micrographs for anneals at 800°C in a pure H_2 atmosphere, for 1, 15, and 60 min. At this elevated temperature, the nanocrystals have already begun to form throughout the oxide after only 1 min. From Figure 4.6(b), it can be seen that the size distribution in the center of the oxide consists of numerous small nanocrystals ($<15\text{ nm}$), in addition to a few very large crystallites with sizes on the order of about 100 nm. The presence of these large crystallites after a only a 1-min anneal

reveals that not only is the reduction reaction rate substantial, but the coarsening process must also be fast.

In addition, there are features present in the nanocrystal distribution similar to those observed for 700°C, including both moderate-size nanocrystals near the surface of the oxide, and irregularly shaped nanocrystals at the oxide/SiGe interface. Nanocrystals at the top of the oxide form due to the fast out-diffusion of water near the surface, which in turn allows the reduction reaction to proceed forward quickly in that region. As in the 700°C case, the irregularly shaped nanocrystals at the semiconductor interface are likely due to deposition of liberated Ge onto the $\text{Si}_{0.75}\text{Ge}_{0.25}$. The shapes are irregular since the deposition rate is rapid and the deposited Ge does not have sufficient time to settle into its final equilibrium shapes. Finally, note that after additional annealing the nanocrystals grow and coarsen, as can be seen in Figure 4.6(c) and Figure 4.6(d).

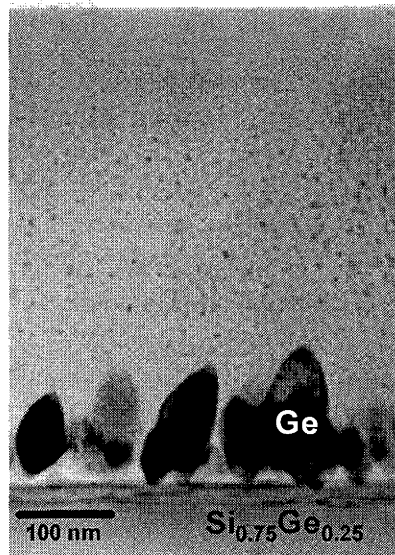


Figure 4.7 XTEM micrograph for anneal at 875-900°C in pure H_2 for 1 min.

4.4.3 875-900°C Anneals

The highest temperature employed for pure H₂ annealing was 875-900°C. Figure 4.7 shows the XTEM for such an anneal after 1 min. All the GeO₂ is quickly reduced after this short time, due to the fast progression of reduction at this elevated temperature. Ge has in turn formed small nanocrystals throughout the oxide, in addition to large, diamond-shaped crystallites at the Si_{0.75}Ge_{0.25} interface. These crystallites are similar to those found in the 800°C anneals, but now they are slightly more regular in shape, likely due to the increased ability of the Ge to flow at 875°C, and therefore attain a shape closer to the ideal equilibrium structure.

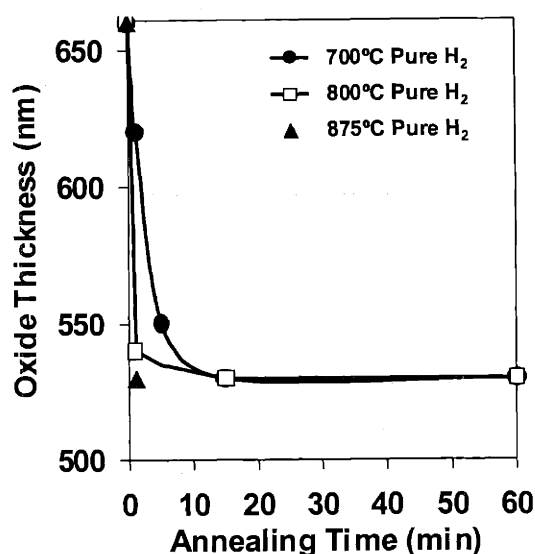


Figure 4.8 Measured oxide thickness as a function of annealing time in pure H₂ for various temperatures.

4.4.4 Oxide Thickness versus Annealing Time

Analysis of the oxide thickness reveals a decrease with increasing annealing times, as presented in Figure 4.8, due to the out-diffusion of the water formed by the reduction

of GeO_2 , which removes oxygen atoms from the oxide. The decrease in thickness is an indicator of the amount of GeO_2 reduced. In particular, for pure H_2 anneals at 700°C , the thickness has decreased substantially after about 10 min. Raising the temperature decreases the amount of time required to reduce the available GeO_2 . For example, at 800°C , the reduction is practically complete after only 5 min, and at 875°C , 1 min is more than sufficient to reduce the oxide.

Another useful indicator of the reduction progression is the oxide etch rate. Due to the thermodynamic instability of SiGe mixed oxides, etch rates in many solutions (including conventional etches for silica and nitride) are significantly increased compared to SiO_2 . In particular, in the case of the 50:1 HF etch at room temperature, the as-oxidized $\text{Si}_{0.75}\text{Ge}_{0.25}\text{O}_2$ has an etch rate of 149 nm/min, whereas the corresponding etch rate for thermal SiO_2 is about 14 nm/min, revealing an enhanced etch rate of about an order of magnitude. After annealing in H_2 for sufficient time to completely reduce the GeO_2 present in the mixed oxide, the etch rate drops to the value expected for SiO_2 . This indicates that the presence of GeO_2 in the mixed oxide is the source of the dramatic increase in etching, and allows us to use the etch rate as a marker for the extent of the reduction.

4.5 Nanocrystal Formation in Dilute Hydrogen

Dilute H_2 anneals (6% H_2) of the $\text{Si}_{0.75}\text{Ge}_{0.25}\text{O}_2$ on $\text{Si}_{0.75}\text{Ge}_{0.25}$ yield drastically different nanocrystal distributions. Due to an increased Ge liberation rate at elevated H_2 concentrations, a 15 min anneal in pure H_2 at low temperatures leads to the reduction of

almost all the GeO_2 . In contrast, for 6% H_2 at low temperatures, 60 min is required to significantly reduce the GeO_2 .

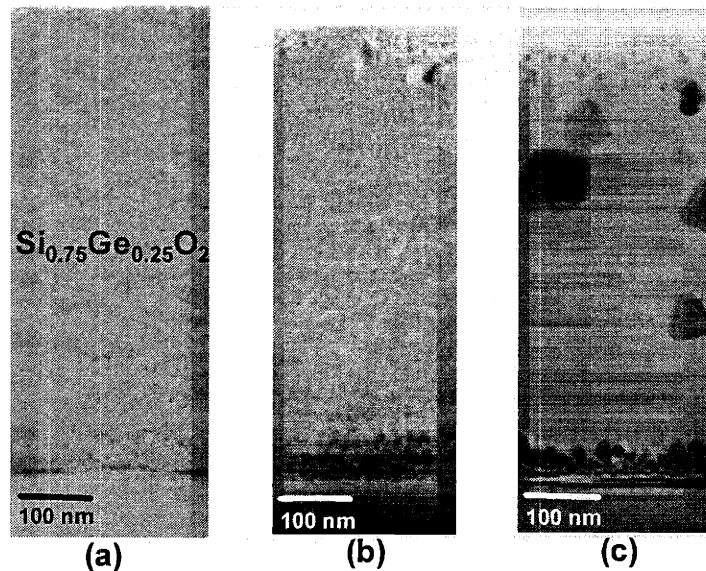


Figure 4.9 XTEM micrographs for anneals at 700°C in 6% H_2 : (a) as-oxidized, (b) 15 min anneal, and (c) 60 min anneal.

4.5.1 700°C Anneals

XTEM micrographs for 700°C anneals, presented in Figure 4.9, reveal the formation of sparse, large crystallites with sizes as large as 100 nm. Sparse distributions of Ge crystallites are caused by a low frequency of nanocrystal nucleation events. We postulate that an immobile nucleus forms when two liberated Ge atoms collide while diffusing in the oxide. Since the reduction reaction proceeds slowly for dilute H_2 concentrations, the amount of liberated Ge is small, and hence, based on simple collision kinetics, the expected Ge-Ge collision frequency (and in turn, the nanocrystal nucleation events) should be significantly lower, as compared to a pure H_2 anneal.

Figure 4.9(c) also shows the presence of small Ge nanocrystals at the oxide/ $\text{Si}_{0.75}\text{Ge}_{0.25}$ interface, similar to those observed for pure H_2 anneals at 700°C , as shown in Figure 4.3(d). The density and size of these interfacial nanocrystals is similar for both pure and dilute H_2 , indicating that their formation must be independent of the Ge-Ge collision rate, unlike nanocrystals in the bulk of the oxide, whose density varies dramatically with H_2 concentration. This observation is consistent with the proposed initial formation mechanism for interfacial Ge nanocrystals, whereby Si-induced reduction of GeO_2 close to the interface forms Ge nuclei, which in turn grow larger via the adhesion of liberated Ge.

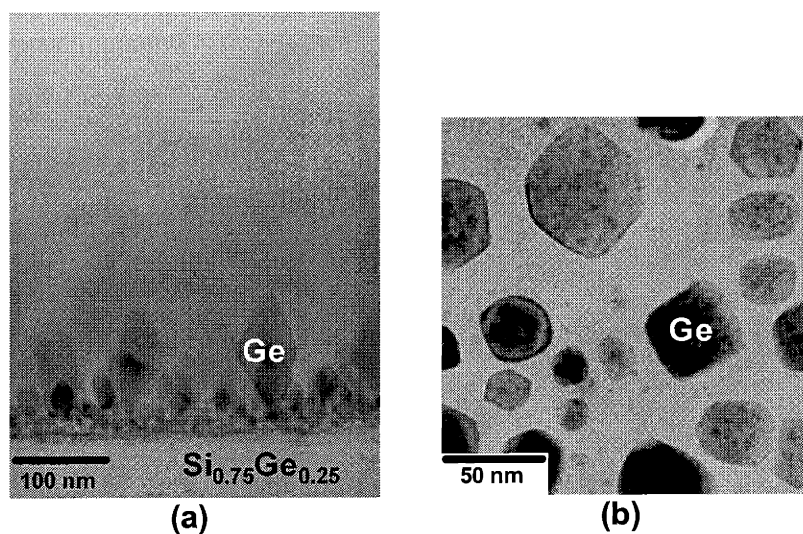


Figure 4.10 Micrographs for anneal at 800°C in 6% H_2 for 60 min: (a) XTEM, and (b) PVTEM through the crystallites.

4.5.2 800°C Anneals

For dilute- H_2 annealing at 800°C , no nanocrystals form in the oxide bulk after a 60-min anneal, and only crystallites on the $\text{Si}_{0.75}\text{Ge}_{0.25}$ interface are present, as shown by the TEM in Figure 4.10. These interfacial Ge crystallites form via solid phase epitaxy,

whereby atoms diffuse through a solid and are deposited at an interface.^{98,99} In the present case, crystallites on the bottom of the oxide have equilibrium, diamond-like shapes (as revealed by the XTEM in Figure 4.10(a)), and are oriented in approximate registry with the underlying $\text{Si}_{0.75}\text{Ge}_{0.25}$.

PVTEM samples of the nanocrystals were created by first performing a timed, dilute HF etch of the oxide, stopping on the nanocrystals. Subsequently, a conventional PVTEM sample was prepared by grinding and ion milling the sample backside. The PVTEM micrograph in Figure 4.10(b) shows that the well-formed nanocrystals have a six-sided, faceted structure; some of the crystallites may not have attained ideal equilibrium faceting due to kinetic limitations present at the annealing temperature and time.

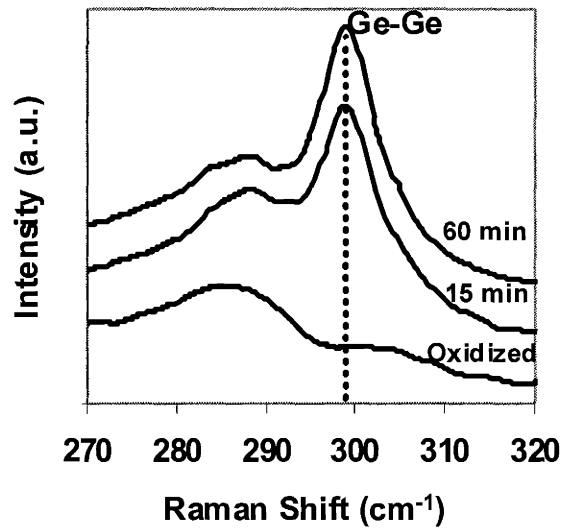


Figure 4.11 Raman spectra for SiGe-oxide annealed in 6% H₂ at 800°C, for 15 and 60 min.

A possible explanation for lack of Ge nanocrystals in the bulk of the oxide is the combination of a very low nucleation frequency for dilute H₂, in addition to increased coarsening at 800°C. Low nucleation frequency arises from a low liberated Ge concentration (due to the dilute H₂ concentration), combined with a high liberated Ge diffusivity at 800°C. In addition, increased coarsening could extinguish any small nuclei that do happen to form in the oxide bulk, allowing all the liberated Ge to eventually diffuse to the SiGe/oxide interface and adhere to the crystallites that have already nucleated. Raman spectra of the sample annealed for various times (Figure 4.11), indicate that Ge crystallites form after 15 min. This was verified with XTEM, which revealed Ge nanocrystals forming only at the interface, as shown in the XTEM of the 60-min anneal in Figure 4.10. The XTEM of the 15-min anneal possesses a similar structure, except the interfacial nanocrystals were smaller.

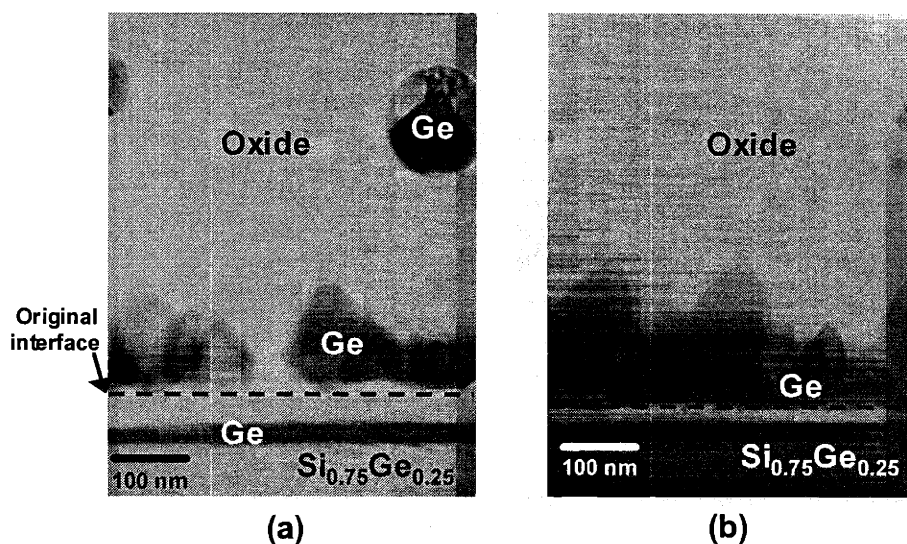


Figure 4.12 XTEM micrographs for 6% H₂ anneals at (a) 900°C for 60 min, and (b) 800°C for 240 min.

4.5.3 900°C Anneals

For completeness, Figure 4.12(a) shows XTEM for a 60-min anneal at 900°C using dilute H₂, revealing the presence of a very low density of large crystallites with sizes larger than 100 nm. Such a small number of nanocrystals in the bulk is consistent with a very low nucleation density, similar to the situation observed for the 800°C case; the appearance of some nanocrystals in the bulk may simply be due to the long annealing time employed. In particular, since reduction proceeds much more quickly at higher temperatures, comparing equal time anneals at 800 and 900°C may not be equivalent. After 60 min at 900°C, our thickness reduction measurements for dilute H₂ annealing indicate that all the GeO₂ should be completely reduced. To allow for a fair comparison based on equivalent amounts of GeO₂ reduction, a prolonged anneal (240 min) at 800°C was performed, which should be sufficient to reduce all of the available GeO₂. Figure

4.12(b) shows the XTEM for this annealing condition, revealing the presence of very few nuclei in the bulk of the oxide, as in the 900°C, hence we conclude that prolonged anneals above 800°C lead to a low (but finite) density of nanocrystals in the oxide bulk.

Finally, also note that in the 900°C case, crystallites also develop near the bottom of the oxide, via the same mechanism that creates interfacial crystallites in the 800°C case. In addition, due to the elevated temperature and long anneal time, the underlying $\text{Si}_{0.75}\text{Ge}_{0.25}$ is inadvertently oxidized by the water vapor byproduct. Due to the high temperature, Ge is expelled to the oxidation front and forms a thin 28-nm layer of highly dislocated Ge on the $\text{Si}_{0.75}\text{Ge}_{0.25}$, as shown in the XTEM in Figure 4.12(a).

4.6 Summary of Nanocrystal Formation Results

4.6.1 Homogenous Nanocrystal Density

Various Ge crystalline microstructures obtained in the bulk of the oxide are illustrated graphically in Figure 4.13 as a function of both H_2 concentration and annealing temperature. In all cases, the time chosen is such that a majority of the GeO_2 originally present in the oxide has been reduced, as was verified by thickness contraction of the oxide film. For example, in the case of the pure H_2 anneals at 700°C, 5 min is required to complete a significant portion the reduction, whereas for 875°C, only 1 min is needed (Figure 4.8).

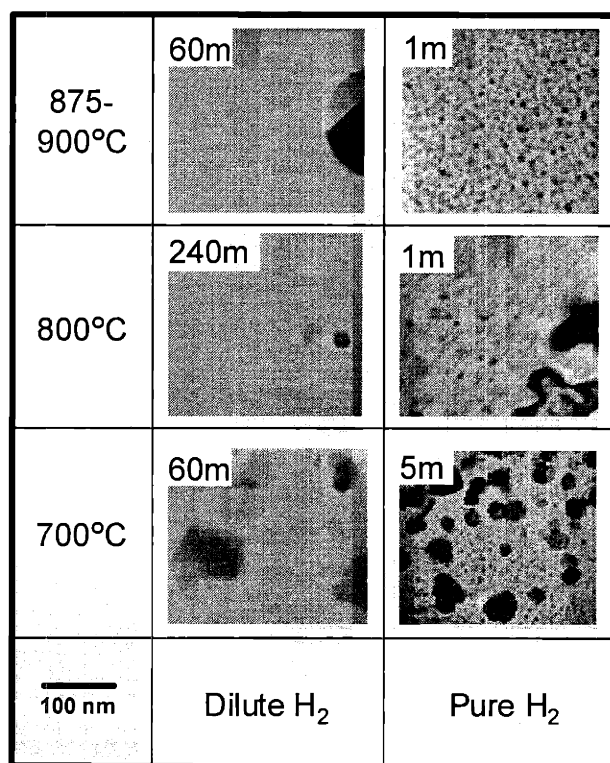


Figure 4.13 Summary of XTEM micrographs showing homogenous Ge nanocrystals for various annealing conditions.

The main effect of increasing the H₂ concentration is an increase in the homogenous nanocrystalline density, namely the density of nanocrystals in the bulk of the oxide. This effect arises from an increase in the rate of the reduction reaction, since a higher H₂ concentration will cause the reduction reaction (Eqn. 4.2) to proceed forward more rapidly. This results in an increase in the elemental Ge concentration diffusing within the bulk of the oxide, which in turn implies an increased Ge-Ge collision rate, and hence more nucleation events, leading to a higher ncGe density.

Another important observation is the low homogenous nanocrystal density present for high temperatures and dilute H₂. Under such conditions, the elemental Ge concentration, at any given instant, is very low, since any small amount of elemental Ge

liberated by the reduction reaction quickly diffuses within the oxide and deposits onto the SiGe interface. Low elemental Ge concentration in the oxide bulk in turn implies that the Ge-Ge collision frequency, and hence the nanocrystal nucleation density, is extremely low, and potentially non-existent for very dilute H₂ concentrations.

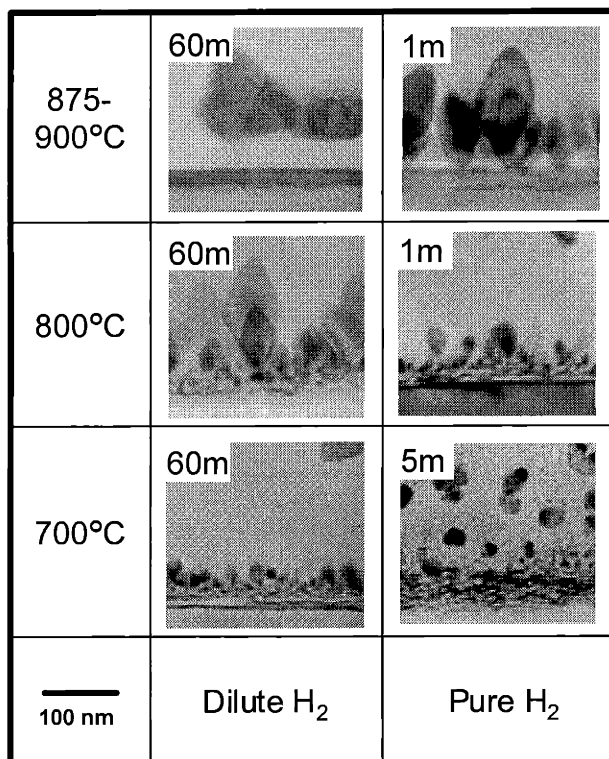


Figure 4.14 Summary of XTEM micrographs showing interfacial Ge for various annealing conditions.

4.6.2 Interfacial Ge Deposition

The amount of observed Ge interfacial deposition is summarized in Figure 4.14, which shows the incredibly fast rate at which Ge deposits on the interface at high temperatures (*e.g.* over 100 nm of Ge deposition after only 1 min at 875°C, using pure H₂). This fast solid phase epitaxy process indicates the existence of an enhanced Ge

diffusion mechanism, as compared to the slow diffusivity of Ge in thermal SiO₂ (0.1 nm diffusion length for 1 h at 900°C).

In the present material system, diffusion of elemental Ge in the oxide could be partially increased by the presence of GeO₂ in the oxide, and possibly enhanced even more by any free volume created during the reduction process. During the GeO₂ reduction reaction, oxygen is removed from the oxide matrix via water out-diffusion. The process eventually leads to a significant volume contraction, but the contraction may possess a relaxation time that is longer than the characteristic diffusion time for elemental Ge in such a porous oxide network. Just like grain-boundary diffusion is known to be many orders of magnitude larger than corresponding lattice diffusion, a similar process may be responsible for the observed fast Ge diffusion in the reducing oxide. Finally, another possible Ge diffusion enhancement mechanism might be the production of GeO gas via $\text{GeO}_2 + \text{Ge} \Rightarrow 2\text{GeO}$.⁶⁸

4.7 Nanocrystal Formation Model

To condense the key physical effects contributing to the observed trends, a model of the nanocrystalline Ge formation process was developed and a simulation was performed to generate T-H₂ maps for nanocrystalline density and Ge interfacial deposition. The model assumes the system is in thermodynamic quasi-static equilibrium, and that the process is entirely described by the reduction equation (Eqn. 4.2). The model is depth independent and is formulated using average concentrations (over the entire oxide depth), hence the system is described using only a few scalar concentrations,

namely $[\text{GeO}_2]$, $[\text{H}_2]$, $[\text{Ge}]$, and $[\text{H}_2\text{O}]$. Also, since H_2 diffusion is extremely fast, $[\text{H}_2]$ is set to a constant value, determined by the percent of H_2 in the ambient and the solubility of H_2 in the mixed oxide. For the purposes of this simulation, the slight temperature dependence of hydrogen solubility in oxide can be overlooked, and the solubility can be approximated as constant over the entire 700-900°C range.¹⁰⁰

At every instant in time, equilibrium concentrations are calculated and new ncGe nuclei are formed based on the instantaneous elemental Ge concentration. The ncGe nucleation rate is proportional to the Ge-Ge collision frequency, which is calculated using mean free path estimates. In particular, assuming that elemental Ge atoms have a cross-sectional area A_{Ge} , based on classical collision theory the collision rate per unit volume can be shown to equal $A_{\text{Ge}}^2 D_{\text{Ge}} [\text{Ge}]^3$, where D_{Ge} is the Ge diffusivity and $[\text{Ge}]$ is the instantaneous elemental Ge concentration. Combining this expression with the probability factor gives the following estimate for the ncGe nucleation rate per unit volume:

$$\text{Nucleation Rate} = A_{\text{Ge}}^2 D_{\text{Ge}} [\text{Ge}]^3 P_n \exp\left(-\frac{E_n}{kT}\right), \quad \text{Eqn. 4.3}$$

where P_n is the prefactor of the Arrhenius exponent, and E_n is the nucleation energy barrier.

In addition to the quasi-static equilibrium concentrations given by the reduction reaction, the model also incorporates the out-diffusion of water vapor via the surface. The amount of water that out-diffuses in a given time dt is estimated using a simple 1-dimensional (1D) model. In general, for any 1D system with an initial step-wise

concentration of length L and diffusivity D , the rate of change of the average

concentration (over the entire depth) is given by $\frac{dc}{dt} \approx -\frac{\pi^2 D}{L^2} c$.⁵⁰ In our case of water

out-diffusion via the surface, this implies that the new water concentration after a time dt is

$$[H_2O]' = [H_2O]_o - A[H_2O]_o dt, \quad \text{Eqn. 4.4}$$

where $A = \frac{\pi^2 D_{H_2O}}{2L^2}$, and D_{H_2O} is the water diffusivity in the oxide, L is the oxide

thickness, and the factor of 2 in the denominator arises from the fact that only the surface leads to water out-diffusion.

Our model also includes the decrease in $[Ge]$ due to fast diffusion of elemental Ge, leading to deposition onto the oxide/SiGe interface and onto pre-existing homogenous ncGe. Using a similar approach employed to calculate the out-diffusion of water, the new elemental Ge concentration after a time dt is

$$[Ge]' = [Ge]_o - B_{int}[Ge]_o dt - B_{nc}[Ge]_o dt, \quad \text{Eqn. 4.5}$$

where $B_{int} = \frac{\pi^2 D_{Ge}}{2L^2}$, and $B_{nc} = \eta \frac{\pi^2 D_{Ge}}{L_{nc}^2}$. The B_{int} term represents deposition of

elemental Ge at the oxide/SiGe interface, and the B_{nc} term accounts for deposition onto homogenous ncGe. L_{nc} is the mean free distance an elemental Ge atom must diffuse

before encountering a Ge nanocrystal, and is given by $L_{nc} = (\rho_{nc} \pi r_{nc}^2)^{-1}$, where ρ_{nc} is the

ncGe density and r_{nc} is the average radius of homogenous Ge nanocrystals. Also note

that the nanocrystal deposition term B_{nc} also contains an additional factor η , which is the sink efficiency of the nanocrystals given by:⁵⁰

$$\eta = \frac{1}{1 + D_{\text{Ge}}/\kappa r_{\text{nc}}}, \quad \text{Eqn. 4.6}$$

where κ is the transfer rate, which is a measure of the incorporation rate of impinging Ge atoms into the growing nanocrystal. When the transfer rate is much faster than the diffusivity, the sink efficiency approaches unity, and ncGe growth is diffusion-controlled. On the other hand, when the diffusivity greatly out-paces the transfer rate, the sink efficiency is much smaller than unity and nanocrystal growth is sink-controlled. In this regime, only a fraction of the Ge atoms colliding onto the Ge nanocrystal are incorporated.

Incremental changes caused by water out-diffusion and Ge deposition can be folded into oxygen and germanium conservation equations. Conservation of oxygen atoms yields

$$\frac{d[\text{H}_2\text{O}]}{dt} + 2 \frac{d[\text{GeO}_2]}{dt} = -A[\text{H}_2\text{O}], \quad \text{Eqn. 4.7}$$

and conservation of germanium atoms implies

$$\frac{d[\text{Ge}]}{dt} + \frac{d[\text{GeO}_2]}{dt} = -B[\text{Ge}], \quad \text{Eqn. 4.8}$$

where $B = B_{\text{int}} + B_{\text{nc}}$.

In addition to these conservation equations, the other differential equation required to model the system is given by differentiating the equilibrium equation

$$K[\text{H}_2]^2[\text{GeO}_2] = [\text{H}_2\text{O}]^2[\text{Ge}]. \quad \text{Eqn. 4.9}$$

After differentiation, and using the conservation equations to replace the derivatives of $[\text{GeO}_2]$ and $[\text{H}_2\text{O}]$, the following expression for the rate of change of $[\text{Ge}]$ can be derived:

$$\frac{d[\text{Ge}]}{dt} = \frac{2A[\text{H}_2\text{O}]^2[\text{Ge}] - KB[\text{H}_2]^2[\text{Ge}] - 4B[\text{H}_2\text{O}][\text{Ge}]^2}{K[\text{H}_2]^2 + 4[\text{H}_2\text{O}][\text{Ge}] + [\text{H}_2\text{O}]^2}. \quad \text{Eqn. 4.10}$$

At every incremental time step dt , this equation is used to calculate the change in $[\text{Ge}]$ based on the previous values of $[\text{Ge}]$, $[\text{GeO}_2]$, and $[\text{H}_2\text{O}]$. Using the new $[\text{Ge}]$ value, new $[\text{GeO}_2]$ and $[\text{Ge}]$ values are calculated using discrete versions of the conservation equations. In addition, the amount of Ge deposited onto the oxide/SiGe interface and onto homogenous nanocrystals (contributing to ncGe growth) is given by $B_{\text{int}}[\text{Ge}]dt$ and $B_{\text{nc}}[\text{Ge}]dt$ respectively. During the course of the simulation, the total amount of Ge deposited on the interface is calculated, and the amount of Ge deposited onto ncGe is used to calculate the ncGe growth rate dr_{nc}/dt .

A major approximation employed in the simulation is the use of only the average nanocrystal radius. Within every time interval dt , 2-atom nuclei form via Ge-Ge collisions. In addition, during every time interval, existing nuclei grow via deposition of elemental Ge. As time progresses this process would lead to a distribution of nanocrystal radii, but to reduce computational complexity, we only carry an average adjusted radius (based on volume conservation) throughout the simulation. This approximation reduces the computational complexity, thereby allowing for a parallel calculation to be carried out for the entire T-H₂ space.

4.8 Nanocrystal Formation Simulation Results

Simulations were carried out using a time step of 0.1 s, an oxide depth $L=500$ nm, and the solubility of H₂ in the mixed oxide was set to 0.00765 nm^{-3} . Based on

thermodynamics data tables,¹⁰¹ the GeO_2 reduction equilibrium constant was given by $K_o=8.3 \times 10^3$ and $E_{\text{red}}=0.8$ eV. The nanocrystal nucleation activation energy was set to $E_n=0.5$ eV with a corresponding pre-factor of $P_n=0.2$, and the transfer rate was set to $\kappa=10^4$ nm/s. Water diffusivity in the mixed oxide was given by $D_{\text{H}_2\text{O}}=D_{\text{oH}_2\text{O}} \exp(-E_{\text{H}_2\text{O}}/kT)$, with $D_{\text{oH}_2\text{O}}=5.1 \times 10^8$ nm²/s and $E_{\text{H}_2\text{O}}=0.69$ eV, chosen to fit the data. These values correspond to a three-fold increase in the pre-factor and an activation energy decrease of about 0.1 eV compared to water diffusivity in thermal SiO_2 . Finally, the Ge diffusivity in the mixed (contracting) oxide was chosen to have an activation energy of 3.0 eV (compared to 3.9 eV in thermal SiO_2), with a pre-factor of 8.4×10^{18} nm²/s, which is about 10^7 times larger than the corresponding pre-factor for Ge diffusivity in high-quality thermal silicon oxide.¹⁰² The large enhancement in diffusivity is required to explain the fast rate of Ge deposition onto the oxide/SiGe interface, and as was noted in the previous section, and is possibly due to the diffusivity enhancement caused by free volume created during the reduction process.

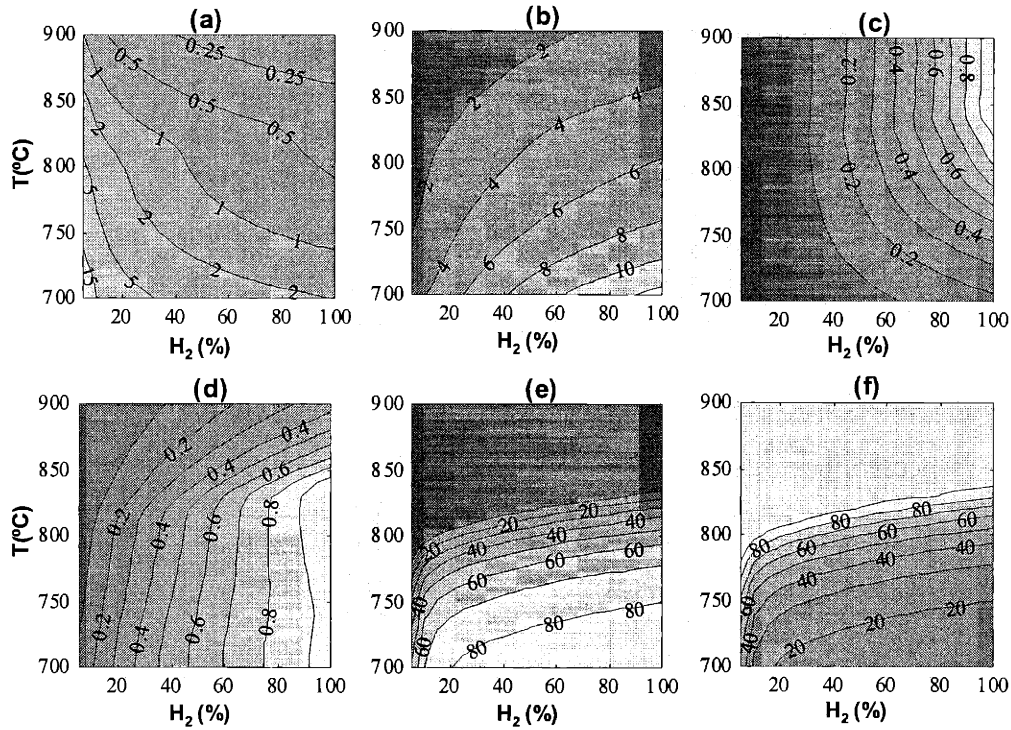


Figure 4.15 Simulation results showing T-H₂ contour maps for (a) time (in min) required to reduce half of the initial GeO₂, (b) time-averaged elemental Ge (in percent of total liberated Ge), (c) time-averaged ncGe nucleation rate (in normalized units), (d) homogenous ncGe density (in normalized units), (e) amount of Ge incorporated into homogenous ncGe (in percent of total released Ge), and (f) amount of Ge deposited onto oxide/SiGe interface (in percent of total released Ge).

Results of calculations using the aforementioned parameters are presented in Figure 4.15. Rather than present results for equal times across the T-H₂ space, a more meaningful comparison for equal amounts of GeO₂ reduction is made. In our results, the simulation time for each T-H₂ point is such that 50% of the initial GeO₂ is reduced. In this way, each system at a given T-H₂ is annealed for enough time such that half of the initial GeO₂ undergoes reduction. A contour map for this required time is presented in Figure 4.15(a). Consistent with our experimental data, the reduction time drops drastically with increasing temperature. The shortest reduction time of less than a quarter

of a minute is required for annealing in pure H_2 at $900^\circ C$, whereas the longest time is about 30 min for annealing in 6% H_2 at $700^\circ C$.

Due to increased Ge diffusivity at elevated temperatures, any liberated Ge quickly diffuses and deposits onto the oxide/SiGe interface, thereby leading to lower elemental Ge concentrations at high temperatures. This stems from a dramatic increase in Ge diffusivity at high temperatures, in conjunction with a slowly increasing Ge production rate (with respect to increasing temperature). Simulation data for the time-averaged elemental Ge concentration (in percent of the total released Ge) is presented in Figure 4.15(b), showing that the highest elemental Ge content occurs at high H_2 concentrations and low temperatures.

Based on the time-averaged elemental Ge content, it might be expected that the nucleation rate is largest for conditions with high elemental Ge concentrations, like low temperature and high H_2 concentrations. But, in addition to this effect, the nucleation rate scales with the Ge diffusivity. Diffusivity varies exponentially with temperature, and therefore this dependence has a larger influence on the nucleation rate. Based on our simulation results, a T- H_2 contour map for the (normalized) time-averaged nucleation rate is presented in Figure 4.15(c).

Although the time-averaged nucleation rate attains a maximum for high temperatures and pure H_2 , the nanocrystalline density (at the halfway reduction time) does not attain a maximum at those exact annealing conditions, as shown in the (normalized) ncGe density plot in Figure 4.15(d). The maximum occurs at a slightly lower temperature along the pure H_2 line, since the ncGe density depends on both the

nucleation rate and the length of the reduction time employed for each T-H₂ condition. Although lower temperatures have a lower nucleation rate, this is offset by the much longer reduction times, as presented in Figure 4.15(a). Based on various simulations conducted, the exact location of the ncGe density maximum varies markedly with parameters like the Ge diffusivity and the sink efficiency, but the general trend of the results for the dilute H₂ region were consistent. In general, it was concluded that the lowest ncGe density occurs for dilute H₂ annealing with temperatures at or above 800°C, which is consistent with the experimental results presented in Figure 4.13. Finally, note that the simulation does not include nanocrystal coalescence and coarsening, hence comparison with the experimental data must be performed with the understanding that coarsening may alter the ncGe density.

The model was also employed to calculate the percent of the total released Ge incorporated into homogenous nanocrystals (shown in Figure 4.15(e)), and the percent deposited onto the oxide/SiGe interface (Figure 4.15(f)). Figure 4.15(f) shows that the majority of the liberated Ge deposits onto the interface for temperatures above about 780°C for dilute 6% H₂, whereas a higher temperature of about 840°C is required to meet this condition for pure H₂ anneals. These results are consistent with the experimental data presented in Figure 4.14, which show substantial interfacial deposition for anneals above the noted temperatures.

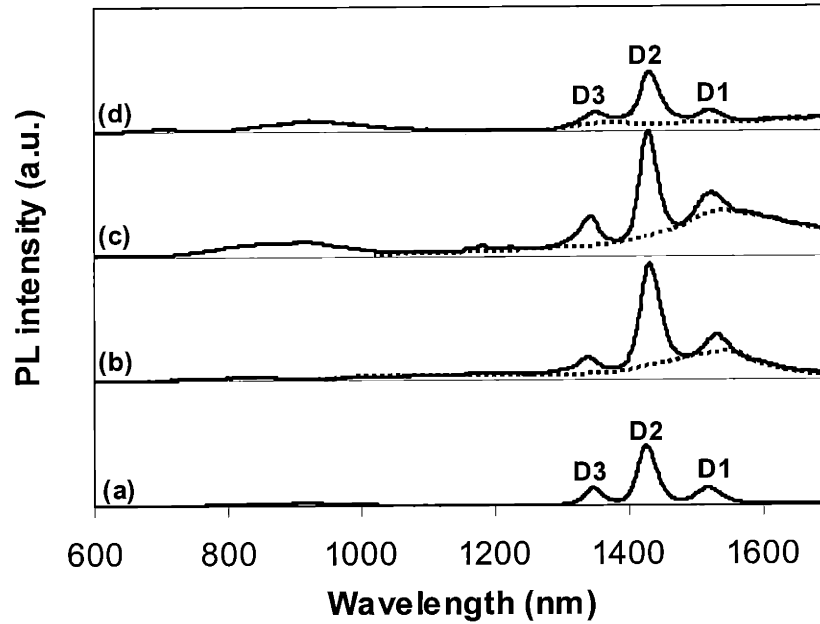


Figure 4.16 Infrared 4K photoluminescence spectra for (a) as-oxidized SiGe-oxide, and after pure H₂ anneals at (b) 875-900°C for 1 min, (c) 700°C for 5 min, and (d) 700°C for 15 min.

4.9 Infrared Photoluminescence of Nanocrystalline Ge

Photoluminescence (PL) measurements at 4K, performed on the samples annealed at 700°C using pure H₂, show a dependence on annealing time, and hence on nanocrystal size. Figure 4.16(a) shows the spectra for the as-oxidized Si_{0.75}Ge_{0.25}O₂ on Si_{0.25}Ge_{0.25}, revealing the presence of a group of peaks (D1-D3) due to dislocations in the underlying SiGe graded buffer on Si,¹⁰³ and a broad peak in the near-infrared range (0.8-1.0 μm). Comparing the H₂ annealed samples (Figure 4.16(b,c,d)) with the unannealed sample (Figure 4.16(a)) reveals an increase in the intensity of the 0.8 μm near-infrared peak.

This broad peak is probably due to interfacial oxygen-deficient defects between the oxide and the Ge nanocrystals.¹⁰⁴

Of more interest, in our case, is the PL caused by the quantum confinement effect. Careful inspection of the spectra reveals a broad background (shown by dotted lines in Figure 4.16) superimposed on the D-lines. PL data collected for the sample annealed at 875-900°C for 1 min, in pure H₂, contains a broad background (Figure 4.16(b)), attributed to the existence of small nanocrystals in the bulk of the oxide, as confirmed by XTEM (Figure 4.7). This background is practically non-existent for the as-oxidized sample, increases dramatically after 5 min of annealing at 700°C (Figure 4.16(c)), then decreases and shifts its edge to higher wavelengths after 15 min, as shown in Figure 4.16(d). Given the presence of numerous small nanocrystals (5-15 nm) in the 5 min sample, we postulate that this infrared background is due to quantum confinement in ncGe, and its broad shape is due to the broad size distribution of the nanocrystals. Within this framework, the shift in the background of 15-min sample is attributed to larger crystallite sizes formed via coarsening.

Recently, tight-binding calculations for Ge nanocrystals have been used to predict bandgap variation with nanocrystal diameter.¹⁰⁵ The results of these calculations were used to determine the absorption edge as a function of ncGe diameter, as presented in Figure 4.17. The results show that spherical Ge nanocrystals with diameters of 5-15 nm have bandgap variations corresponding to wavelengths ranging from 1.0-1.7 μm , which is consistent with the range of our background peak for the 5-min, 700°C annealed sample and the 1-min, 900°C annealed sample. From the inset in Figure 4.17, the

nanocrystal absorption edge (and hence bandgap) has practically approached the bulk Ge limit ($1.87 \mu\text{m}$) past a diameter of 30 nm.

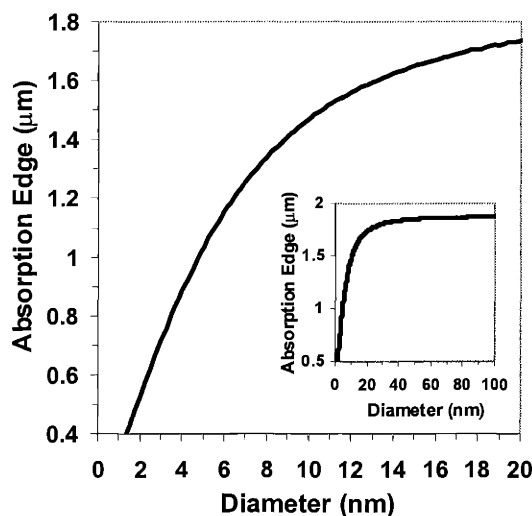


Figure 4.17 Calculated absorption edge versus Ge nanocrystal diameter. Inset shows absorption edge variation for larger diameters up to 100 nm.

Finally, although the broad background observed in this work is likely due to quantum confinement effects in Ge nanocrystals, the exact mechanism is still not absolutely verified. In addition to recombination of both confined holes and electrons in nanocrystals, previous research has indicated that the process could also involve the recombination of an electron (hole) trapped on a nanocrystal surface defect, with a free hole (electron) confined in the nanocrystal.¹⁰⁵

4.10 Summary

Process kinetics for Ge nanocrystal formation during the reduction of $\text{Si}_{0.75}\text{Ge}_{0.25}\text{O}_2$ on $\text{Si}_{0.75}\text{Ge}_{0.25}$ via H_2 was found to comprise a rich group of processes, including nucleation, growth, and coarsening. Dilute H_2 , high-temperature annealing produced few

nanocrystals in the bulk of the oxide, and large crystallites on the SiGe interface. To condense the relevant physical effects, an approximate model describing the reduction process was developed and used to generate T-H₂ maps for various observed quantities like reduction time, nucleation rate, ncGe density, and interfacial deposition. Finally, photoluminescence of samples with dense Ge nanocrystals exhibited a broad background in the infrared region attributed to the quantum confinement effect.

Chapter 5 Fabrication of Nanocrystalline Ge Waveguides on Si

In the previous chapter, nanocrystalline Ge (ncGe) was formed via the reduction of SiGe-oxide on SiGe. Previous work has shown that such Ge nanocrystals possess interesting and potentially useful optical properties, including (1) a shifted absorption edge, due to carrier quantum confinement,^{105,106} and (2) a large nonlinear response originating mainly from excited carrier absorption.¹⁰⁷⁻¹¹⁰ Nonlinear materials have an index of refraction that possesses a slight dependence on light intensity, and as such are suitable materials for all-optical switches, such as nonlinear directional couplers. Since the building blocks of these devices are planar waveguides with a nonlinear core material, waveguide fabrication using our ncGe is the first crucial step required for the application of ncGe to nonlinear photonic devices.

In addition, since Si substrates possess significant benefits, including large wafer size and well-established processing tool-sets, they serve as the ideal platform for the low-cost fabrication of photonic devices. Fortunately, nanocrystalline Ge waveguides and devices can be fabricated on Si substrates, as long as care is taken to ensure that the optical mode is properly isolated from the underlying Si. In general, to create a confined optical mode, lower index materials must clad a higher index waveguide core. Since ncGe in an oxide matrix has a larger index than pure oxide, a ncGe waveguide core clad by SiO₂ serves as an ideal structure for ncGe waveguides on Si substrates, as shown in Figure 5.1(a). The SiO₂ layer between the waveguide core and the Si substrate, dubbed the “standoff”, must be thick enough to provide adequate isolation and minimal leakage into the substrate.¹¹¹

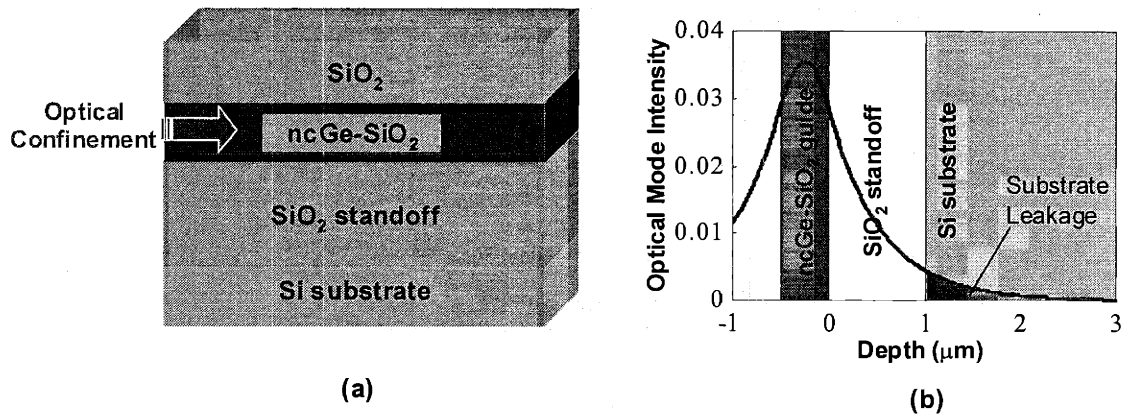


Figure 5.1 (a) Cross-section of ncGe waveguide structure on Si substrate, and (b) calculated mode intensity, showing substrate leakage.

5.1 Optical Power Substrate Leakage

A calculated optical mode intensity profile for a typical ncGe waveguide on Si is presented in Figure 5.1. The integrated area in the substrate region is proportional to the power lost via substrate leakage. Estimations of substrate loss were carried out for a ncGe waveguide formed by the reduction of $\text{Si}_{0.75}\text{Ge}_{0.25}\text{O}_2$. The employed approach combines ray optics propagation with the integrated power loss (as calculated from the modal intensity profile). Specifically, at every reflection at the core/standoff interface, the integrated power is assumed to leak into the substrate.

Substrate power leakage depends strongly on both the standoff thickness and the ncGe core index. If the entire mixed-oxide were reduced, the ncGe core index should approach 1.8, based on volume arguments using $n_{\text{Ge}}=4.0$ and $n_{\text{SiO}_2}=1.45$. In contrast, the index for $\text{Si}_{0.75}\text{Ge}_{0.25}\text{O}_2$ is only 1.49, as measured experimentally with a prism coupler assembly. A fully reduced layer has a large amount of ncGe (hence a higher Ge volume in the oxide), yielding a higher index, thereby implying greater optical confinement and

less substrate leakage. For our substrate loss estimates, the ncGe core is assumed to be partially reduced, with an index of 1.65, yielding conservative overestimates for the substrate loss.

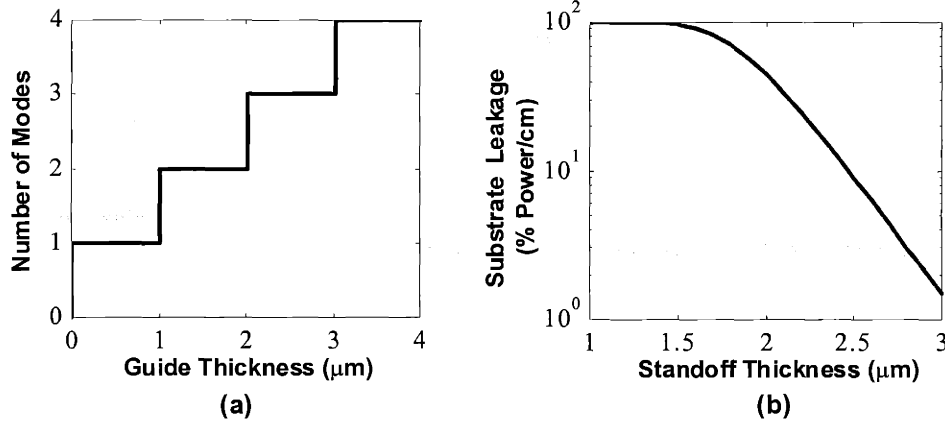


Figure 5.2 Calculation results for ncGe waveguide with index $n=1.65$, (a) number of modes versus ncGe layer thickness, (b) substrate leakage versus standoff thickness for a single-mode, 500-nm thick ncGe waveguide.

As shown in Figure 5.2(a), the waveguide is single mode below 1 μm thickness, therefore a 500-nm thick ncGe layer is used for the substrate leakage estimate presented in Figure 5.2(b). Power loss due to leakage is less than 2% per cm for a 3-μm thick SiO₂ standoff, which is more than acceptable for most applications.

5.2 Waveguide Structure Fabrication

In the previous chapter, ncGe/SiGe/Si structures were easily created by oxidizing epitaxial SiGe on Si at low temperatures, followed by hydrogen annealing to reduce the mixed oxide. Since low-temperature oxidation would require too much time (on the order of days to weeks) to form microns of underlying SiO₂, that process cannot be used to produce thick standoff layers required for waveguide isolation. In this section, various

approaches used to create the standoff are outlined, where the first few methods presented failed to integrate a high-quality, SiGe-oxide with a thick SiO₂ standoff. In all cases, the instability of SiGe-oxide during oxidation processes was a major obstacle.

5.2.1 Oxidation of Polycrystalline SiGe on SiO₂/Si Approach

The obvious method for ncGe waveguide fabrication via SiGe-oxide reduction, begins with CVD deposition of SiGe on an oxidized Si substrate, followed by low-temperature oxidation of the deposited SiGe layer and hydrogen annealing to reduce the SiGe-oxide and form Ge nanocrystals. This approach allows for very thick SiO₂ standoff layers. Since standoff layer formation is performed before any SiGe-oxide is present, very high-temperature, wet oxidation can be used to create thick layers of SiO₂ on Si. In particular, in our work, 2.7- μ m thick SiO₂ layers were formed via wet oxidation of Si wafers at 1100°C for 15 h, and even thicker standoff layers can be created via deposition process, like PECVD. Although seemingly simple, this approach is plagued by two major problems, both stemming from the polycrystalline nature of the SiGe that forms when depositing crystalline material on oxide surfaces.

The first problem is the large surface roughness of the polycrystalline SiGe as deposited via UHVCVD. Such roughness would increase optical scattering losses, and must be minimized. Other obstacles encountered while depositing SiGe on oxide via CVD include a low nucleation rate for moderate-to-high Ge contents, in addition to long incubation times before growth initiation. These factors make layer thickness difficult to estimate, since film thickness depends on the growth rate and the incubation time, which

varies inversely with growth pressure and temperature. In addition, low nucleation rate also implies large grain size and very rough surfaces.

To both minimize the incubation time and increase the nucleation rate, a low growth temperature of 650°C was used. In addition, the SiGe growth was initiated with a high pressure (20 mT) phase to seed the surface with a high density of SiGe nuclei, followed by a low pressure (4 mT) phase to grow the SiGe film. This technique was used to grow about 100 nm of poly-Si_{0.75}Ge_{0.25} on SiO₂/Si substrates, with a peak-to-valley height of about 20-30% of the deposited thickness. Another technique for SiGe growth on oxide requires a Si nucleation layer (using SiH₄), followed by SiGe growth around Si nuclei; unlike SiGe, pure Si has a very low incubation time, making it an ideal seed layer on oxide.

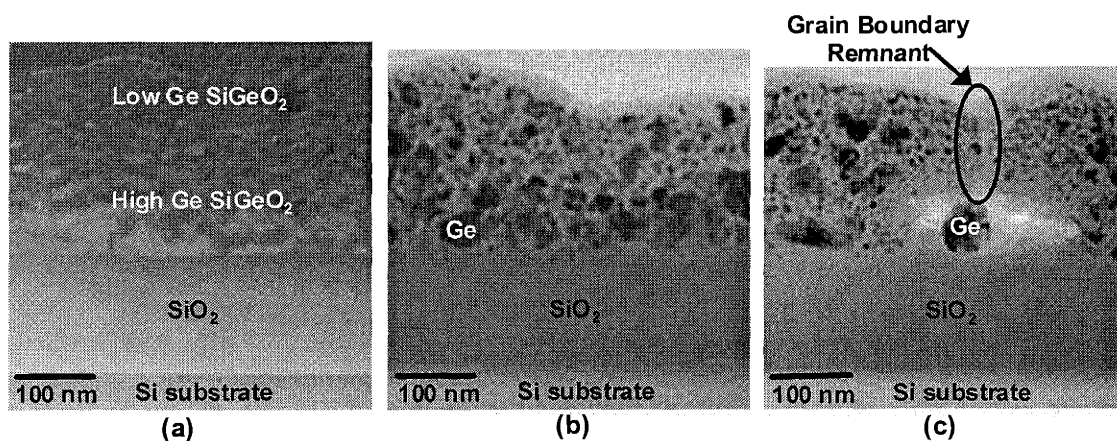


Figure 5.3 XTEM micrographs of polycrystalline SiGe approach: (a) after 5h 700°C wet oxidation, (b,c) after oxidized sample is annealed for 15 min in pure H₂ at 700°C, showing presence of more Ge at SiO₂ interface, and large voids.

Other than surface roughness, polycrystalline SiGe is incompatible with high-quality SiGeO₂ formation using 700°C wet oxidation, due to the fast diffusion of expelled Ge along grain boundaries. This leads to significant Ge pile-up even at low temperatures,

as discussed in Section 3.5.^{52,71} As shown in Figure 5.3(a), 700°C wet oxidized poly-SiGe has non-uniform contrast striations, due to higher GeO₂ content near the SiO₂ interface (forming via Ge pile-up). Non-uniformity of GeO₂ is problematic with regards to the nanocrystal formation process, since huge Ge nanocrystals form near the oxide interface, as illustrated in Figure 5.3(b). Even more catastrophic, regions with higher Ge content experience more volume contraction, and hence voids form near the oxide interface and along former grain boundaries containing high GeO₂ concentrations (Figure 5.3(c)).

Our experiments have shown that polycrystalline SiGe is unacceptable for the proposed ncGe waveguide fabrication due to Ge pile-up during oxidation. The results imply that monocrystalline SiGe on Si is essential for the fabrication of high-quality SiGe-oxide via 700°C wet oxidation. Both approaches that follow satisfy this requirement by employing epitaxial SiGe deposited directly on Si.

5.2.2 Two-step Oxidation of Epitaxial SiGe on Si Approach

Beginning with an epitaxial SiGe layer deposited on a Si substrate, this next approach continues with a conventional 700°C wet oxidation, thereby transforming the SiGe/Si into SiGeO₂/Si. The underlying Si oxidizes very little (a few tens of nanometers) at 700°C, hence this low-temperature oxidation cannot be used to create the microns of required SiO₂ standoff. In response, the SiGeO₂/Si structure is wet oxidized at high temperatures to generate a 3-μm thick SiO₂ standoff.

Wet oxidation of Si at 1100°C for 20 h generates the desired SiO₂ standoff thickness, unfortunately, the pre-existing SiGeO₂ surface layer does not survive the high-

temperature oxidation step. A serious problem with the process is the inadvertent generation of H_2 from the wet oxidation of the underlying Si. The H_2 in turn reacts with GeO_2 (in the $SiGeO_2$) forming volatile GeO gas (via $GeO_2 + H_2 \rightarrow 2GeO$). At low temperatures (below $900^\circ C$), this reaction is not a dominant reduction pathway, due to its small equilibrium constant, with an activation energy of 2.85 eV (yielding only $K=0.08$ at $900^\circ C$). In contrast, the high activation energy implies that at higher temperatures, this reaction is a viable competing pathway to the nanocrystal formation reaction $GeO_2 + 2H_2 \rightarrow Ge + 2H_2O$ (which possesses an activation energy of 0.8 eV, yielding $K=3.0$ at $900^\circ C$).

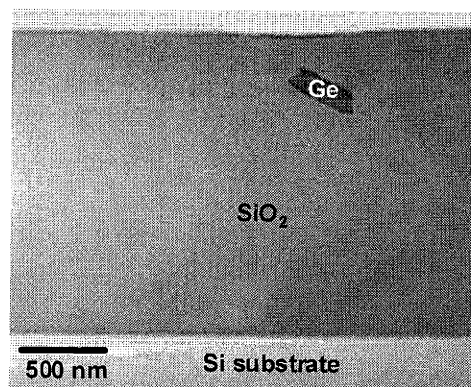


Figure 5.4 XTEM of two-step wet oxidized $Si_{0.75}Ge_{0.25}/Si$, using first $700^\circ C$ and then $1100^\circ C$ wet oxidation. A single large Ge nanocrystal was observed over the entire XTEM sample area, and the entire $SiGe$ -oxide was reduced to SiO_2 .

For example, at $1100^\circ C$, the equilibrium rate constant for the GeO generation by H_2 reduction is about $K=5$, implying that this reaction proceeds forward generating a large amount of volatile GeO , which can out-diffuse via the surface, reducing the Ge content of the $SiGe$ -oxide. To determine if this is truly a problem, a simple experiment was performed by oxidizing a $Si_{0.75}Ge_{0.25}/Si$ sample in two steps. First, a 5h wet oxidation at $700^\circ C$ oxidized the $Si_{0.75}Ge_{0.25}$ layer, producing $Si_{0.75}Ge_{0.25}O_2/Si$. Next, a 5h wet

oxidation at 1100°C was performed to oxidize a portion of the underlying Si. Figure 5.4 shows that the entire SiGe-oxide formed during the first 700°C oxidation step is reduced to SiO₂ during the 1100°C wet oxidation. Also, only one large Ge nanocrystal formed over the entire XTEM sample area. These results prove that the two-step (700°C, 1100°C) wet oxidation process is inadequate for our purposes, likely due to GeO out-diffusion, which causes a loss of Ge.

A possible solution requires the use of high pressure oxidation (HIPOX) at low temperatures to replace the 1100°C step. Wet oxidation rates scale with the square root of the oxidant pressure, and since typical HIPOX uses a pressures of 15-25 atm, this implies that the oxidation rate can be increased by a factor of about 3, reducing oxidation times and allowing for the use of lower temperatures. Unfortunately, HIPOX equipment is expensive, and is not available in most fabrication facilities.

5.2.3 Oxidation of Epitaxial SiGe on Thinned SOI Approach

To avoid the many problems encountered in the previous approaches, a starting substrate of thick buried oxide SOI (silicon-on-insulator) was used to develop a feasible technique for ncGe waveguide structure fabrication. Although more expensive than Si, SOI substrates are composed of a monocrystalline Si surface layer and a buried oxide layer of any desired thickness, thereby eliminating the standoff fabrication step.

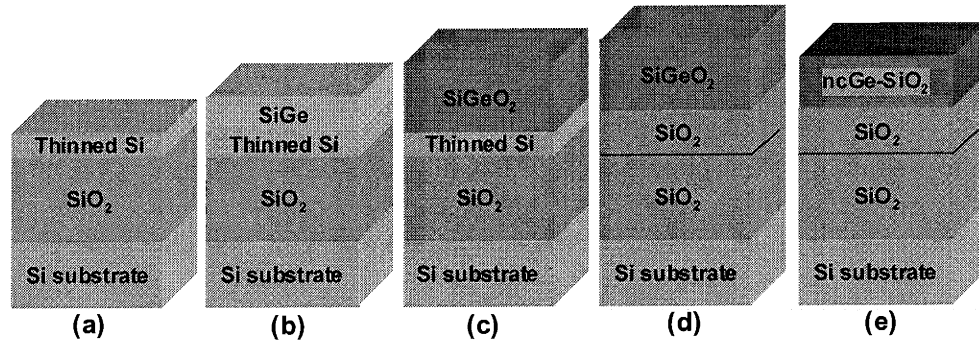


Figure 5.5 Process flow of SOI approach for ncGe waveguide structure fabrication using reduction of SiGe-oxide: (a) thinning of surface Si layer, (b) deposition of epitaxial SiGe, (c) low-temperature oxidation, forming SiGeO₂, (d) moderate temperature oxidation of underlying thinned Si layer, and (e) hydrogen reduction forming ncGe waveguide core.

Figure 5.5 shows the flow for the proposed fabrication process. In our case, the process began with a commercial (wafer bonded) SOI wafer, with a 3- μm buried oxide layer, and 340 nm of surface Si. First, the Si layer is thinned down to about 50 nm by oxidation followed by buffered oxide etching (BOE). Low pressure UHVCVD at 650°C was used to deposit about 250 nm of Si_{0.75}Ge_{0.25} on the thinned Si, with a resulting peak-to-valley surface roughness of about 20 nm (as determined via cross-sectional SEM). Low-temperature (700°C) wet oxidation for 8 h completely oxidized the entire deposited Si_{0.75}Ge_{0.25}, and a moderate temperature (900°C) wet oxidation for 6 h ensured that the underlying thinned Si was totally oxidized. Nanocrystalline Ge was then formed by pure H₂ annealing of the Si_{0.75}Ge_{0.25}O₂/SiO₂/Si structure. Finally, a SiO₂ surface cladding layer can then easily be deposited onto the ncGe-SiO₂/SiO₂, yielding the desired waveguide structure, with none of the problems encountered using previous approaches.

5.3 Nanocrystalline-Ge Slab-Waveguide Structure Characterization

Nanocrystalline-Ge slab waveguides were fabricated using the SOI approach presented in the previous section. After oxidation, the $\text{Si}_{0.75}\text{Ge}_{0.25}\text{O}_2/\text{SiO}_2/\text{Si}$ structure was annealed in 100% H_2 using two different annealing conditions: 700°C for 5 min, and 875°C for 1 min. These conditions were chosen to generate the highest density of Ge nanocrystals, as was determined from the nanocrystal formation study (for $\text{Si}_{0.75}\text{Ge}_{0.25}\text{O}_2/\text{Si}_{0.75}\text{Ge}_{0.25}$) presented in Chapter 4. Characterization of the waveguide structures included XTEM and photoluminescence.

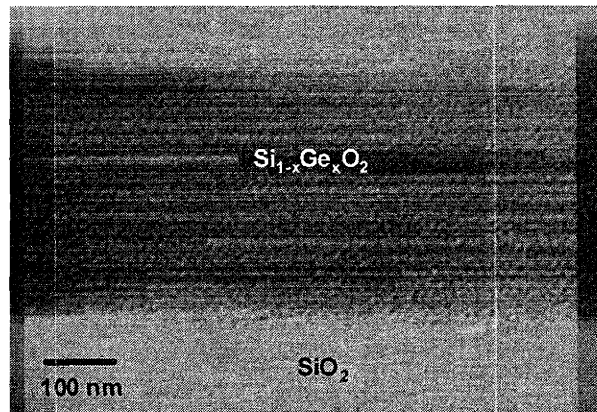


Figure 5.6 XTEM micrograph of as-oxidized slab waveguide structure, fabricated using the SOI approach.

5.3.1 Microstructure Characterization

XTEM of the as-oxidized slab waveguide structure on SOI, shown in Figure 5.6, reveals the absence of contrast striations in the SiGe-oxide. In comparison, in Section 3.5, it was found that low-temperature (700°C) oxidized SiGe had horizontal contrast striations, likely due to variations in the local GeO_2 concentration. During the slab

waveguide fabrication approach, stratified variations in the GeO_2 content may have been eliminated during the second wet oxidation step at 900°C (used to oxidize the underlying Si on the buried SiO_2). During that step, GeO_2 may diffuse more readily, thereby leading to a uniform redistribution of GeO_2 . In addition, the 900°C wet oxidation (of the underlying Si) generates H_2 , which can in turn influence the composition of the $\text{Si}_{0.75}\text{Ge}_{0.25}\text{O}_2$. In particular, some volatile GeO gas can form via $\text{GeO}_2 + \text{H}_2 \rightarrow 2\text{GeO}(\text{g})$, thereby decreasing the GeO_2 content of the mixed oxide. H_2 can also lead to elemental Ge formation via the usual GeO_2 reduction reaction, but the presence of H_2O , fed in for wet oxidation, should re-oxidize most of the elemental Ge.

Raman spectroscopy indicated negligible nanocrystalline Ge presence in the as-oxidized sample, based on a very weak signal at the Ge-Ge peak location (299 cm^{-1}). Unfortunately, the existence of a weak two-phonon Si substrate signal, centered at about the same location as the Ge-Ge peak, makes the exact determination of minute amounts of nanocrystalline Ge difficult. For this reason, Raman spectroscopy of the slab waveguide samples was only able to detect the presence of a substantial amounts of ncGe.

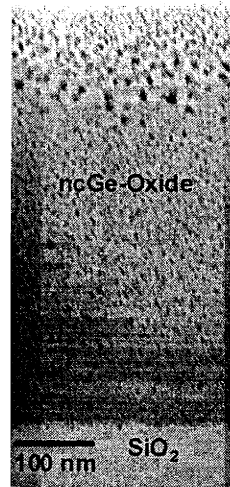


Figure 5.7 XTEM micrograph of slab waveguide structure formed using SOI approach, and annealed in pure H_2 at $700^\circ C$ for 5 min. Very small Ge nanocrystals exist throughout the entire bulk of the oxide.

Hydrogen annealing of the as-oxidized slab waveguide structure led to the reduction of the SiGe-oxide and the formation of Ge nanocrystals. Figure 5.7 shows very small nanocrystals formed using pure H_2 annealing at $700^\circ C$ for 5 min. The largest sized nanocrystals form near the top of the oxide, with diameters ranging from about 10-15 nm. Throughout the rest of the oxide, Ge nanocrystals with an average size of about 5 nm form during annealing.

Such diameters are much smaller than those observed for similar annealing of $Si_{0.75}Ge_{0.25}O_2$ on $Si_{0.75}Ge_{0.25}$ in Figure 4.3(c) of Section 4.4.1. In that case, a background of small ncGe (with diameters as small as 5 nm) form near the bottom of the oxide, but numerous much larger nanocrystals (with sizes as large as 30 nm) are also present. In the present case, such large nanocrystals are not present, possibly due to a decreased amount of GeO_2 in the mixed oxide. As noted earlier, the second $900^\circ C$ wet oxidation step used in the SOI approach may be creating volatile GeO gas, which can out-diffuse and thereby

lead to a decrease in GeO_2 concentration. Upon annealing, such lower GeO_2 concentrations might produce smaller ncGe sizes simply due to the lower content of available Ge. Using this observation in conjunction with results from the previous chapter, a similar mechanism explains why nanocrystals formed by the 875°C annealing of $\text{Si}_{0.75}\text{Ge}_{0.25}\text{O}_2$ on $\text{Si}_{0.75}\text{Ge}_{0.25}$ (as shown in Figure 4.7) had very small diameters. In that case, most of the available Ge is deposited on the semiconductor interface, thereby reducing the available Ge content and leading to very small nanocrystals in the bulk of the oxide. Irrespective of mechanism, any process which depletes or decreases the amount of GeO_2 available for homogenous nanocrystal growth tends to lead to a decrease in ncGe size.

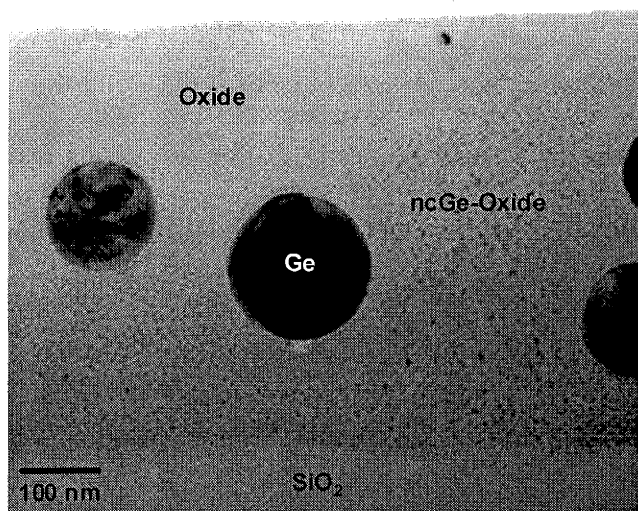


Figure 5.8 XTEM micrograph of slab waveguide structure formed using SOI approach, and annealed in pure H_2 at 875°C for 1 min. A region with small nanocrystals is denoted by “ncGe-oxide”.

Unlike 700°C annealing, 875°C annealing of the slab waveguide structure for 1 min forms huge Ge crystallites throughout the bulk of the oxide, as shown in Figure 5.8. These crystallites have diameters ranging from about 100 to 200 nm, and are surrounded

by regions with almost no small nanocrystals. Also, in regions between huge crystallites, many small nanocrystals with diameters smaller than 10-20 nm are present. The formation of large nanocrystals is partly due to the high elemental Ge diffusivity at 875°C compared to 700°C. Since diffusing Ge cannot deposit on the SiO₂ substrate, this Ge is available for growth of existing crystallites. This is in stark contrast to the case of a semiconductor substrate (like Si_{0.75}Ge_{0.25} in the previous chapter), which acts as a Ge sink and decreases the amount of available Ge for homogenous ncGe growth, inhibiting the formation of huge crystallites (Figure 4.7).

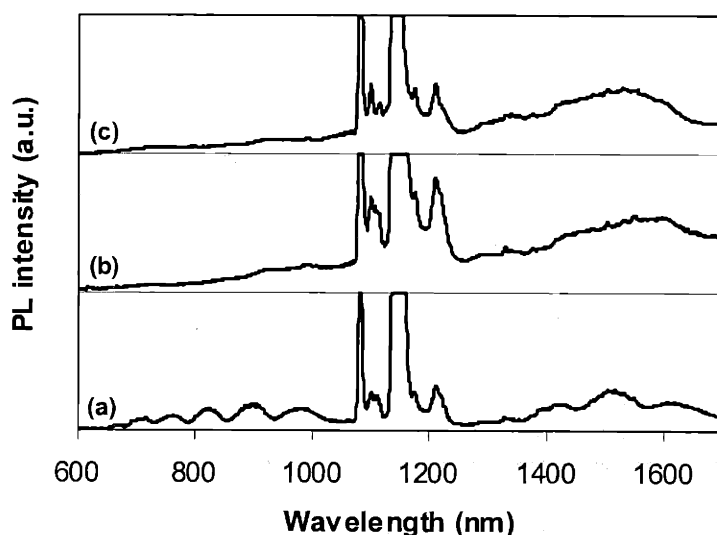


Figure 5.9 4K infrared photoluminescence spectrum for slab waveguide structure formed using SOI approach: (a) as-oxidized, and after pure H₂ anneals at (b) 700°C for 5 min, and (c) 875°C for 1 min.

5.3.2 Photoluminescence

From the XTEM characterization, it was concluded that both the 700°C and 875°C pure H₂ anneals yielded ncGe with sizes less than 15 nm. Based on results presented in the previous chapter, such small nanocrystals experience carrier confinement, which

should shift their absorption edges into the 0.6 to 1.7 μm wavelength range, as indicated by Figure 4.17. Photoluminescence spectra were obtained for the slab waveguide structures using an Ar^+ ion laser ($\lambda=488\text{ nm}$) at an output power of 260 mW. As can be seen from the 4K photoluminescence spectra in Figure 5.9, all the samples have PL peaks due to the Si substrate in the 1.05-1.25 μm range. For comparison, note that the samples in the previous chapter had no signal from the Si substrate due to absorption of light in the graded buffer. All the annealed slab waveguide structures contain broad backgrounds extending from 1.7 μm into the near infrared region, as shown in Figure 5.9(b) and (c). It is thought that this background is mainly due to quantum carrier confinement in small ncGe, which causes a downshift in the wavelength.

In the previous chapter, it was found that a small, broad peak in the near-infrared range (0.8-1.0 μm), is probably due to interfacial oxygen-deficient defects in the oxide.¹⁰⁴ In turn, the tail of the near-infrared broad background may be partially due to this effect, superimposed onto any confinement signal from Ge nanocrystals with diameters smaller than 5 nm. According to theoretical calculations performed by Niquet, ncGe with such diameters correspond to absorption edges of less than 1 μm .¹⁰⁵

Also note that the PL of the as-oxidized sample, presented in Figure 5.9, had a weaker background, compared to the annealed samples. XTEM did not show any obvious ncGe, nor did Raman analysis reveal the presence of an obvious Ge phonon peak, hence the weak PL background is likely not due to nanocrystal confinement. This background may simply be due to defects in the mixed SiGe-oxide, possibly created during the 900°C second wet oxidation step used in the SOI approach. An interesting

observation is the sinusoidal variation of the background, present only in the as-oxidized sample. As a control, the PL spectrum of a thermally oxidized Si substrate, consisting of 2.8 μm of SiO_2 on Si, was collected. The 2.8 μm SiO_2/Si sample had an extremely weak background, likely due to a relative absence of defects in the oxide, but interestingly, this control sample also had a sinusoidal variation in PL intensity versus wavelength.

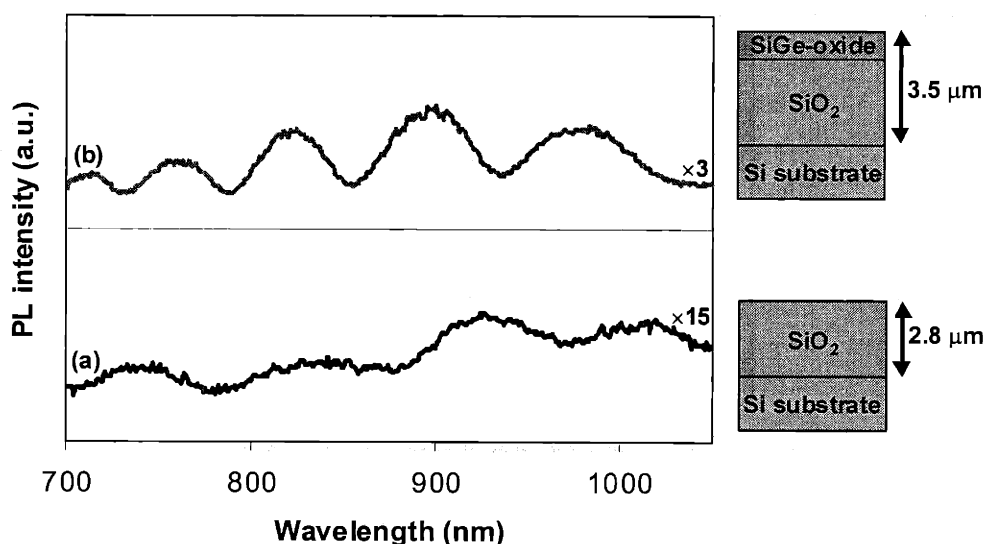


Figure 5.10 4K infrared photoluminescence spectrum for (a) thermally oxidized 2.8 μm SiO_2/Si , and (b) as-oxidized, slab waveguide structure formed using SOI approach. Diagrams to the right show the Fabry-Perot cavities in each sample.

As shown in Figure 5.10, the period of the sinusoidal variation varies inversely with the thickness of the total oxide. This behavior indicates that the sinusoidal character is due to thin film interference between the SiO_2/Si and the oxide/free-space interface. In effect, the oxide layer in these samples acts like a Fabry-Perot (FP) cavity, where the reflecting surfaces are the oxide interfaces. It is well known that a FP cavity has transmission peaks in wavelength space, due to constructive interference from the interfaces.¹¹²

Finally, also note that the annealed samples containing ncGe in the top of the oxide do not exhibit any interference effects in their PL spectra. In these samples, the ncGe present in the top layer of the oxide can absorb any light in the cavity thereby disrupting interference effects. To recap, ncGe slab waveguide samples exhibited a broad PL background, postulated to result from the quantum confinement downshift of the Ge absorption edge.

5.4 Fabrication of Nanocrystalline Ge Strip Waveguides and Nonlinear Directional Couplers

Once a slab waveguide structure of SiGe-oxide/SiO₂/Si was fabricated, a few final processing steps allowed for the fabrication of nanocrystalline Ge strip waveguides and associated photonic devices, as shown in Figure 5.11. After a lithography step, to define the waveguides, dry etching was used to etch the oxide, producing the desired photonic structures. Finally, any desired cladding was deposited onto the surface of the unclad waveguides; for example SiO₂ or silicon oxy-nitride can be deposited via PECVD. Note that the hydrogen annealing step required to reduce the SiGe-oxide can either be performed at the beginning of the process, or after the strip waveguide patterning step, but not after the cladding is deposited. Given that the SiGe-oxide layer experiences a volume reduction, if the ncGe formation step is performed after the cladding is deposited, voids will form between the waveguide cores and the clad.

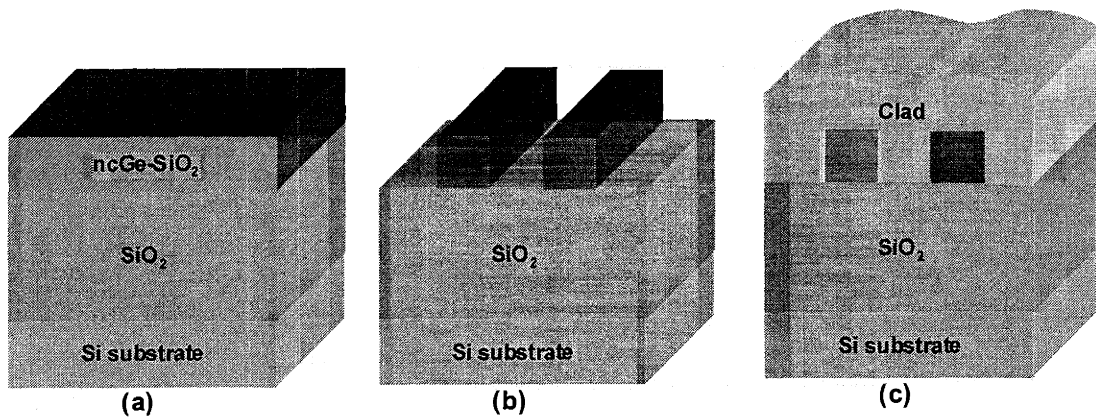


Figure 5.11 Process flow for ncGe strip waveguide fabrication: (a) initial slab waveguide structure, (b) patterning and dry etching of strip waveguides and desired photonic devices, and (c) deposition of desired waveguide clad.

In the present embodiment, the SOI approach was used to create the SiGe-oxide/SiO₂/Si structure, and then patterning was used to create strip waveguides and devices. After this step, annealing in pure H₂ at 700°C for 5 min, created ncGe in the waveguide cores. In addition to strip waveguides containing ncGe cores, a series of nonlinear directional couplers (NLDC) were also fabricated on the same die. An optical image of the fabricated waveguides and nonlinear directional couplers is shown in Figure 5.12.

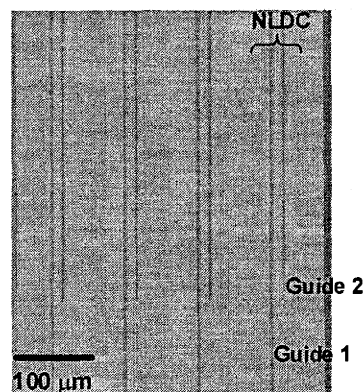


Figure 5.12 Optical image of ncGe waveguides and nonlinear directional couplers fabricated using the SOI approach.

A nonlinear directional coupler consists of a pair of coupled waveguides with cores composed of nonlinear material. In a common version of this device, the length of the waveguides is set to be the transfer length, which is the length required for low intensity light in one guide to completely transfer into the other waveguide via evanescent mode coupling, as shown in Figure 5.13(a). If the index of the waveguides does not exhibit any nonlinear behavior, then the two-waveguide system merely acts as a coupler, but the situation is quite different for nonlinear waveguides. In such a case, as the light intensity is increased and the critical power is exceeded, the signal no longer transfers completely and most of the light remains in the input guide (Figure 5.13(b)). This makes the NLDC an ideal device for all-optical switching, as shown in Figure 5.14.¹¹³

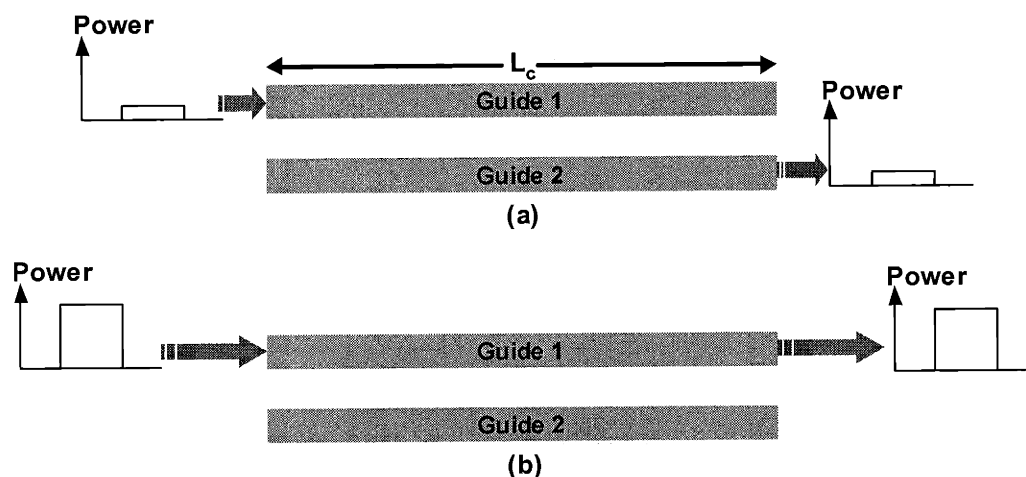


Figure 5.13 Nonlinear directional coupler operating as an all-optical switch, where (a) low intensity light is coupled into guide 2, and (b) high intensity light remains in the input guide.

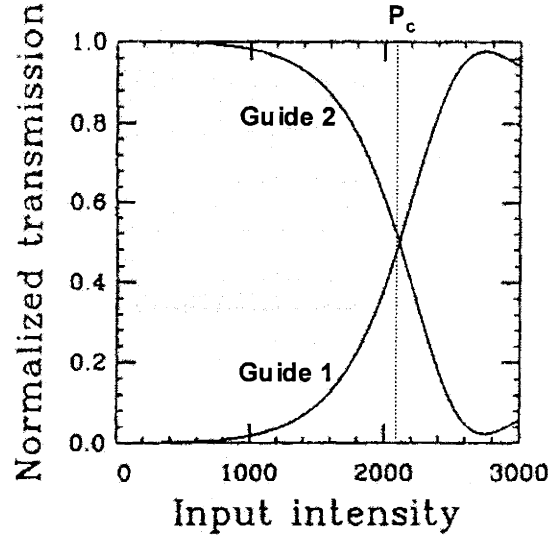


Figure 5.14 Predicted switching behavior of a nonlinear directional coupler, showing the switching power, P_c .¹¹⁴

The switching behavior of a NLDC is due directly to the variation of the index with intensity, and careful electromagnetic analysis yields an expression for the critical power:

$$P_c = \frac{\lambda A_{\text{eff}}}{n_2 L_c}, \quad \text{Eqn. 5.1}$$

where A_{eff} is the effective mode area (determined by the amount of power in the core), L_c is the length of the waveguide coupler, and n_2 is the nonlinear coefficient of the index.¹¹³

This equation implies that the switching power varies inversely with the nonlinear coefficient, which explains why a material with a large nonlinear response is desired for all-optical devices like NLDCs.

Since nanocrystalline Ge has been shown to exhibit a large nonlinear response, the fabricated NLDCs are ideal devices to explore possible real-world applications. Based on n_2 for ncGe in oxide obtained by previous researchers,¹⁰⁹ a quick calculation using Eqn. 5.1 shows that the critical power required for switching could be as high as 1 W.

Unfortunately, since input coupling from an external laser into a waveguide core is typically around 10%, and table-top laser powers are at most about 1-2 W, the power coupled into the waveguide core will be at most 0.1-0.2 W. This may not meet the required critical power for nonlinear switching in our devices, therefore preparations for testing our ncGe-NLDCs with an ultra-short pulse (high instantaneous power) laser are currently in the works.

5.5 Summary

The required standoff thickness needed to isolate a ncGe waveguide was determined to be about 3 μm , and three approaches were explored to fabricate such thick buried oxide compatible with our Ge nanocrystal formation process. The polycrystalline-SiGe approach failed due to Ge pile-up during oxidation, and the two-step oxidation method vaporized Ge in guide layer. The final, successful approach used thinned SOI starting wafers to deposit epitaxial SiGe, followed by low-temperature oxidation steps. After H_2 annealing which formed Ge nanocrystals in the guide layer, the resultant slab waveguide was then used to fabricate ncGe strip waveguides and NLDCs that are currently awaiting testing using a ultra-short pulse laser.

In fabricating ncGe waveguides on Si substrates, a substantial number of process steps could have been eliminated if SiGe-on-insulator substrates were commercially available. In addition, such substrates would also be of great importance to future high-speed, low-power microelectronics applications. To address this demand, the next

chapter deals with our pioneering work in fabricating SiGe-on-insulator and strained-Si-on-insulator using wafer bonding.

Chapter 6 SiGe-On-Insulator and Strained-Si-on-Insulator

High mobility strained-Si on relaxed SiGe virtual substrates¹⁹ are being incorporated into mainstream CMOS, and future generations may incorporate the benefits of both strained-Si and SOI. The substrate that will accomplish the above goal consists of low-defect, monocrystalline, relaxed SiGe-on-insulator (SGOI), with a surface strained-Si device layer, yielding strained-Si-on-insulator (SSOI). Equivalently, SSOI substrates can also be fabricated such that the strained-Si lies directly on the insulating substrate, with no underlying relaxed SiGe layer.

6.1 Benefits of SSOI for CMOS

CMOS technology has undergone, exponential performance improvements over the last few decades due to the continual scaling of gate lengths. Unfortunately, minimum feature sizes have been scaled down to such an extent that technical and capital expenditure obstacles will soon make continued scaling infeasible. In response, improvements imparted by SGOI and SSOI are appearing on the horizon at the most opportune time to find wide-spread industry acceptance. By combining the benefits of a semiconductor-on-insulator structure and high-mobility strained-Si, SGOI and SSOI substrates provide dramatic improvements that the CMOS industry will surely not overlook.

6.1.1 Benefits of Semiconductor-on-Insulator

Si-on-insulator (SOI) provides numerous performance benefits compared to bulk Si CMOS, and has emerged as a key technology for a number of applications requiring high speeds and low power. Technologies like SOI are of key importance, since they are compatible with tool-sets and processes currently available in bulk Si CMOS foundries.

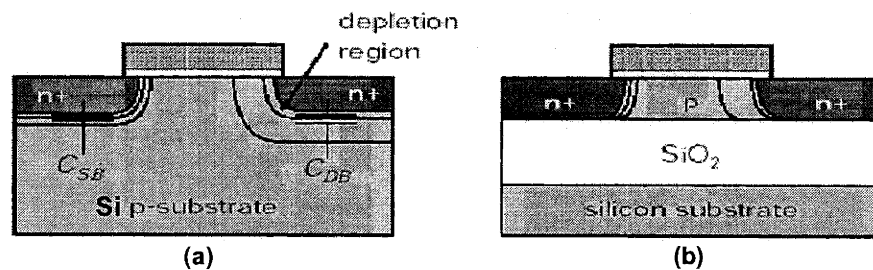


Figure 6.1 Cross-section of MOSFET using (a) bulk Si, and (b) SOI substrate.

SOI substrates consist of a high-quality, defect-free, monocrystalline Si layer on an insulating, buried oxide layer. Many benefits of SOI stem from the reduction of parasitic junction capacitances, as shown in Figure 6.1. Bulk Si MOSFETs possess a depletion region between their junction regions and the underlying Si substrate, which contributes to the MOSFET's internal capacitance, thereby increasing the device's RC delay. In contrast, an insulating layer underneath the device reduces the junction capacitance, and allows for higher speed operation of digital circuits. High-speed digital chips using SOI have been fabricated and tested, yielding a 25-35% performance improvement over bulk Si, which is equivalent to about two years of improvement in bulk Si CMOS scaling, as shown Figure 6.2.

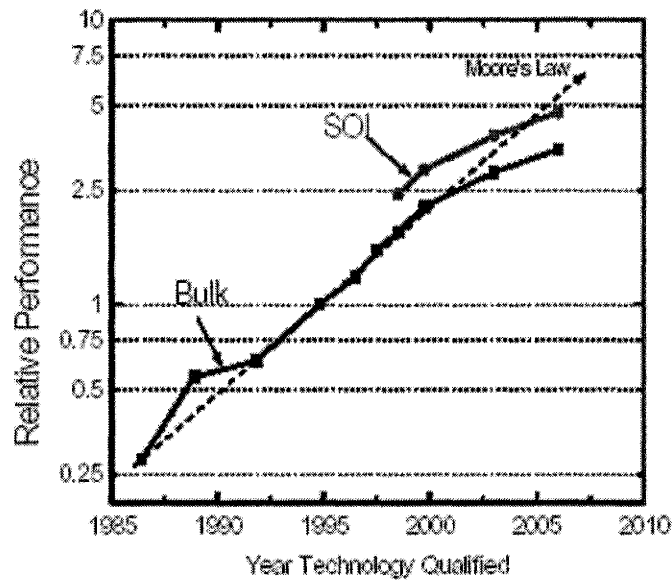


Figure 6.2 SOI versus bulk Si CMOS performance.

In addition, by reducing parasitic capacitances, power consumption of digital circuits is also decreased since (dynamic) power scales as $C_{\text{tot}} V_{\text{dd}}^2$, where V_{dd} is the supply voltage set by the technology. Also, dropping the supply voltage of the SOI technology can yield even more of a decrease in power, while keeping performance constant. For example, a 4 Mb SRAM circuit built with SOI CMOS reduces power consumption by a factor of 1.7-3 (40%-70% reduction), compared to a bulk Si CMOS circuit with the same performance, and operating at a higher voltage.

SOI CMOS also provides significant improvements (over bulk Si CMOS) for analog and RF circuits. For example, circuit blocks built using SOI exhibit reduced cross-talk between digital and RF circuits, low DC power consumption, and higher quality passive on-chip components.¹¹⁵ These features, in addition to monolithic integration of digital and RF blocks, provide a cost-effective alternative to GaAs MESFET technology, which has been the main technology of choice for 2-6 GHz RF front-end circuits.

Other than the aforementioned performance benefits, SOI CMOS has other benefits which contribute to significant revenue gain compared to bulk CMOS. These include: (1) 20% fewer process steps due to simpler well isolation (6% revenue gain), (2) 3 fewer masking steps, and (3) increased circuit density due to tighter isolation and reduction in latch-up (45% revenue gain).¹¹⁵ Finally, one of the early applications of SOI was for circuits with reduced soft-error rates (SER). SER is caused by the upset of MOSFET memory operation by cosmic rays or other radiation, making SOI ideal for high-altitude and space applications.

6.1.2 Benefits of Strained-Si

Strained-Si CMOS (ϵ CMOS) is a greatly improved version of conventional bulk Si CMOS, and hence retains all the benefits of conventional CMOS. Some of these benefits include mature processing technology, low-cost and large (hence high die yield) substrates, low power consumption, and the ability to monolithically integrate digital and analog functions. The key enabling technology for ϵ CMOS, is the growth of high-quality, relaxed SiGe on Si via CVD deposition, which allows for the creation of strained-Si. Strained-Si layers can either be located on the surface or buried beneath a SiGe layer. Due to energy band alignment, surface strained-Si acts as a carrier channel for both n- and p-MOSFETs operating in enhancement mode. In contrast, buried strained Si only allows for depletion mode MOSFETs.

ϵ CMOS possesses enhanced electron and hole mobility, caused by both a lower electron and hole effective mass, and reduced inter-valley scattering due to the strain-

induced energy splitting between various sub-bands. In particular, for a relaxed SiGe substrate with $x_{\text{Ge}}=0.3-0.4$, one finds a mobility enhancement (and hence a current enhancement) of about 1.8 for NMOS and 1.6 for PMOS.¹⁹ Increased mobility translates into enhanced drive currents and hence improved circuit and system performance.

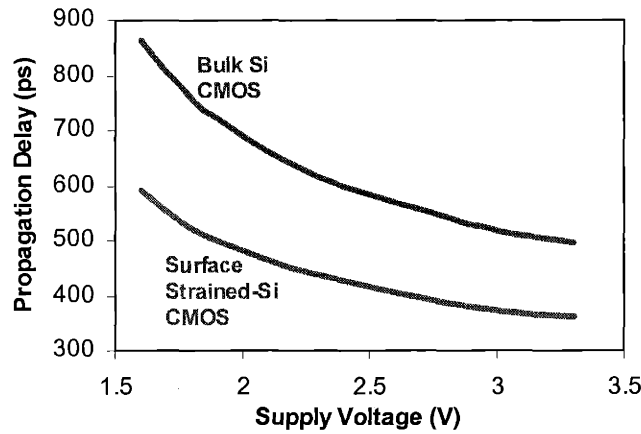


Figure 6.3 Calculated static inverter propagation delay dependence on power supply voltage, for 0.35 μm bulk Si CMOS and surface channel strained-Si CMOS, with a capacitive load of 50 pF.

In the case of digital logic gates, increased drive current can be used to increase performance and hence decrease the propagation delay (by keeping the supply voltage constant and moving from bulk Si to surface strained-Si) as shown in Figure 6.3. For low power mobile applications, the dynamic power can be reduced by lowering the supply voltage (as shown in Figure 6.4). Dropping the supply voltage from 3.3V down to 2V allows strained-Si to retain the same performance of 3.3V bulk Si CMOS, but reduces dynamic power consumption by over 60% (since dynamic power scales as CV_{dd}^2).

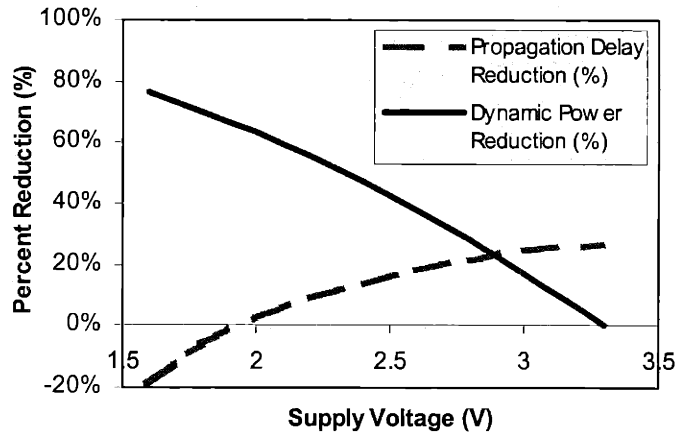


Figure 6.4 Calculated inverter dynamic power and propagation delay reduction for 0.35 μm surface strained-Si CMOS with a given supply voltage compared to 0.35 μm bulk Si CMOS with supply of 3.3V. Dynamic power comparison based on same clock frequency for both systems, with a capacitive load of 50 pF.

With respect to analog circuit blocks, mobility enhancement can be used to keep current constant (compared to bulk Si CMOS) while reducing the widths of the MOSFETs, which in turn leads to area reduction (and hence smaller die sizes and increased yield). More importantly, reduction of device widths implies smaller device capacitances and therefore larger bandwidths.

In addition, buried strained-Si also exhibits reduced noise, since the inversion layer is not adjacent to the gate oxide, hence carriers are not scattered by the oxide interface. Elimination of scattering leads to lower device noise and therefore is of great importance for wireless circuits (*e.g.*: Low Noise Amplifiers) which demand low noise figures. Another advantage of buried channels is a higher mobility at low perpendicular fields (*i.e.*: low gate voltages), due to the absence of mobility degradation by carrier scattering at the surface, which occurs for both conventional CMOS and surface channel ϵCMOS .

An advanced buried channel device includes a threshold implant or modulation doping (with dopants outside the conduction channel, thereby reducing dopant scattering) which increases the low-field mobility to around 1000-3000 cm²/Vs, depending on the doping level.¹¹⁶ This buried device is a depletion mode MOSFET, and hence conducts current for zero gate-source bias. Although problematic for digital applications, which are fabricated using enhancement-mode devices (*e.g.*: surface strained CMOS, or conventional CMOS), the buried channel MOSFET may rival high-speed analog technologies, like GaAs MESFETs and SiGe HBTs (hetero-junction transistors).

One final, but important advantage of ϵ CMOS is the potential to integrate both digital logic and high-speed, high-power, low-noise analog circuits. This integration is made possible because analog devices could employ buried channel devices, whereas digital elements will consist of enhancement-mode, surface channel devices. Compared to other high-speed, analog integration technologies (*e.g.* SiGe-BiCMOS, which are SiGe HBTs integrated with Si CMOS) ϵ CMOS offers distinct advantages, including ease of integration (since all devices are MOSFETs), improved scaling characteristics, and reduced costs for equal performance.

6.2 SGOI and SSOI Fabrication Techniques

Due to all the advantages of both SOI and strained-Si, many methods have recently been explored to achieve their merger. In this section, we present a short review of these attempts, setting the stage for our novel process.

6.2.1 SIMOX

SIMOX (separation-by-implanted-oxygen) is a popular technique for SOI fabrication, but unfortunately is only applicable to SGOI fabrication for low Ge fractions. The technique involves the implantation of oxygen ions to a desired depth, followed by a high-temperature anneal to remove implant damage from the crystalline layer, and to form the buried oxide layer. Fukatsu *et al.* have successfully applied the approach to SGOI fabrication for a Ge content of 18%.¹¹⁷ Unfortunately, Ishikawa *et al.* have shown that the technique is only applicable for Ge fractions less than 30%, due to SiGe surface oxidation and thermal instability during post-implant annealing.¹¹⁸

6.2.2 Ge Interdiffusion via High-Temperature Oxidation

Another method demonstrated by Tezuka *et al.* requires high-temperature oxidation of a low Ge content SiGe layer on a thinned SOI substrate.¹¹⁹ The oxidation leads to the expulsion of Ge ahead of the oxidation front, and the high temperature causes Ge interdiffusion into the underlying thinned Si. It is believed that at the high temperatures employed, the buried oxide flows, and that the strain relaxation mechanism for the Ge-rich layer (forming due to Ge expulsion from the oxide) is lattice expansion and not dislocation nucleation. On a wafer-scale, this hypothesis is invalid, since it implies that the relaxing film must expand beyond the edge of the substrate, which is not observed. It is important to note that, to date, the approach is plagued by high defect densities when applied to blank wafers, or imposes restrictions on the size of low defect areas via the use of micron-sized mesas (3 μm or less).¹²⁰

6.2.3 Wafer Bonding

In addition to the above approaches, wafer bonding is another technique for the fabrication of SGOI or SSOI, involving the bonding of wafers with SiGe films to insulating substrates, like oxidized Si wafers. This method was originally presented and developed by our group.^{121,122}

A few widely employed techniques exist for accomplishing layer transfer of a thin film onto any desired substrate: (1) backside grinding, (2) backside grinding combined with etch back, (3) and delamination via implantation. In the grind and etch-back approach, wafers are bonded, the backside of the wafer containing the transfer layer is thinned by grinding,¹²³⁻¹²⁵ possibly followed by an etch to remove the remaining excess material,¹²⁶⁻¹²⁸ leaving the transferred layer. In contrast, delamination via implantation (dubbed Smart-Cut), employs ion implantation prior to wafer bonding, possibly followed by annealing to cause delamination and layer transfer at the implant depth.^{129,130} In all of these methods, a final polishing step is usually employed to smooth and thin the transferred layer.

Research by our group demonstrated the feasibility of wafer bonding for the fabrication of SGOI substrates.¹²¹ Cheng *et al.*¹²² demonstrated SGOI fabrication using a grind and Si etch-back process, where their KOH etch stopped on the 20% Ge region of the relaxed SiGe graded buffers. Chemical mechanical polishing (CMP) was then used to both smooth and thin the SiGe transferred layer. Similarly, Huang *et al.*¹³¹ used Smart-Cut to transfer relaxed SiGe onto oxidized Si substrates, followed by CMP to thin and smooth the thick transferred layer. The limitation of the above methods is the lack of

control over the final SiGe thickness on the handle wafer, and the wafer-scale uniformity of the transferred layer, making the fabrication of ultra-thin transferred layers impossible.

Smart-Cut approaches incorporating epitaxial etch-stop layers have been proposed and demonstrated for SOI fabrication.^{132,133} The process requires ion-implantation to a depth below an epitaxial SiGe etch-stop layer on a Si substrate. After wafer bonding and layer transfer via delamination, a selective Si etch is used to remove material until the SiGe etch-stop is reached. The approaches allow for uniform transferred layer thickness by employing an etch-stop in conjunction with Smart-Cut. Previous work presented this concept within the context of Si layer transfer and not for low defect density, relaxed SiGe or strained-Si layer transfer. To our knowledge, no previous research has described a detailed process, etch-stop, or selective material removal that can be employed to create SGOI or SSOI.

Some of our preliminary work presented the basic process for SGOI creation based on wafer bonding, combined with backside grinding, Si etching, and SiGe etching.¹³⁴ The approach incorporated a strained-Si etch-stop layer (thinner than the critical thickness) and a relaxed SiGe transfer layer. After wafer bonding to an oxidized Si handle wafer, and annealing at 800°C to strengthen bonding, the backside of the donor wafer was thinned via wafer grinding, followed by a KOH etch that etches Si while stopping on the 20% Ge region. Since the surface of the transferred layer was still quite rough, a SiGe etch was used to remove excess SiGe and stop on the strained-Si layer, leaving a much smoother surface, with precise control over the transferred layer thickness. In that work, a SiGe etch consisting of hydrogen peroxide, hydrofluoric and

acetic acid was found to have a high selectivity while stopping on the strained-Si. A remaining challenge was the appearance of micron-sized depressions in the etch-stop that formed once the strained-Si surface was exposed to the etching solution. It is now thought that preferential threading dislocation core etching may have been the source of that phenomenon.

6.3 SGOI and SSOI via Wafer Bonding

We now present our novel, general process for SGOI and SSOI fabrication using wafer bonding combined with stop layers. Significant improvements had been incorporated into our preliminary work,¹²¹ providing solutions to increase the robustness of the stop layer, improve wafer bonding at low temperatures, and eliminate preferential etching of threading dislocations. In particular, to improve etch-stop robustness, the use of thicker metastable stop layers was studied, and a detailed analysis was conducted to determine dislocations nucleation as a function of layer thickness. Since metastable layers relax when exposed to high temperatures, a low-temperature wafer bonding process was developed. With regards to preferential threading dislocation etching due to the selective etching, a solution of HNO₃, dHF (dilute HF), and HAc (acetic acid) was found to have an acceptable selectivity, etch rate, and with no preferential dislocation etching. Also, within the context of the above improvements, a method for transferring SiGe layers using delamination via implantation was also developed, providing an alternative layer transfer method in addition to our grind and etch-back process. Finally, note that the process was used for ultra-thin transfer layers (50 nm or less), demonstrating

that the wafer bonding process is applicable to the fabrication of ultra-thin SGOI and SSOI for fully depleted CMOS (requiring ultra-thin semiconductor layers on oxide).

6.4 Process Flow and Experiments

Two methods for fabricating relaxed, monocrystalline SiGe (and strained-Si) layers on insulating handle substrates were developed in this work: (1) backside grinding and etch-back, and (2) delamination via implantation. These processes were combined with a stop layer allowing for the fabrication of thin transferred layer(s) with uniform thickness across the wafer.

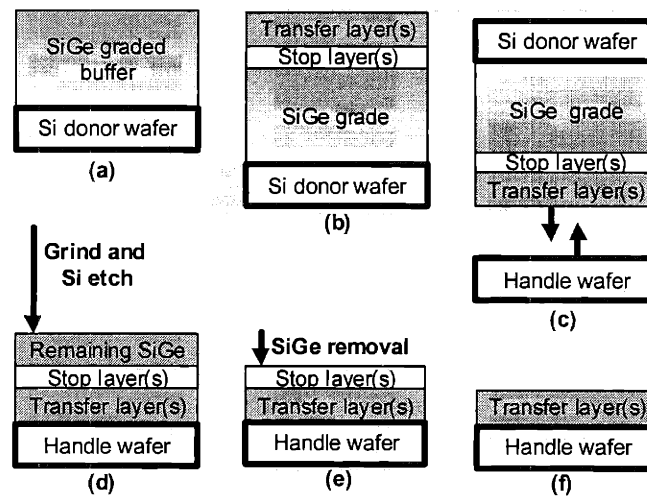


Figure 6.5 Process flow for generic grind and etch-back layer transfer method combined with stop layer(s): (a) UHV-CVD growth of SiGe graded buffer and surface planarization, (b) regrowth of stop and transfer layers, (c) wafer bonding to handle wafer, (d) backside grinding and Si etch stopping in graded layers, (e) selective SiGe removal, and (f) optional removal of stop layer.

6.4.1 Grind and Etch-back Approach

Figure 6.5 shows the process flow diagram for the generic grind and etch-back layer transfer method, combined with additional etch-stop steps. A monocrystalline Si donor wafer is used as a substrate for the deposition of a relaxed SiGe graded buffer to attain the desired lattice constant.⁴⁴ Since the deposited SiGe is lattice mis-matched to Si, dislocations form to relieve strain when the thickness of the SiGe exceeds the critical thickness. To minimize the number of threading dislocations (which terminate at the surface and interfere with device performance), the deposited SiGe can be graded, and the Ge content gradually increased in each subsequent layer. This process allows the SiGe to relax and attain its equilibrium lattice constant by reusing existing threading dislocations to relieve additional strain.⁴⁵ The method is crucial for creating relaxed SiGe layers with low threading dislocation density. A wafer containing a relaxed SiGe surface layer, dubbed a SiGe virtual substrate, is an ideal starting substrate for the fabrication of advanced electronic devices, including strained surface and buried channel MOSFETs.^{30,135}

In the present experiments, relaxed $\text{Si}_{0.75}\text{Ge}_{0.25}$ graded buffers were deposited on 100 mm Si substrates using ultrahigh-vacuum chemical vapor deposition (UHV-CVD). The graded buffer growths were performed at 900°C, with a grading rate of 10% Ge per μm , and capped with a 3 μm relaxed $\text{Si}_{0.75}\text{Ge}_{0.25}$ layer. The high growth temperature has been found to allow for rapid dislocation motion, and hence promote efficient re-use of existing threading dislocations to relax the graded layers.⁴⁵ In addition, deposition rates

are reasonably fast at this high temperature, requiring a couple of hours for a $\text{Si}_{0.75}\text{Ge}_{0.25}$ graded buffer deposition.

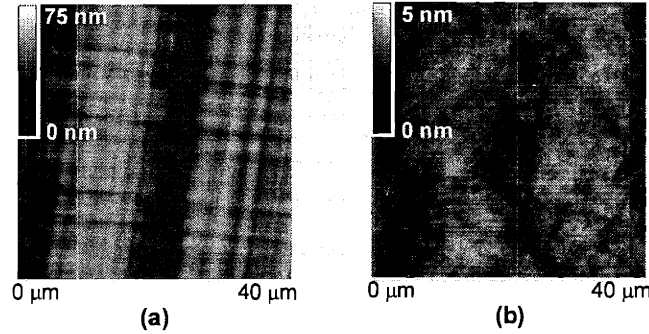


Figure 6.6 AFM images showing surface planarization of $\text{Si}_{0.75}\text{Ge}_{0.25}$ graded buffer using CMP: (a) as-grown surface with drastic cross-hatch roughness, and (b) after CMP.

The draw-back of the SiGe grading process stems from the creation of surface roughness, termed cross-hatch roughness because of the characteristic orthogonal pattern (on 100 Si) created by strain fields from orthogonal misfit dislocations in the graded buffer.¹³⁶ Since substantial roughness inhibits wafer bonding, a surface planarization step is required consisting of any method that reduces roughness, while not removing too much material. In this work, CMP¹³⁷ was employed to remove cross-hatch roughness, as shown in Figure 6.6, using a 15-min CMP step removed about 2 μm from the $\text{Si}_{0.75}\text{Ge}_{0.25}$ cap. Typically, as-grown $\text{Si}_{0.75}\text{Ge}_{0.25}$ graded buffers have a root-mean-square (rms) roughness of about 9 nm with a peak-to-valley height of about 60 nm; a 15-min CMP reduces the rms down to 0.4 nm and the peak-to-valley height to less than 4 nm over a 40×40 μm² scan area.

After surface planarization, thin etch-stop and transfer layers are deposited onto the SiGe virtual substrate at low temperatures (650°C or lower) to minimize Ge

segregation^{138,139} and inter-diffusion,^{49,140,141} and hence maintain sharp interfaces. The stop can consist of one or more layers composed of SiGe alloys, Si, Ge or combinations thereof. Transfer layers, although depicted as one layer in Figure 6.5, could consist of multiple layers containing device and bonding layers. The desired layer thicknesses depend on the intended application. For example, for fully-depleted MOSFETs the combined thickness of the transfer layers should be 30 nm or less.¹⁴² In such a case, it may be advantageous to use the device layer as the bonding layer; for example, a strained-Si device layer could be bonded directly to the handle wafer, allowing for the layer transfer of strained-Si directly onto an insulating substrate.

In the present work, the transfer layer was an ultra-thin (50 nm or less) layer of relaxed $\text{Si}_{0.75}\text{Ge}_{0.25}$, with a stop layer composed of strained-Si. Usually, pseudomorphic strained layers are thinner than the equilibrium critical thickness, thereby ensuring that misfit dislocations are not generated leading to relaxation.¹⁴³ For deposition of Si on relaxed $\text{Si}_{0.75}\text{Ge}_{0.25}$, the equilibrium critical thickness is about 16 nm. Beyond this thickness (and at elevated temperatures), relaxation occurs via misfit generation at the strained layer interface. At lower temperatures, typically less than 650°C, misfit generation is kinetically limited, and therefore fully strained, metastable layers can be thicker than the equilibrium critical thickness.³⁹ Metastable strained-Si layers can serve as improved stop layers since they can be grown thicker than the critical thickness. In particular, in the case of higher Ge content SiGe, the equilibrium critical thickness can become prohibitively thin to serve as an etch-stop layer. Thicker metastable layers may be useful for the creation of more robust, thicker etch-stop layers, or multiple device

layers. In our experiments, the etch-stop and transfer layer deposition was performed at 650°C or lower, which not only allowed for metastable layers, but also limited Ge interdiffusion between ultra-thin layers.

To explore the use of metastable strained-Si etch-stop layers, experiments were conducted to determine threading and misfit dislocation densities as function of strained-Si thickness for deposition at 650°C on $\text{Si}_{0.75}\text{Ge}_{0.25}$ virtual substrates. The samples consisted of strained-Si with thicknesses of 10, 20, 40, and 80 nm, with a 20 nm $\text{Si}_{0.75}\text{Ge}_{0.25}$ cap layer. Threading dislocation densities were determined using the etch-pit density (EPD) method.⁵⁴ For these measurements, the as-grown samples were first stripped of the 20 nm $\text{Si}_{0.75}\text{Ge}_{0.25}$ cap layer by etching for about 50 s in a selective SiGe etch consisting of $\text{H}_2\text{O}_2\text{:HF:HAc}$ (2:1:3). Before etching, the solution was allowed to equilibrate for a minimum of 2 h, yielding a relaxed $\text{Si}_{0.75}\text{Ge}_{0.25}$ etch rate of 38 nm/min, and a Si etch rate of 0.3 nm/min. To determine the influence of temperature on dislocation generation and relaxation, a set of (cap-stripped) samples were annealed in a N_2 ambient at 800°C for 1 h. EPD was then performed by dipping in a Schimmel⁵⁴ etch for times ranging from 30 to 50 s. Threading densities versus strained-Si layer thickness were then measured by etch pit counting using Nomarski microscopy. In addition to the threading density measurements, plane-view transmission electron microscopy (PVTEM) was used to determine misfit densities for selected samples before and after annealing. The misfit densities then allowed for the calculation of the relaxation for the metastable Si layers.

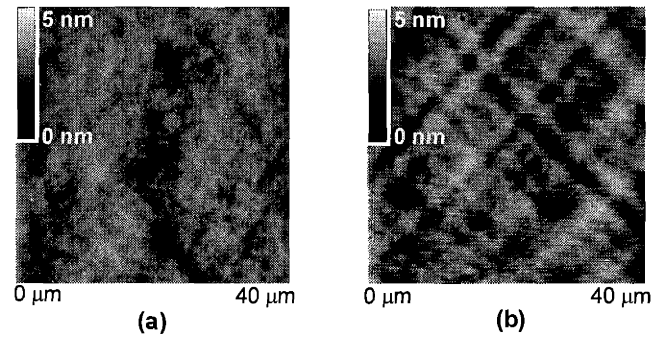


Figure 6.7 AFM showing re-appearance of mild cross-hatch roughness on a planarized $\text{Si}_{0.75}\text{Ge}_{0.25}$ virtual substrate after low-temperature (550°C) deposition of the etch and transfer layers: (a) planarized virtual substrate, (b) after 550°C deposition.

In addition to the aforementioned metastable test samples grown at 650°C , the structure used to demonstrate the entire SSOI fabrication process was grown at a lower temperature of 550°C , allowing for even greater metastability. The 550°C structure consisted of a 100-nm $\text{Si}_{0.75}\text{Ge}_{0.25}$ layer (that buried the CMP interface), followed by a 30-nm, metastable, strained-Si stop layer, and a 30-nm $\text{Si}_{0.75}\text{Ge}_{0.25}$ transfer layer. Since deposition re-introduced mild cross-hatch, due to the presence of strain fields from misfits in the graded buffer, surface roughness was characterized before and after the 550°C deposition. As can be seen from Figure 6.7, mild cross-hatch with a peak-to-valley height of about 5 nm and rms roughness of 0.5 nm is introduced after the etch-stop growth. The mild roughness does not inhibit bonding significantly, but from a manufacturing perspective, it may be advisable to deposit an intermediate material (e.g.: oxide, oxy-nitride) after the low-temperature growth, which could then be planarized to remove the mild cross-hatch roughness. The resulting surface with the exposed

intermediate material would then be bonded to the handle wafer, and the remainder of the process would continue as outlined.

Proceeding with the description of the process depicted in Figure 6.5, the substrate onto which the SiGe is transferred, dubbed the handle wafer, can consist of any material with the needed electrical, optical, or thermal properties for the final application. For example, the handle wafer can be a Si substrate, oxide on Si, quartz, or glass. In the current SGOI fabrication demonstration, the handle wafers consisted of thermally oxidized, monocrystalline, 100 mm Si substrates.

As shown in Figure 6.5(c), the SiGe virtual substrate and the handle wafer are then cleaned and bonded. Cleaning can be performed using a variety of chemistries, including a standard RCA,⁴⁶ or piranha clean (1:3 H₂O₂:H₂SO₄).⁴⁸ If the surface has exposed SiGe, the ammoniac bath (SC1) of the RCA clean etches the SiGe surface,⁴⁷ and hence it is typically replaced with a piranha solution. The cleaning process can leave the surface of the wafers either hydrophilic or hydrophobic, largely dependent on whether a final HF dip is employed to passivate semiconductor surfaces. Hydrophilic wafer surfaces lead to stronger bonding when a low-temperature anneal (<600°C) is employed, and hence are preferred for low-temperature bonding.¹⁴⁴ For our purposes, the pre-bond clean employed left the wafer surfaces hydrophilic and consisted of two steps: (1) a 3 min piranha clean followed by a 1 min 50:1 HF dip on the SiGe virtual substrates, and (2) a 6 min piranha clean on both the SiGe virtual substrates and the oxidized Si handle wafers.

In addition to cleaning, wafer surfaces were also activated in an oxygen plasma prior to bonding to improve bond strength and reduce intrinsic interfacial bubble

nucleation.¹⁴⁵ Plasma activation creates dangling bonds on exposed oxide surfaces (native, grown or deposited oxide) and increases bonding strength, allowing for a low-temperature post-bond anneal (less than 400°C). In stark contrast, conventional high-temperature (900°C or greater) annealing is typically used to attain strong Si to oxide bonding.

Since plasma activation can damage the first few monolayers of a surface,¹⁴⁶ a substantial improvement in bond strength can still be obtained by only activating the oxidized handle wafer and not the SiGe virtual substrate.¹⁴⁷ In addition to enabling strong low-temperature wafer bonding, plasma activation is effective in eliminating the appearance of intrinsic interface bubbles upon annealing, likely due to the elimination of hydrocarbon contamination on surfaces.¹⁴⁸ These interfacial hydrocarbons are thought to act as nucleation sites for gases (*e.g.* H₂ and desorbed hydrocarbons) leading to interfacial bubbles.¹⁴⁹

In this work, after plasma activation for 1 min in a 1 Torr oxygen plasma, wafers were immersed in a water bath for 5 min, directly bonded at room temperature, and then annealed at 400°C for 2 h. Extending the annealing time to 14 h led to no significant increase in bond strength, as measured by the Maszara technique. Low-temperature bonding was crucial since the metastable strained-Si etch-stop layers relax when exposed to high annealing temperatures.

As shown in Figure 6.5(d), the backside of the SiGe virtual substrate is then removed leaving behind only a portion of the SiGe graded buffer. In the present implementation, the backside of the first substrate was thinned by wafer grinding,

reducing the thickness of the donor substrate down to a thickness of about 100 μm . The remaining Si on the backside was removed by employing a Si etch, for example, either KOH or TMAH. These etches remove Si and stop on the 20% Ge region of the relaxed SiGe graded layers, with a very high selectivity.^{150,151} For our work, the Si etch consisted of an aqueous KOH solution (30% at. KOH) at 80°C; at this temperature, the required etch time was only 2 h, whereas at 65°C a total of about 6-8 h was needed to completely remove the remaining backside material and stop on the 20% Ge region of the graded buffer.

Next, selective SiGe removal can be used to strip the remaining SiGe and stop on the strained-Si with a selectivity of about 10 or more, as depicted in Figure 6.5(e). For our demonstration, since the stop layer consisted of strained-Si on a relaxed $\text{Si}_{0.75}\text{Ge}_{0.25}$ virtual substrate, any SiGe etch that does not attack Si appreciably was acceptable. Since Ge tends to oxidize faster than Si, a generic SiGe selective removal process consists of an oxidizer and an oxide stripping agent. For example, in Section 3.3, it was shown that low-temperature steam oxidation of SiGe proceeds at a much faster rate than Si low-temperature oxidation, hence this process could be used to oxidize the SiGe layer while stopping on the strained-Si, and then the SiGe-oxide could be stripped.¹⁵²

In addition, possible wet etches that can be used to etch SiGe are: (1) hydrogen peroxide, hydrofluoric acid, and a dilutant,^{153,154} (2) nitric acid, hydrofluoric acid, and a dilutant,^{155,156} (3) ammonium hydroxide, hydrogen peroxide, and a dilutant (RCA clean), (3) ammonium hydroxide, hydrogen peroxide, hydrofluoric acid, and a dilutant,⁴⁷ (4) ammonium hydroxide, hydrogen peroxide, hydrofluoric acid, ammonium fluoride, and a

dilutant, or (5) sulfuric acid, nitric acid, hydrofluoric acid, and a dilutant. The dilutant can consist of water, acetic acid (HAc) or a mixture of both. Optimally, the etch chosen for SiGe removal should have a high enough selectivity (typically 10 or greater) to effectively stop on the stop layer(s), and should possess a fast SiGe removal rate, implying a reasonable etch time.

Detailed studies for two of the aforementioned etches are presented here. The first etch presented for SSOI fabrication is a H_2O_2 :HF:HAc solution in ratios of 2:1:3 and 12:1:2. In both cases, the solutions were allowed to equilibrate for a minimum of 2 h before use. The other etch consists of HNO_3 :dHF:HAc, where dHF represents dilute HF (100:1 water:HF). Various constituent proportions were examined to optimize the selectivity and the $\text{Si}_{0.75}\text{Ge}_{0.25}$ etch rate. Etch rates for various samples were determined by masking a section of the wafer surface with wax, etching the sample, removing the wax, and measuring the step height using a profilometer.

Finally, as presented in Figure 6.5(f), the stop layer(s) can be optionally removed, if they interfere with the desired application. In the case of a strained-Si etch-stop, an etch that is selective to Si and does not etch the underlying transfer layer can be used. If the transfer layer is composed of SiGe, a low-temperature KOH or TMAH etch is ideal for controllably removing the strained-Si etch-stop, leaving behind the transferred layers on the second substrate. If required by the intended application, additional device layers could then be deposited onto the transferred layer. In the current implementation, the strained-Si stop layer was not removed, and characterization was performed on the SSOI

structure composed of a strained-Si stop layer and an ultra-thin, relaxed $\text{Si}_{0.75}\text{Ge}_{0.25}$ layer on buried oxide.

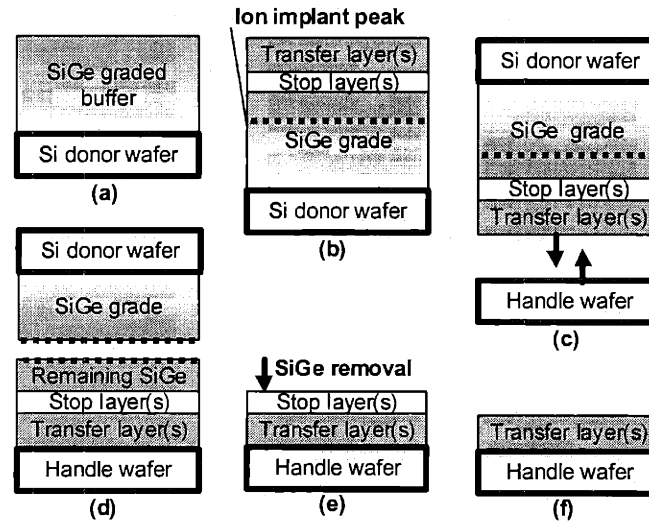


Figure 6.8 Process flow for generic delamination via implantation layer transfer method combined with stop layer(s): (a) UHV-CVD growth of SiGe graded buffer and surface planarization, (b) regrowth of stop and transfer layer(s) followed by ion implantation (c) wafer bonding to handle wafer, (d) delamination of the wafer pair at the implant depth, (e) selective SiGe removal stopping on stop layer, and (f) optional removal of stop layer.

6.4.2 Delamination via Implantation Approach

Figure 6.8 shows the process flow for our delamination via implantation method, combined with stop layer(s). All the material layers and the majority of the process steps are identical to the process described for the grind and etch-back technique. The first difference appears in the step shown in Figure 6.8(b), where after the deposition of the stop and transfer layers, ions are implanted into the surface of the substrate, with an implant peak below or on the stop layer(s). In our case, H_2^+ ions were implanted with an energy of 200 keV, and a dose of $4 \times 10^{16} \text{ H}_2/\text{cm}^2$, yielding an implant depth of about 870 nm.

The wafers are then bonded using the same procedure as described for the grind and etch-back process, except that the post-bond anneal causes the wafers to split at the ion implant peak. In the case of hydrogen ion implantation, the splitting is due to pressure exerted by the formation of hydrogen bubbles upon annealing.¹⁵⁷ The temperature required for splitting depends on the Ge content, and ranges between room temperature and 600°C. The higher the Ge content, the lower the required delamination temperature. For the present experiment, an anneal at 550°C for 3 h was sufficient to cause wafer splitting.

The SiGe virtual substrate is split at the implant peak transferring a portion of SiGe graded buffer, stop layer(s), and transfer layer(s) onto the handle wafer. In our case, the thickness transferred was about 870 nm, but a shallower implant depth would have been preferable, since it would have reduced the etch time required in the subsequent selective etching shown in Figure 6.8(e). In addition, if the excess SiGe had been thinner, then slower etch chemistries with extremely high selectivity could have been used, while still achieving reasonable etch times.

6.5 Metastable Strained-Si Layers

As noted earlier, strained-Si layers thicker than the equilibrium critical thickness are useful in our fabrication process, since they serve as thicker stop layers, and allow for greater flexibility in layer design. Beyond the critical thickness, strained-Si layers relax via the introduction of misfit dislocation segments at the strained-Si/Si_{0.75}Ge_{0.25} interface.³⁹ Misfit segments can either be created by the glide of existing threading

dislocations, or by the nucleation of two new threads, as shown in Figure 6.9. Nucleation possesses a larger kinetic barrier than glide, and hence is substantially reduced at low temperatures.¹⁴³ Irrespective of magnitude, both processes are suppressed by kinetic barriers at low temperatures, allowing for the deposition of metastable strained-Si layers that are thicker than the equilibrium critical thickness. These metastable layers do not contain significant misfit dislocations and hence exhibit little or no strain relaxation.

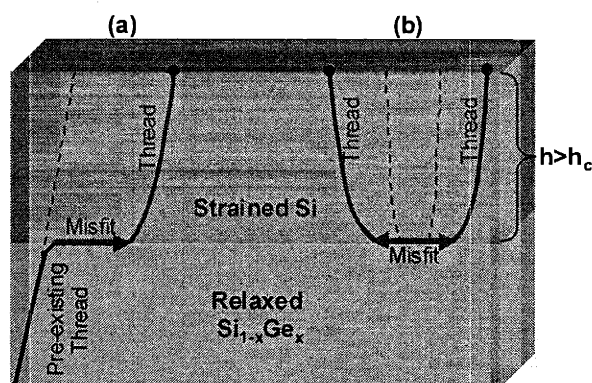


Figure 6.9 Strained-Si layer thicker than the critical thickness h_c , relaxing via misfit formation, by (a) the glide of existing threading dislocations, (b) the nucleation of new threading segments.

The goal of the low-temperature strained-Si growth experiments was to determine the thickness (beyond the equilibrium critical thickness) that a strained-Si layer can be grown before misfit and additional threading dislocations form. As was explained in detail in the experimental description section, the strained-Si for these experiments was grown on $\text{Si}_{0.75}\text{Ge}_{0.25}$ virtual substrates using UHV-CVD at 650°C ; samples with strained-Si thicknesses of 10, 20, 40 and 80 nm were deposited. To test the thermal stability of the metastable layers, samples were annealed for 1 h at 800°C in a N_2 ambient.

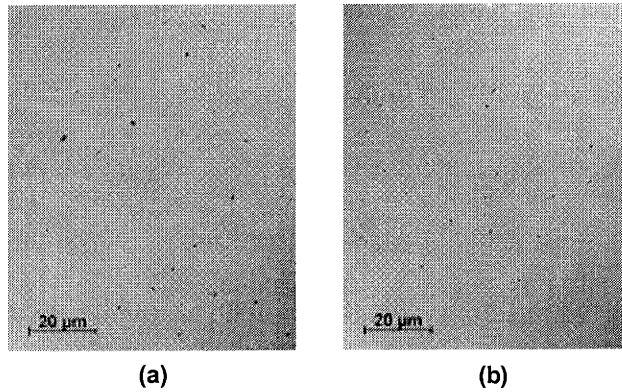


Figure 6.10 Nomarski optical images of EPD results for the 40-nm strained-Si depositions: (a) as-grown, and (b) after a 1h 800°C anneal in N₂.

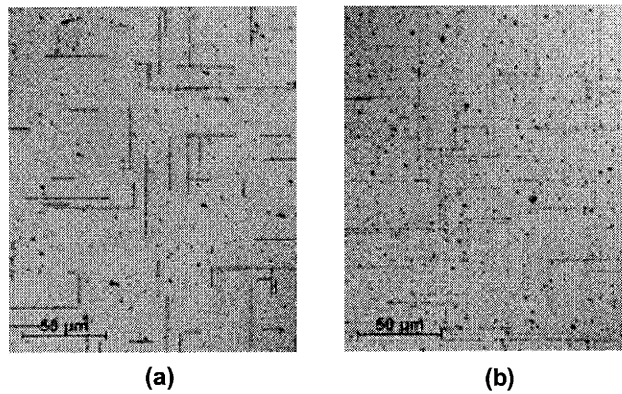


Figure 6.11 Nomarski optical images of EPD results for the 80-nm strained-Si depositions: (a) as-grown, and (b) after a 1h 800°C anneal in N₂.

EPD was used to determine the threading dislocation density for each sample.

Figure 6.10 and Figure 6.11 show surface optical images (representative of the entire surface) of the 40 nm and 80 nm strained-Si samples, as-grown and after annealing. EPD results for the 40 nm strained-Si sample reveal an increase in threading dislocations, but did not reveal any misfit dislocations. The absence of visible misfit via EPD is not conclusive, and needs to be verified with PVTEM, which in our case revealed a multitude of misfit at the strained-Si interface. In contrast, Figure 6.11 shows that, after annealing, the 80-nm strained-Si sample experienced increases in both threading and misfit

dislocations. Again, since EPD may not reveal all misfit, the absolute magnitude of misfit density cannot be deduced from this data.

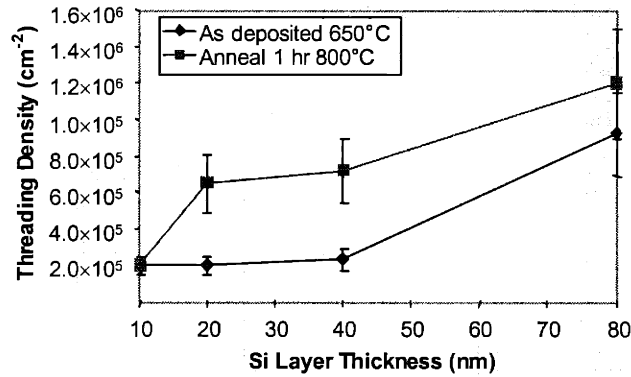


Figure 6.12 Threading dislocation density versus strained-Si thickness, for 650°C as-grown samples, and after the 800°C anneal.

A summary of the EPD threading densities for as-grown and annealed samples is presented in Figure 6.12. For comparison, note that the threading density for the underlying $\text{Si}_{0.75}\text{Ge}_{0.25}$ virtual substrates was found to be $2.0 \times 10^5 \text{ cm}^{-2}$, dubbed the baseline threading density. The as-deposited 10, 20 and 40 nm strained-Si samples had threading densities that were almost identical to that of the underlying virtual substrate. This observation indicates that at the growth temperature of 650°C and for thicknesses less than 40 nm, the thermal energy (for the given strain energy) is not high enough to nucleate a substantial number of new threading dislocations. In contrast, the 80-nm strained-Si had a large number (about $6 \times 10^5 \text{ cm}^{-2}$) of newly nucleated threading dislocations due to the larger strain energy present in the structure (since strain energy scales with layer thickness).

Annealing the samples at 800°C for 1 h dramatically influenced the threading dislocation density. In all samples with strained layers greater than the critical thickness

(16 nm), annealing led to the nucleation of new threading dislocations. Based on this data, we conclude that from the perspective of threading dislocation density, metastable strained layers with thicknesses less than 40 nm are acceptable as long as they are not exposed to temperatures greater than the growth temperature.

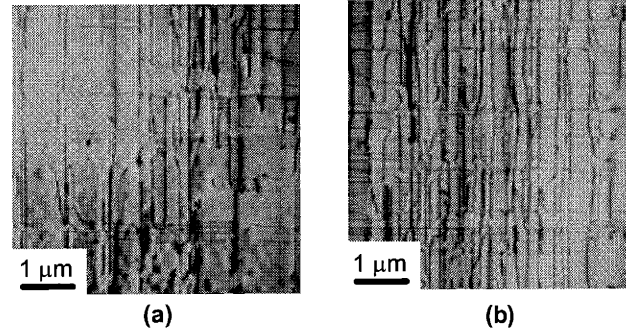


Figure 6.13 PVTEM micrographs for the 40 nm strained-Si samples: (a) as-grown, and (b) after the 800°C anneal.

Although low threading densities are important for high-quality device material, an additional concern is the degree of relaxation in metastable layers. The relaxation is directly related to the misfit density, and can be estimated using PVTEM. PVTEM images for the 40-nm strained-Si samples are presented in Figure 6.13, showing the presence of misfit dislocation in the as-grown sample, and the introduction of a substantial number of additional misfit segments after the 800°C anneal. The misfit density ρ_{md} can be used to calculate the plastic strain δ (*i.e.* strain relieved due to misfits) using

$$\delta = b_{eff} \rho_{md} \quad \text{Eqn. 6.1}$$

where b_{eff} is the effective Burgers vector given by $b_{eff} = a_{Si}/\sqrt{2}$ and a_{Si} is the lattice constant of Si.¹⁵⁸ In addition, the strain due to the difference in lattice mismatch is given by

$f=(a_f-a_s)/a_f$, where a_f and a_s are the film and substrate lattice constants, and in our case $f=0.0106$. The total mismatch strain is in turn distributed between elastic strain ϵ , and the plastic strain due to misfit dislocations δ , hence $f=\epsilon+\delta$. Finally the degree of strained-Si relaxation is calculated using

$$\text{Relaxation Factor} = \frac{\delta}{f}. \quad \text{Eqn. 6.2}$$

	Strained Si Thickness (nm)	ρ_{md} (cm ⁻¹)	S (μm)	δ	Relaxation Factor (%)
As Grown	10	0	0	0	0%
Annealed	10	0	0	0	0%
As Grown	20	0	0	0	0%
Annealed	20	$<9.1 \times 10^2$	>11	$<3.5 \times 10^{-5}$	$<0.3\%$
As Grown	40	2.2×10^4	0.45	8.5×10^{-4}	8%
Annealed	40	5.2×10^4	0.19	2.0×10^{-3}	19%

Table 6.1 Misfit densities and strain relaxation factors for various strained-Si samples, where the misfit density ρ_{md} is measured using PVTEM, and the misfit spacing S, plastic strain δ , and relaxation factors are calculated based on ρ_{md} . The as grown samples were deposited at 650°C, and then annealed for 1 h at 800°C.

Results of calculations based on PVTEM results are presented in Table 6.1. No dislocations were detected for the as-grown and annealed 10-nm strained-Si samples, nor for the as-grown 20-nm strained-Si sample, indicating the absence of any significant strain relaxation. In contrast, after annealing, the 20-nm strained-Si sample had a sparse number of misfit dislocations, yielding a relaxation factor of at most 0.3%, hence the layer is still almost fully strained even after 1 h at 800°C. In contrast, the as-grown 40 nm strained-Si sample already had a detectable misfit array, with a corresponding

relaxation of about 8%. The presence of a misfit array with no significant increase the threading density above the base-line density (as shown in Figure 6.12) implies that during growth at 650°C, relaxation for the 40 nm strained-Si occurs via glide of existing threading dislocations, rather than by nucleation of new threads. Annealing the 40 nm strained-Si sample at 800°C for 1 h, led to a doubling of the misfit density, implying a relaxation factor of 19%.

The results allow for the establishment of limits on the thickness of metastable strained-Si layers grown at low temperatures. In the case of as-grown metastable layers, it was found that thicknesses less than 40 nm experienced almost no strain relaxation or increase in threading densities. As a conservative rule of thumb, we deduced that for depositions at a temperature of 650°C or lower, metastable strained-Si layers with thickness up to double the critical thickness show no significant increase in threading density, and no substantial misfit density or strain relaxation. Annealing at elevated temperatures can introduce a measurable amount of strain relaxation via misfit formation, and the nucleation of additional threading dislocations.

Given the above conclusions, the structure chosen for SSOI and SGOI process development and characterization had a 30-nm metastable strained-Si stop with a 30-nm relaxed $\text{Si}_{0.75}\text{Ge}_{0.25}$ transfer layer. The structure was grown at 550°C, and had no detectable misfit or additional threading dislocations. After wafer bonding and delamination via implantation (which transferred a 870 nm thick layer), the sample was die-sawed into $1 \times 1 \text{ cm}^2$ squares, and various SiGe etches were used to remove excess $\text{Si}_{0.75}\text{Ge}_{0.25}$, while stopping on the strained-Si.

6.6 SiGe Selective Etching using $\text{H}_2\text{O}_2\text{:HF:HAc}$

Previous work by Carns *et al.*¹⁵⁴ showed that solutions of $\text{H}_2\text{O}_2\text{:HF:HAc}$ selectively etch SiGe alloys over Si. In particular, they found that a solution of $\text{H}_2\text{O}_2\text{:HF:HAc}$ in proportions of 2:1:3 etched bulk, n-type Si at a rate of about 0.1 nm/min, whereas we measured a slightly larger bulk Si etch rate of 0.3 nm/min. Irrespective of the precise value of the bulk Si etch rate, relaxed $\text{Si}_{0.75}\text{Ge}_{0.25}$ etches at a rate that is about two orders-of-magnitude larger. In particular, etch measurements on our undoped $\text{Si}_{0.75}\text{Ge}_{0.25}$ virtual substrates indicate an etch rate of 38 nm/min, which in combination with the Si etch rate, yields a selectivity of about 130. High selectivity would seem to make this etch ideal for the purposes of removing excess SiGe and stopping on a strained-Si stop layer.

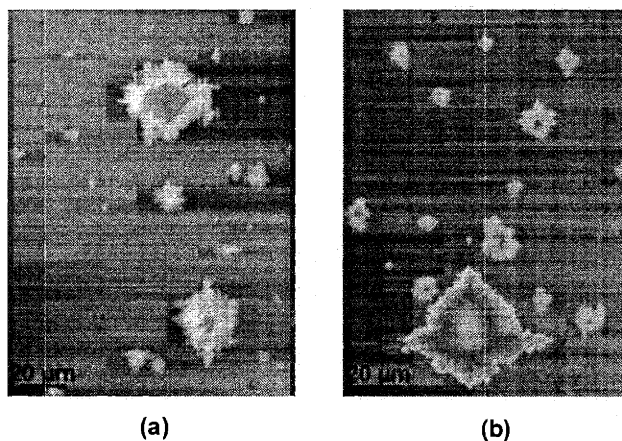


Figure 6.14 Optical images of the strained-Si stop surface after etching in $\text{H}_2\text{O}_2\text{:HF:HAc}$ with constituent ratios (a) 2:1:3 and (b) 12:1:2.

To test the solution, a $1 \times 1 \text{ cm}^2$ die of the layer transferred sample was etched in a $\text{H}_2\text{O}_2\text{:HF:HAc}$ (2:1:3) solution for a total of about 28 min until all the $\text{Si}_{0.75}\text{Ge}_{0.25}$ was removed and the strained-Si was totally exposed. Although the etch stopped on the strained-Si layer, upon immediate exposure to the solution, tiny depressions (with

diameters ranging from one to tens of microns) developed on the strained-Si surface, as shown in Figure 6.14(a). The density of these depressions ranged from about 7×10^4 to $1 \times 10^5 \text{ cm}^{-2}$, and since the dislocation density in the samples was $2 \times 10^5 \text{ cm}^{-2}$, the similarity in densities points to the possibility that the depressions are related to preferential dislocation etching. Various ratios of $\text{H}_2\text{O}_2\text{:HF:HAc}$ were tested in an attempt to modify the etch and minimize the formation of depressions. Figure 6.14(b) shows that surface of the strained-Si etch-stop after etching in a $\text{H}_2\text{O}_2\text{:HF:HAc}$ (12:1:2) solution, revealing the persistent presence of the depressions.

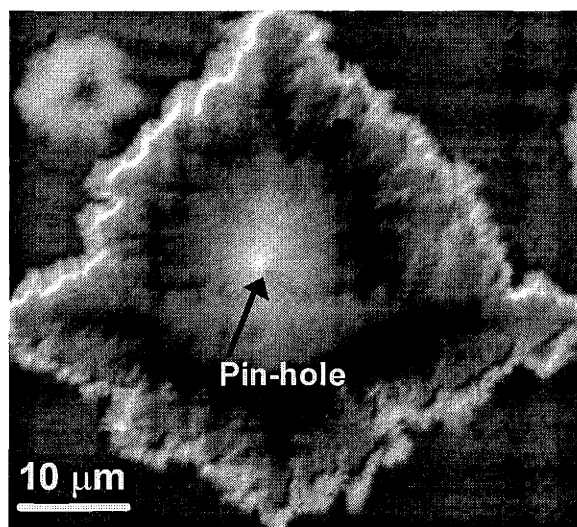


Figure 6.15 Optical image of a large depression in strained-Si stop surface, showing pin-hole in the center.

Figure 6.15 is a magnified optical image of one of the largest depressions, illustrating the existence of a pin-hole at the center; in this particular case, the pin-hole diameter was estimated to be about 0.1 μm based on the optical images. Sub-micron sized pin-holes may originate from the preferential etching of threading dislocation cores. Once formed, pin-holes could conceivably act as channels for the etching solution, thereby allowing the etching of the underlying $\text{Si}_{0.75}\text{Ge}_{0.25}$, and leading to under-cutting,

as shown in Figure 6.16. In turn, this under-cutting causes the formation of the observed surface depressions in the strained-Si.

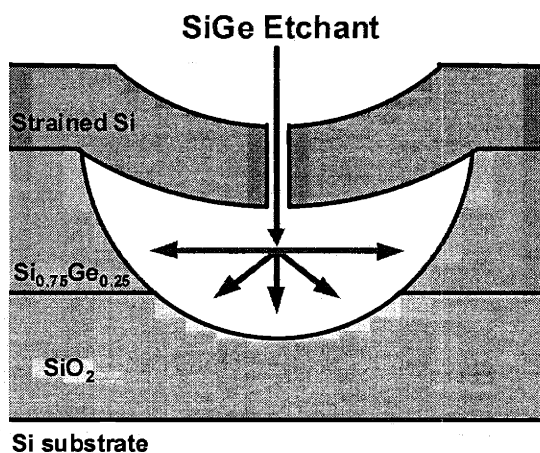


Figure 6.16 Diagram illustrating the $\text{Si}_{0.75}\text{Ge}_{0.25}$ under-cutting process due to the presence of a pin-hole through the strained-Si stop layer.

It is conjectured that the $\text{H}_2\text{O}_2\text{:HF:HAc}$ etch may cause preferential dislocation core etching due to the highly concentrated HF content of the etch. As is known from EPD techniques used for measuring dislocation densities, etches that preferentially etch dislocations typically have very high HF concentrations (combined with one or more oxidizing agents).⁵⁴

In conclusion, although the $\text{H}_2\text{O}_2\text{:HF:HAc}$ etch chemistry had excellent selectivity for SiGe etching over Si, the appearance of pin-holes that leading to under-cutting (due to preferential dislocation etching) was a serious problem. Varying the constituent ratios (with the constraint of retaining a reasonably fast SiGe etch rate) did not resolve the problem, therefore a different etch chemistry with good SiGe selectivity and no preferential dislocation etching was sought.

6.7 SiGe Selective Etching using HNO₃:dHF:HAc

By changing both the oxidizing agent and drastically reducing the HF concentration, we devised a HNO₃:dHF:HAc etch that had the necessary selectivity, with a fast SiGe etch rate, and with no indication of preferential dislocation etching.¹⁵⁶ The etch rate depended strongly on the constituent concentrations. Based on chemical kinetics, etch rates follow a power-law dependence given by¹⁵⁶

$$\text{Etch Rate} = k [\text{HNO}_3]^A [\text{dHF}]^B \quad \text{Eqn. 6.3}$$

where [HNO₃] and [dHF] represent respective HNO₃ and dHF concentrations, and k, A and B are materials dependant constants. To optimize the SiGe etch rate and selectivity, we measured the etch rate of Si(100) and relaxed Si_{0.75}Ge_{0.25} virtual substrates for various concentrations of HNO₃ and dHF. The data was then plotted and fit to determine k, A, and B for Si and Si_{0.75}Ge_{0.25}.

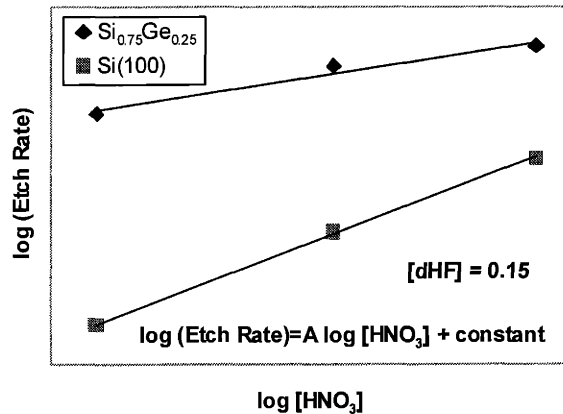


Figure 6.17 Log-log plot of etch rate versus HNO₃, at constant [dHF]=0.15, used to fit the A constant for Si and Si_{0.75}Ge_{0.25}.

Figure 6.17 is a log-log plot of etch rates versus [HNO₃] for constant [dHF]=0.15; fitting a line to each data set allows for the extraction of A for Si and Si_{0.75}Ge_{0.25} etching.

about 820 nm thick, implying a projected etch time of about 32 min. A higher selectivity could have been attained by using lower HNO_3 concentrations, but the corresponding etch rate would have been too slow for the removal of the thick $\text{Si}_{0.75}\text{Ge}_{0.25}$ layer. For example, solutions with $[\text{HNO}_3]=0.25$ are projected to have a selectivity of about 110, but a $\text{Si}_{0.75}\text{Ge}_{0.25}$ etch rate of only 5 nm/min. For the current $\text{Si}_{0.75}\text{Ge}_{0.25}$ removal thickness of 820 nm, the etch rate implies an estimated etch time of almost 3 h, which is unreasonably long. On the other hand, since the excess $\text{Si}_{0.75}\text{Ge}_{0.25}$ thickness is determined solely by the implant depth, the thickness can easily be reduced. Given that after delamination, the peak-to-valley roughness is on the order of 100 nm, a safe lower limit would be an implant depth no closer than 100 nm from the etch-stop layer. In that case, the etch time using the highly selective $[\text{HNO}_3]=0.25$ etch would only be about 20 min, allowing for the effective use of etching solutions with a very high selectivity of 100 or greater.

6.8 SGOI and SSOI Fabrication Results

Detailed material characterization is now presented for ultra-thin SSOI fabricated using the delamination via implantation method, with a metastable strained-Si stop layer. Results using the grind and etch-back approach were similar and the final structure was the same as for the delamination via implantation method. Results presented here used $\text{HNO}_3:\text{dHF}:\text{HAc}$ (45:15:40) etching after layer transfer, to create high-quality SSOI with a 30-nm strained-Si stop layer and a 30-nm relaxed $\text{Si}_{0.75}\text{Ge}_{0.25}$ transfer layer. One liter batches of the etch solution were employed for the etching of $1\times 1\text{ cm}^2$ layer transferred dies. The potency of the etch was found to decrease significantly after 20 min of etching,

A similar plot where [dHF] was varied and [HNO₃] was kept constant allowed for the extraction of the B and k constants. For Si and relaxed Si_{0.75}Ge_{0.25}, the fitting parameters were A_{Si}=6.6, B_{Si}=3.7, k_{Si}=5.0×10⁵ nm/min, A_{SiGe}=2.7, B_{SiGe}=1.4, and k_{SiGe}=3.2×10³ nm/min.

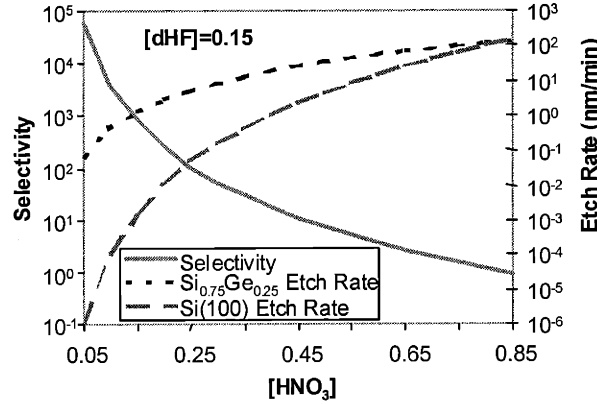


Figure 6.18 Calculated Si and Si_{0.75}Ge_{0.25} etch rates and selectivity for varying [HNO₃] and constant [dHF]=0.15.

Fitting parameters then allowed for the calculation of both the etch rates and the selectivity over a wide range of constituent concentrations. The selectivity was given by the relaxed Si_{0.75}Ge_{0.25} etch rate over the Si etch rate. Figure 6.18 shows the calculated selectivity and etch rates with dHF held constant at 15%, [dHF]=0.15. As a function of decreasing [HNO₃], the Si etch rate decreases faster than the Si_{0.75}Ge_{0.25} etch rate (as can also be seen from Figure 6.17), leading to increased selectivity for lower HNO₃ concentrations. The trade-off for attaining this extremely high selectivity (for low [HNO₃]) is a significant decrease in the Si_{0.75}Ge_{0.25} etch rate.

For the purposes of the current work, [HNO₃]=0.45 and [dHF]=0.15 was predicted to yield an acceptable selectivity of about 11 and a relaxed Si_{0.75}Ge_{0.25} etch rate of 26 nm/min. In the current case, the amount of Si_{0.75}Ge_{0.25} that needed to be removed was

hence a fresh batch was used after every 20 min; the reduction in etch rate versus time is possibly due to the consumption of the dilute HF. Etch rates could be stabilized by replacing the HF in the solution with BOE (buffered oxide etch), which contains a buffering agent consisting of ammonium fluoride (NH_4F), that maintains the HF concentration. In such a stabilized etch, one batch of solution could be employed for prolonged etch times, without the need to replenish the HF.

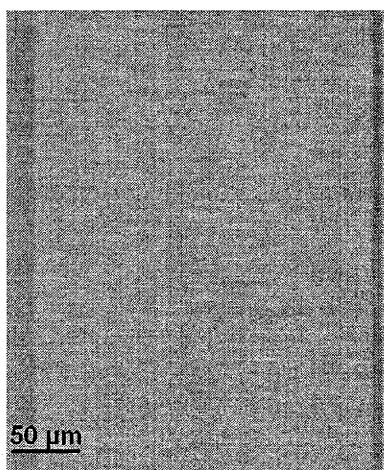


Figure 6.19 Optical image of the SSOI surface after the etch stops on the strained-Si.

Based on visual observations of sample surface color, it appears that the strained-Si etch-stop surface was exposed after about 40-45 min of etching, since after an additional 10 min, the surface appearance was not altered. Figure 6.19 is a Nomarski optical image of the sample surface after 45 min, with no indication of any etch-related defects, like depressions or pin-holes. The image also shows the presence of a mild cross-hatch pattern, but this is not due to roughness at the etching surface. Rather, this cross-hatch is present at the bonding interface and is the mild surface roughness that was re-introduced

during the post-CMP, low-temperature growth of the stop layer, as was shown in Figure 6.7(b).

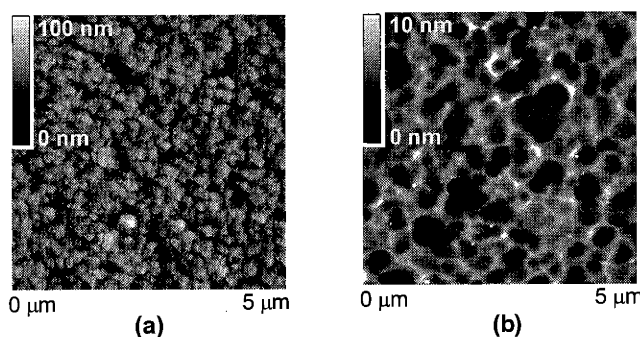


Figure 6.20 AFM images of (a) very rough delaminated $\text{Si}_{0.75}\text{Ge}_{0.25}$ surface, (b) smooth surface after SiGe etching stops on strained-Si layer.

An AFM scan of the strained-Si stop surface after SiGe selective etching, Figure 6.20(b), proves that the surface has no cross-hatch pattern, and a peak-to-valley height of only 9 nm, with a rms roughness of about 1.1 nm. In contrast, immediately after delamination, the surface was extremely rough, as shown in Figure 6.20(a), with a peak-to-valley height of about 135 nm and rms roughness of 12 nm. Etching removed excess SiGe until the strained-Si stop was reached at valleys. Then, the etch slowly attacks the strained-Si at the valleys, until the peaks of the remaining SiGe are totally removed. Finally, the etch attacks the strained-Si evenly across the entire sample surface. This process leads to a transference of any surface roughness after delamination (scaled by the inverse of the selectivity) to the etch-stop surface. In the present case, the selectivity is about 11, which is consistent with an order of magnitude reduction in roughness, as indicated by the AFM data in Figure 6.20. Transference of delamination surface roughness to the etch-stop layer could easily be reduced even further by utilizing a higher selectivity, as could be obtained by the use of low HNO_3 concentrations.

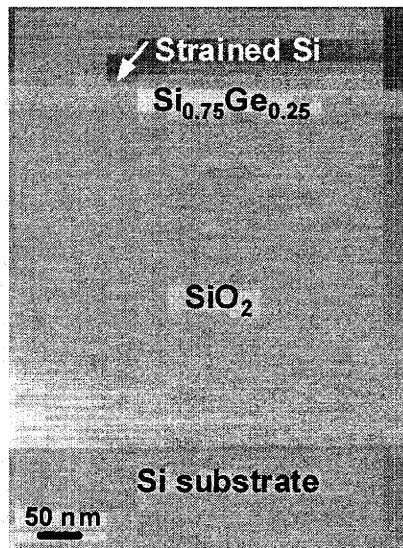


Figure 6.21 Cross-section transmission electron micrograph (XTEM) of SSOI sample consisting of a strained-Si stop and a ultra-thin SiGe transfer layer.

A cross-section transmission electron microscopy (XTEM) image of the SSOI structure shows that the etch has indeed stopped on the strained-Si stop, as shown in Figure 6.21. To conclusively prove that the 30 nm $\text{Si}_{0.75}\text{Ge}_{0.25}$ layer was fully relaxed, and the strained-Si was fully strained, Raman spectroscopy using a 514.5 nm laser was performed on the final SGOI structure. As shown in Figure 6.22, the spectrum has three peaks in the 500 to 520 cm^{-1} range, corresponding to Si-Si bonds in the relaxed $\text{Si}_{0.75}\text{Ge}_{0.25}$ bonding layer, the strained-Si etch-stop layer, and the Si substrate.

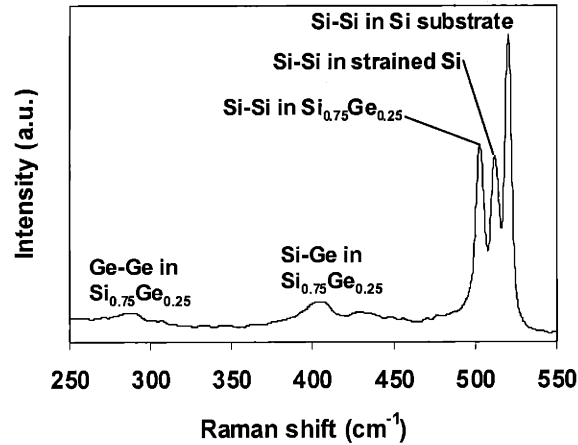


Figure 6.22 Raman spectrum of a SSOI sample with strained-Si on the surface.

Using the location of these peaks, the concentration and strain of the $\text{Si}_{0.75}\text{Ge}_{0.25}$ bonding layer and the strain in the Si etch-stop layer were deduced.⁵⁶ The strained-Si layer was found to be tensile, with a strain of 1.02%, corresponding to a fully strained-Si layer on a relaxed $\text{Si}_{0.75}\text{Ge}_{0.25}$ virtual substrate. This proves that although the original strained-Si layer thickness of 30 nm was greater than the equilibrium critical thickness by about two-fold, relaxation via misfit generation did not occur. Since the strained-Si was fully strained, this in turn implies that the 30-nm $\text{Si}_{0.75}\text{Ge}_{0.25}$ layer (deposited on the strained-Si) was fully relaxed, as verified by our Raman spectra calculations.

Finally, although the results presented above were for ultra-thin layers, the method can just as easily be applied to the fabrication of thick, uniform, smooth, relaxed SiGe layers on oxide, which could be of great use for optical waveguide applications. XTEM of a thick SGOI substrate fabricated using grind and KOH etch-back combined with a strained-Si stop is presented in Figure 6.23.

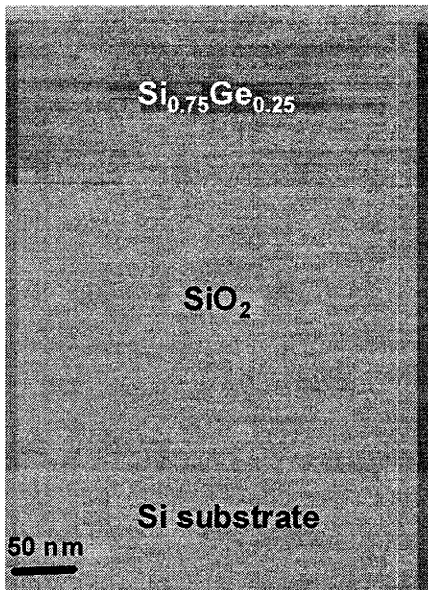


Figure 6.23 XTEM of SGOI structure possessing a thick, uniform, smooth, relaxed $\text{Si}_{0.75}\text{Ge}_{0.25}$ layer.

6.9 Other Versions of SGOI and SSOI Fabrication using Wafer Bonding

The demonstration presented in this work is but one possible embodiment of SGOI and SSOI fabrication using our proposed wafer bonding method. The process used a strained-Si stop layer and a relaxed $\text{Si}_{1-x}\text{Ge}_x$ transfer layer, where the strained-Si was not removed after SiGe selective etching, as shown in Figure 6.24(a). The process could have been just as easily executed without the relaxed $\text{Si}_{1-x}\text{Ge}_x$ transfer layer, which would have led to the fabrication of a SSOI substrate with strained-Si directly on oxide, as shown in Figure 6.24(b). That embodiment could be of use in the fabrication of ultra-thin (<30 nm) transferred layers, for fully depleted MOSFETs on SSOI substrates.¹⁵²

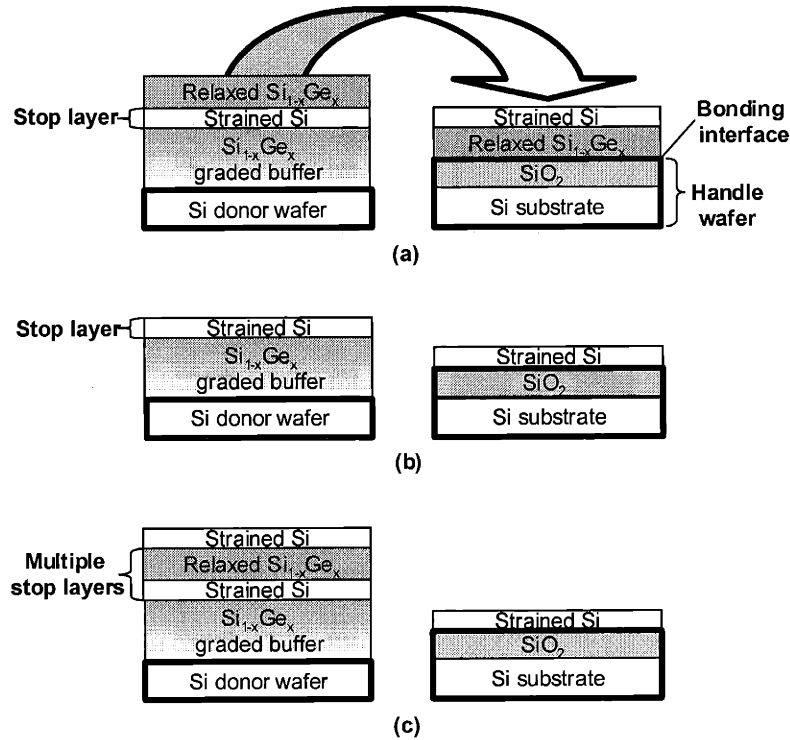


Figure 6.24 Other versions of the SGOI and SSOI fabrication method (a) using of a strained-Si stop layer and a Si_{1-x}Ge_x transfer layer, where the stop is not stripped after SiGe removal, (b) directly bonding strained-Si stop to insulator, and (c) using multiple stop layers that are stripped after bonding and SiGe removal.

Another version of our SGOI and SSOI fabrication process is the use of multiple stop layers; for example, a multiple stop could consist of a strained-Si layer followed by a thin, relaxed Si_{1-x}Ge_x layer, as shown in Figure 6.24(c). This structure is then capped with the desired device layers, for example, a strained-Si layer. After bonding and layer transfer, SiGe selective removal stops on the strained-Si. A low-temperature KOH (or TMAH) etch is then employed to remove the strained-Si etch-stop while stopping on the ultra-thin Si_{1-x}Ge_x. Any selective SiGe removal process is then used to strip the thin Si_{1-x}Ge_x layer. The final structure consists of only the device layers (e.g.: strained-Si) on the insulating substrate.

6.10 Summary

Two novel processes were pioneered, developed, and optimized for fabricating high-quality, low-defect, monocrystalline, relaxed SiGe-on-insulator (SGOI) and strained-Si-on-insulator (SSOI) substrates. An extensive study was conducted on the use of metastable strained-Si etch-stop layers, low-temperature wafer bonding, and various selective SiGe etches, as applied to the improved fabrication of ultra-thin SSOI and SGOI using wafer bonding. The technique may serve as an enabling technology for the fabrication of future substrates used in upcoming CMOS generations, in addition to advanced photonic and opto-electronic applications.

Chapter 7 Conclusions and Suggestions

SiGe-on-insulator (SGOI) and strained-Si-on-insulator (SSOI) substrates provide a multitude of new avenues for novel device applications, and both microelectronics and photonics stand to benefit greatly from the commercialization of SSOI and SGOI substrates. If production yields for wafer bonded SSOI and SGOI can be increased to meet the huge demand required by chip manufacturers, these substrates stand to become the future platforms of choice for advanced high-speed, low power CMOS microelectronics.

7.1 Summary of Experimental Work

Many of the SiGe structures and devices considered in this thesis revolved around the fabrication of SiGe-on-insulator. In the case of ϵ CMOS applications, a buried oxide structure combined with a strained-Si surface layer (potentially on relaxed SiGe) is an ideal substrate for high-speed, low power circuits. Also, with regards to ncGe optical waveguides on Si substrates, a surface layer of SiGe is required for the nanocrystalline-Ge formation process, and a thick buried oxide layer is essential for optical confinement.

Before delving into the creation of a general fabrication method for SGOI, a detailed study of SiGe oxidation was carried out. Wet oxidation of SiGe with various Ge content was performed and oxidation rates were determined. At temperatures considered, SiGe was found to hydro-thermally oxidize much faster than Si. At 700°C, the oxidation rate enhancement of $\text{Si}_{0.75}\text{Ge}_{0.25}$ compared to Si is about a factor of 60 for a 1 h oxidation, making wet oxidation a useful selective SiGe oxidation mechanism. For short times, at

700°C, wet oxidation rate enhancements were attributed to an increase in the Ge oxidation reaction rate constant and water solubility, whereas for long times, the enhancement was mainly attributed to an increase in the water diffusivity and solubility in SiGe mixed oxide. In addition, 800°C wet oxidation led to Ge expulsion from the oxide, creating a Ge pile-up at the oxidation front. The Ge pile-up effect creates pure SiO₂ and was attributed to fast Ge diffusion (away from the front) as compared to the oxidation rate. In contrast, at 700°C, a mixed oxide containing both SiO₂ and GeO₂ formed due to a slower rate of Ge diffusion as compared to the faster SiGe oxidation rate present at this temperature. Finally, selective, lateral oxidation of buried SiGe, sandwiched between Si, was performed in the hopes of developing a novel method for buried oxide fabrication. Results of these experiments revealed that the lateral oxidation of buried SiGe is rate-limited by stress effects due to volume expansion. A similar approach was employed to demonstrate how selective lateral oxidation could still be of use in the fabrication of isolation layers for vertical transistor structures. The knowledge gleaned from the SiGe oxidation experiments was crucial in understanding the limits of a multitude of previously proposed methods for SGOI fabrication, including the use of Ge expulsion via high-temperature oxidation, and SIMOX.

Once the properties of SiGe oxidation were mapped out, and it was concluded that 700°C wet oxidation led to the formation of a mixed oxide containing SiO₂ and GeO₂, with no Ge pile-up, the stability of mixed oxide was examined. It was found that the presence of a reducing agent, like Si or H₂ led to the reduction of the GeO₂, and the precipitation of elemental Ge. Based on this approach, hydrogen annealing was

employed to form Ge nanocrystals in an oxide matrix. The microstructure and size of the Ge crystallites was found to vary greatly with annealing temperature and hydrogen concentration. To understand the kinetics involved and hence better control the nanocrystal size distribution, a detailed transmission electron microscopy study was conducted, mapping out nanocrystalline Ge formation kinetics for the entire annealing parameter space. The general trends of these experiments indicated that the diffusivity of the elemental Ge in reducing oxide is orders of magnitude faster than Ge diffusion in thermal SiO₂. To concisely summarize the results of the annealing experiments, a mathematical model was developed which included the effects of reduction thermodynamics, product diffusion, nanocrystalline Ge nucleation and growth. With respect to optical properties, previous theoretical research indicated that Ge nanocrystals with diameters less than 15 nm should experience substantial quantum confinement, leading to a downshift of their absorption edge into the telecom infrared wavelength regime. Photoluminescence (at 4K) spectra of our nanocrystalline Ge samples had a broad background ranging from 1-1.7 μm , attributed to quantum confinement and consistent with a broad nanocrystal size distribution. The location of this PL peak in the telecom wavelength region pointed to the potential use of nanocrystalline Ge for telecom applications, in particular for nonlinear devices in the 1.3-1.5 μm window.

To explore the use of nanocrystalline Ge (ncGe) (formed via SiGeO₂ reduction) for potential optical applications, a method for integrating such material into waveguides was required. To address this issue, the task of fabricating ncGe-oxide waveguides on Si was undertaken. Integration of this waveguide structure onto Si substrates required the use of

a thick SiO₂ standoff layer to provide adequate optical confinement and minimize power loss into the Si substrate. Modal calculations indicated that a 3 μm thick SiO₂ standoff was required to attain negligible leakage, and to address this requirement, a variety of standoff fabrication methods were attempted. The naive approach of polycrystalline SiGe deposition onto SiO₂/Si, followed by low-temperature (700°C) oxidation to form the SiGe mixed oxide, was disastrous due to Ge pile-up formation during the poly-SiGe oxidation. Grain boundaries in the polycrystalline SiGe acted as fast diffusion channels, thereby allowing the expelled Ge to escape re-incorporation into the fast moving oxidation front. Due to this effect, further approaches were constrained to the use of epitaxial SiGe. A brute force approach employing a two-step wet oxidation, first (at 700°C) to create the SiGeO₂, and then (at 1100°C) to create the thick SiO₂ standoff, was determined to be unacceptable due to Ge out-diffusion, possibly via GeO gas formation. A final, successful approach using SiGe deposition on thinned SOI starting wafers was developed. Using this approach, a slab waveguide structure consisting of a nanocrystalline Ge layer with a thick, 3-μm SiO₂ standoff was fabricated, and was found to have broad photoluminescence in the telecom wavelength range. Strip waveguides and nonlinear directional couplers (NLDCs) were fabricated on the substrate, and plans are underway for ultra-fast pulse laser measurements to probe the nonlinear properties of the strip waveguides and NLDCs. During all the work on ncGe waveguides, it was concluded that SiGe-on-insulator (SGOI) substrates were the key to facilitating the process flow, providing more motivation for our final project on SGOI substrate fabrication.

Although a variety of techniques were explored to fabricate SiGe-on-insulator (SGOI) for various applications, it was determined that a versatile, generic process was in great demand. In particular, such a method also needed to allow for the formation of strained-Si-on-insulator (SSOI), which will likely become the ultimate low power, high-speed CMOS platform. To satisfy this demand, a generic SGOI and SSOI fabrication method was pioneered and developed in detail. This generic process addressed the issues present in the use of wafer bonding for layer transfer onto handle wafers, where the biggest problem was the lack of control over transferred layer thickness, making ultra-thin layers (needed for fully depleted CMOS) impossible to fabricate. To solve this problem, stop layers were incorporated into the structure and a multitude of SiGe etches were presented, tested, and optimized. In addition, the use of metastable, strained-Si stop layers was pioneered, in combination with low-temperature wafer bonding, which allowed for the metastable layers to remain fully strained, thereby inhibiting dislocation nucleation and relaxation via misfit extension. This novel SSOI and SGOI fabrication process will hopefully serve as the critical enabling technology for a new era in semiconductor-on-insulator microelectronics, opto-electronics and photonics.

7.2 Suggestions for Future Directions

Currently, the progression of SSOI and SGOI is actually inhibited by further research in the academic setting. The path required for further improvements involves movement of the fabrication technology into an industrial setting. In particular, the largest problem with academic wafer bonding experiments is the low wafer yields due to

particulate contamination present at the bonding surface. The use of industrial, fully-enclosed, automated wafer bonding equipment would alleviate this problem. Such fully-enclosed equipment has the capability to perform many of the critical steps in the wafer bonding process, namely cleaning, surface activation, wafer bonding, and annealing, thereby greatly reducing particulate contamination and improving wafer yields.

An area of SSOI and SGOI fabrication which could benefit from more research is the use of the patented wafer bonding approach for the fabrication of relaxed, high Ge content SiGe layers on oxide, namely 60-70% Ge. Such Ge concentrations have been shown to allow for large improvements in hole mobility, hence allowing for the fabrication of high-speed p-MOSFETs. Since high Ge, relaxed layers have very thin strained-Si critical thicknesses, the fabrication of SSOI and SGOI for these high Ge content structures will very likely employ the metastable stop technique proposed and demonstrated in this work.

With respect to nanocrystalline Ge formed via SiGe-oxide reduction, future structures that would yield less nanocrystal size broadening should be explored. Such structures could consist of multilayered, periodic SiGe/Si with film thicknesses on the order of the desired nanocrystal size, namely 5-10 nm. These structures would then be oxidized and annealed in a hydrogen environment, similar to the process employed in this thesis, but high-pressure oxidation might be used to quickly oxidize the intermediate Si layers at 700°C. In addition, ultra-fast (short pulse, high instantaneous power) laser testing of nonlinear ncGe waveguides and NLDCs fabricated in this work should also be completed.

Finally, once SGOI substrates are commercialized, future research into ncGe waveguides and nonlinear devices would be worthwhile. The existence of commercial SGOI would greatly simplify the fabrication process for ncGe waveguides formed using the reduction of SiGe-oxide, thereby liberating researchers from the delicate task of substrate fabrication and allowing them to focus on novel device applications.

===Let the era of SGOI and SSOI begin===

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